# Introduction to Virtualization



# In this lecture

- **Topics to be Covered**:
  - What is Virtualization?
  - Why Virtualization?
  - How Virtualization is done?

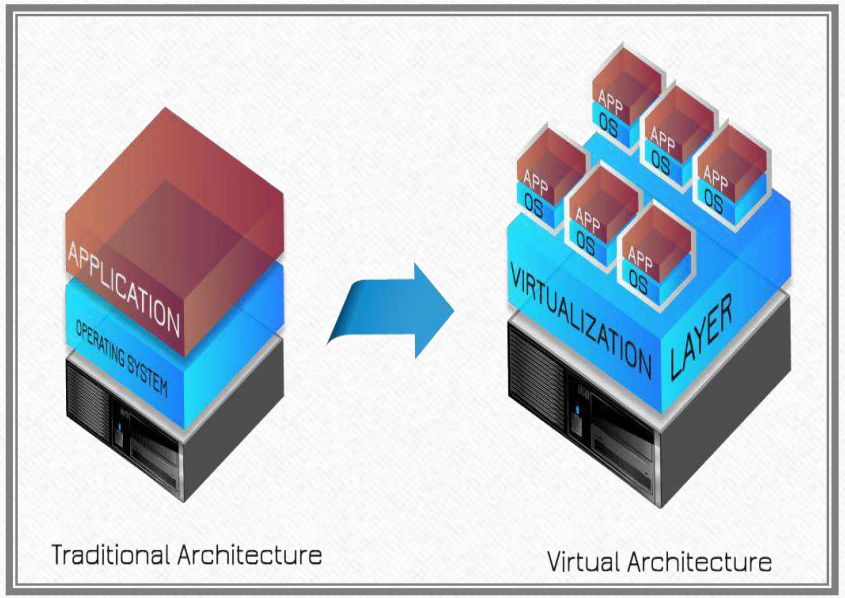


Virtualization is the process of using software to create virtual versions of physical hardware, like computers, servers, or networks, allowing multiple systems to run on one machine efficiently.

Virtual University

# What is Virtualization?

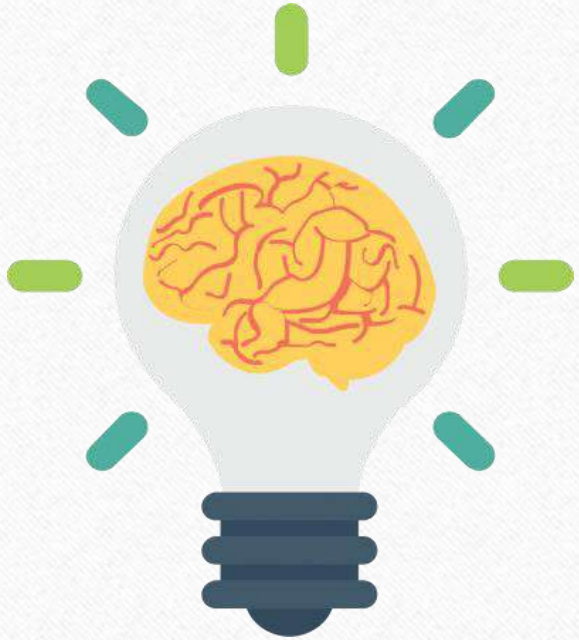
- Pretty old but to critical to understand
- A system pretends to be two or more
- Simulation of hardware devices or features using software
- Used to create virtual version of something (like network)



# Why Virtualization?

Benifits

- Best use of resources
- Flexibility and scalability
- Increased agility
- Disaster Recovery
- Increased Security
- Cost saving



# How Virtualization

- Achieved through the use of virtualization software called Hypervisor
- Hypervisor creates virtualized versions of physical resources
- Hypervisor allocates these resources to virtual machines (VMs)
- VM operates as if it were a physical machine
- Run different applications, operating systems, and services using VMs

# Benefits of Virtualization



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# Benefits of Virtualization

- **Topics to be Covered:**
  - Resource utilization
  - Scalability and flexibility
  - Improved disaster recovery and business continuity
  - Increased security



# Resource Utilization




- Create virtualized versions of physical resources
- Use the same hardware more effectively
- Leading to better utilization of resources

# Scalability and flexibility

- Easier to scale resources up or down as needed
- No need to make physical changes to the underlying infrastructure



# Improved Disaster Recovery and Business Continuity

Copies

- Create replicas of their production environment
- Quickly activated in the event of a disaster
- Does not affects the end user



# Increased Security

- Isolate and secure sensitive applications and data
- It becomes harder for malicious actors to access them
- Minimal to no chances of virus attack or security breach



# Types of Virtualization

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# 6 Types of Virtualization

Virtual University

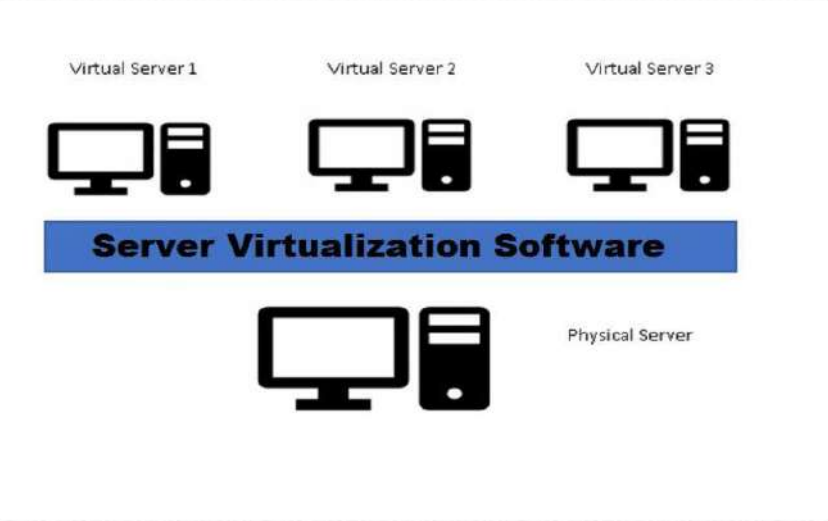
- **Topics to be Covered:**
  - Server Virtualization
  - Desktop Virtualization
  - Application Virtualization
  - Storage Virtualization
  - Network Virtualization
  - Hardware Virtualization

| Virtualization Type | Focus | Uses |
|---|---|---|
| **Server Virtualization** | Physical servers ko virtual servers mein divide karna | Server resources ko efficiently use karna |
| **Desktop Virtualization** | Desktop environments ko remote access ke liye central server pe run karna | Remote work aur centralized management |
| **Application Virtualization** | Applications ko bina install kiye run karna | Easy deployment aur compatibility |
| **Storage Virtualization** | Physical storage devices ko ek virtual pool mein combine karna | Storage ko efficiently manage karna |
| **Network Virtualization** | Physical network components ko virtualize karna | Network scalability aur flexibility |
| **Hardware Virtualization** | Physical machine pe multiple virtual machines chalana | Hardware ka efficient use karna |

# Server Virtualization

Dividing a physical server into multiple virtual servers, each running its own operating system and applications.

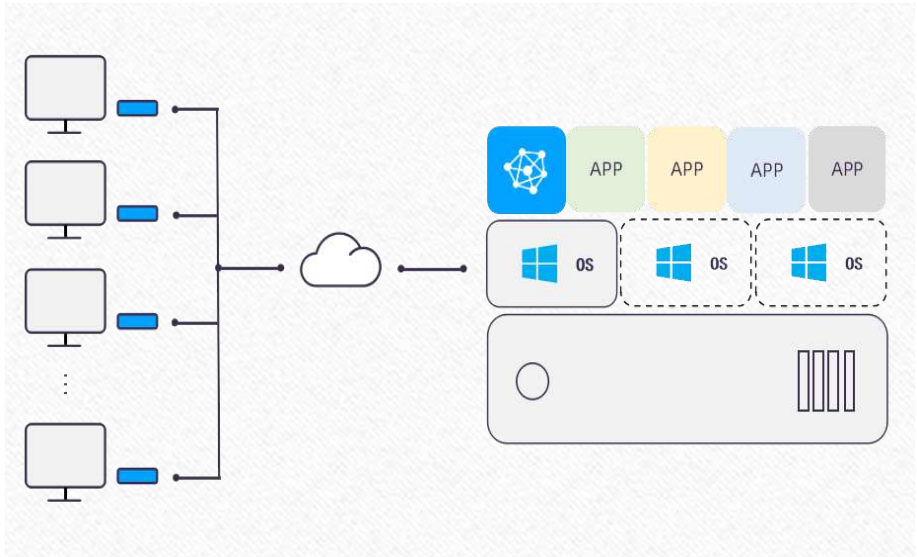
- Allows multiple virtual servers to run on a single physical server
- By sharing the underlying hardware resources



# Desktop Virtualization

Provides a virtual desktop environment, allowing users to access their desktops remotely.

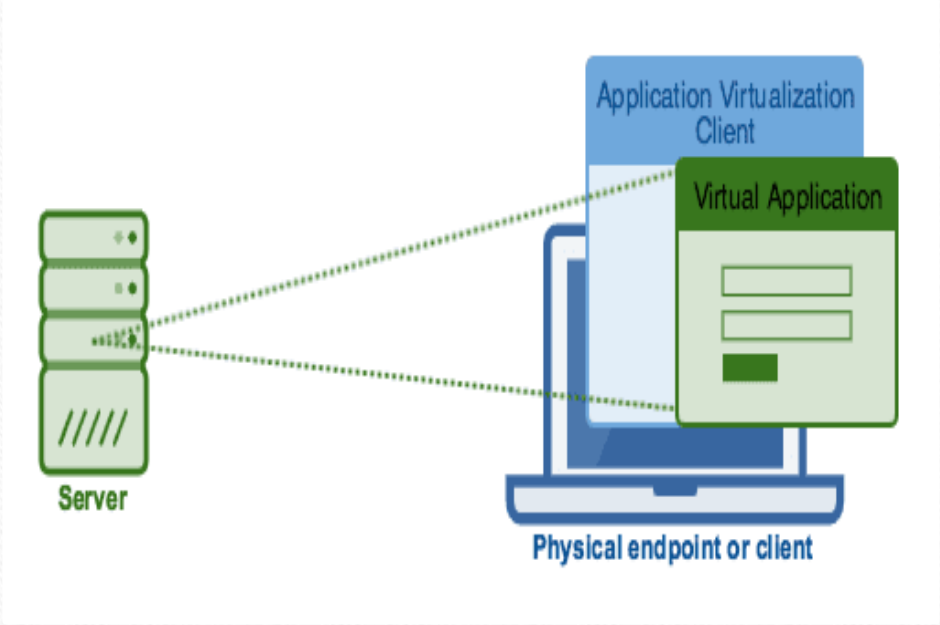
- Multiple virtual desktops to run on a single physical machine
- Each user have his separate desktop environment



# Application Virtualization

Allows multiple applications to run on a single machine without conflicts or compatibility issues.

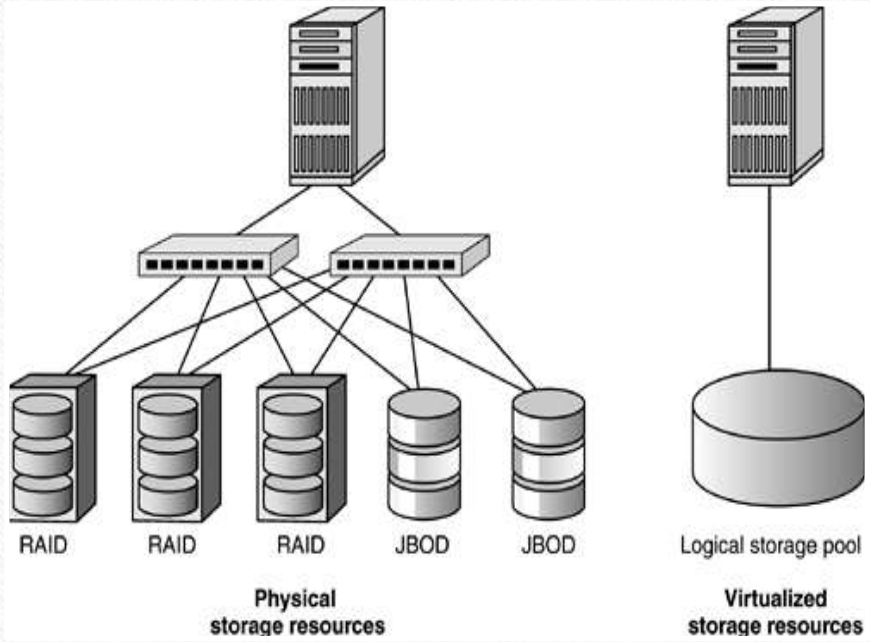
- Allows applications to run on a virtualized operating system
- Separate from the host OS



# Storage Virtualization

Combining multiple physical storage devices into a single virtual storage pool for easier management and better resource utilization.

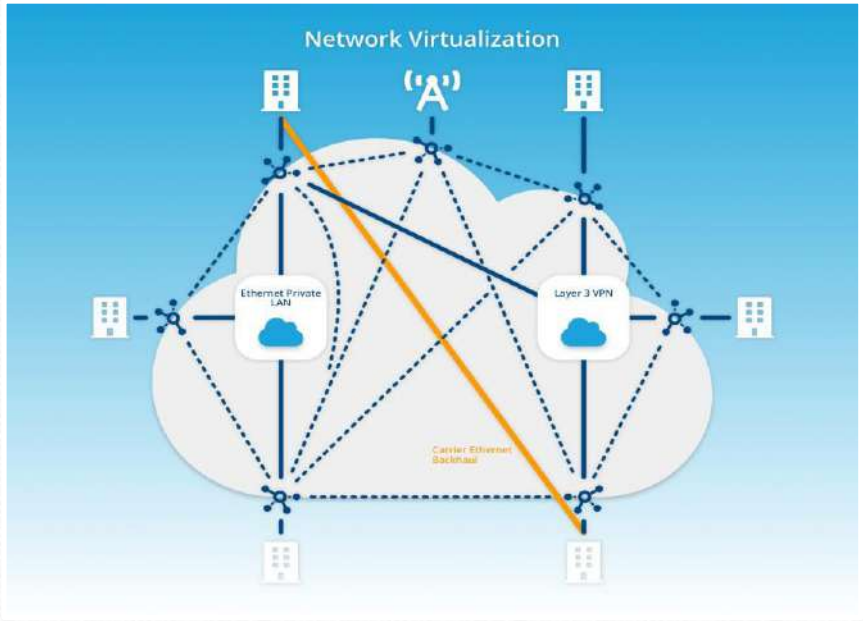
- Pools storage resources from multiple physical devices into a single virtual storage device
- Making it easier to manage storage and improve utilization



# Network Virtualization

Network Virtualization is a technology that allows multiple virtual networks to coexist on a single physical network infrastructure.

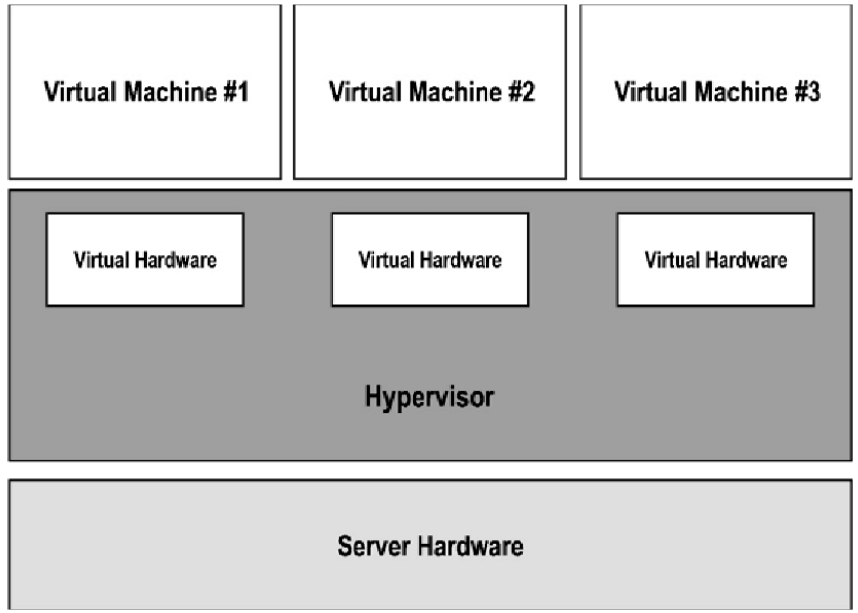
- Creates virtualized versions of physical network devices as such switches and routers
- Multiple virtual networks to run on a single physical network



# Hardware Virtualization

it allows multiple operating systems to run on a single physical machine.

- Creates virtualized versions of physical hardware devices, such as GPUs
- Allows multiple virtual devices to run on a single physical device



# Application level Virtualization



# Application Virtualization



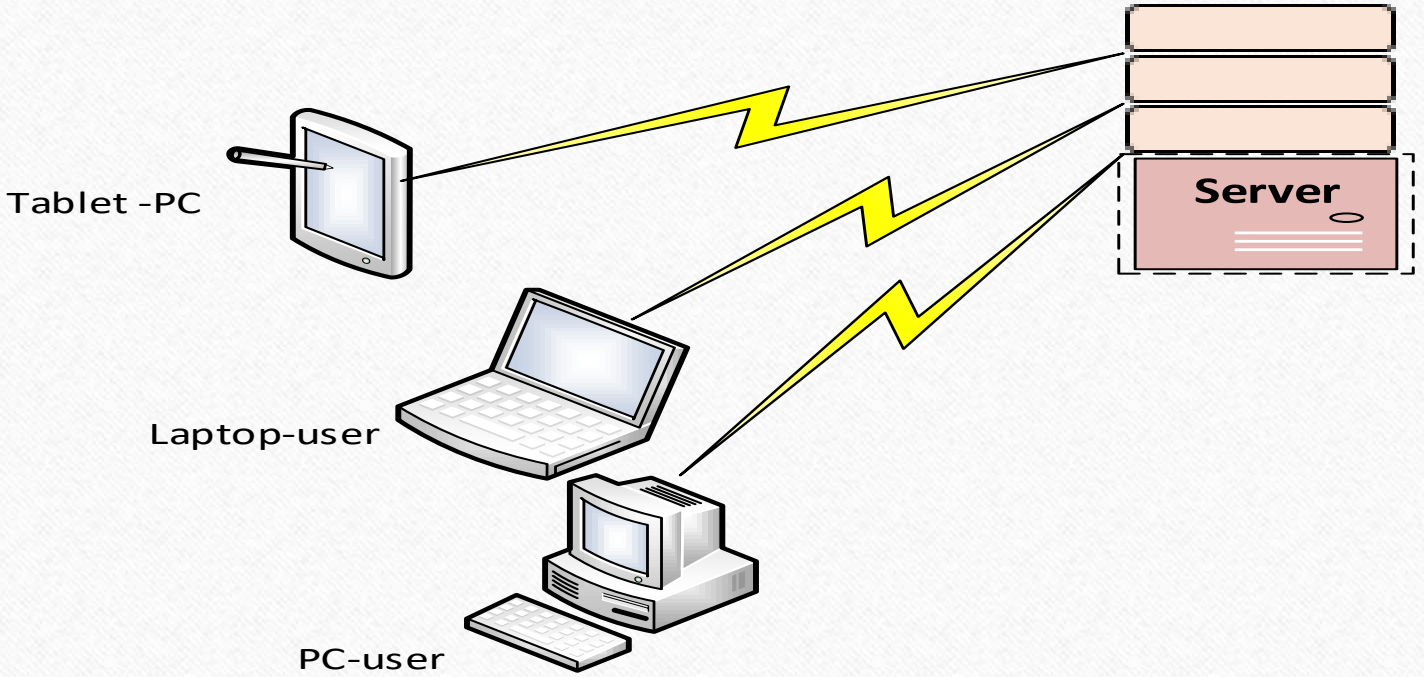

- **Topics to be Covered**:
  - What is Application Virtualization?
  - Benefits of Application Virtualization
  - Limitations of Application Virtualization

# What is Application Virtualization?




- Application level Virtualization is a technique in which client applications are hosted remotely instead of installing them on clients local devices.
- In this type of technology, the local devices provide physical resources like CPU and RAM that may be required to run the software, but nothing is installed locally on the device itself.
- This technology separates the application from the underlying OS that is execution of application is independent of OS
- In this environment, the application actually runs remotely but virtually runs locally.

# Application level Virtualization



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# Benefits of Application Virtualization




- **Reduce cost**
- **Enable Cross Platform Operations**: (Windows applications run on LINUX and vice versa) Cross-platform operations" ka matlab hai ke application virtualization ke zariye ek platform ke liye banai gai a pplication (jaise Windows) doosre platform (jaise Linux) par bhi chal sakti hai, bina us platform ke liye specific coding ya re-development ki zarurat ke.
- **Simplified Management**
- Application Virtualization allows an IT admin to manage and maintain applications across an organization rather than manually installing applications to every user's machine.
- Security and Stability Improved by isolating the application from the OS.

# Benefits of Application Virtualization

- Prevent conflicts with other Virtualized Apps
- Permits users to run multiple App Instances: Yeh feature ka matlab hai ke ek hi software ko ek se zyada dafa (instances) e k hi device par ek hi waqt chalaya ja sakta hai, bina kisi problem ke. Jesy Watsapp, Google tabs

Limitations:-

Single Point of Failure: One server problem affects all apps.
Bandwidth Requirements: Needs strong internet for smooth app performance.
Problematic Applications: Some apps don't work well in virtualized environments.
High Initial Investment: Setting up the infrastructure can be costly.
Challenging Quick Scalability: Scaling up virtualized apps quickly is complex and resource-intensive.

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# Limitations of Application Virtualization




- **Single Point of Failure**: launching all Apps on servers increases the risk of failure.
- **Bandwidth Requirements**: sufficient bandwidth required for Streaming Apps to end-users
- **Problematic Applications**: Some Apps need system drivers and don't work as virtualized applications.
- **High Initial Investment**: Initially migrating data to the Data Center or cloud
- **Challenging Quick Scalability**: to ensure all the requisite software, enough storage, security and resource availability is a tedious job.

# Solutions of Application Virtualization

# Solutions of Application Virtualization

- Different vendors provide methods or solutions to achieve Applications Virtualization.
- VMware Thin App
- Microsoft App-V
- Citrix Virtual Apps and Desktops
- Parallel Remote Application Server (RAS)

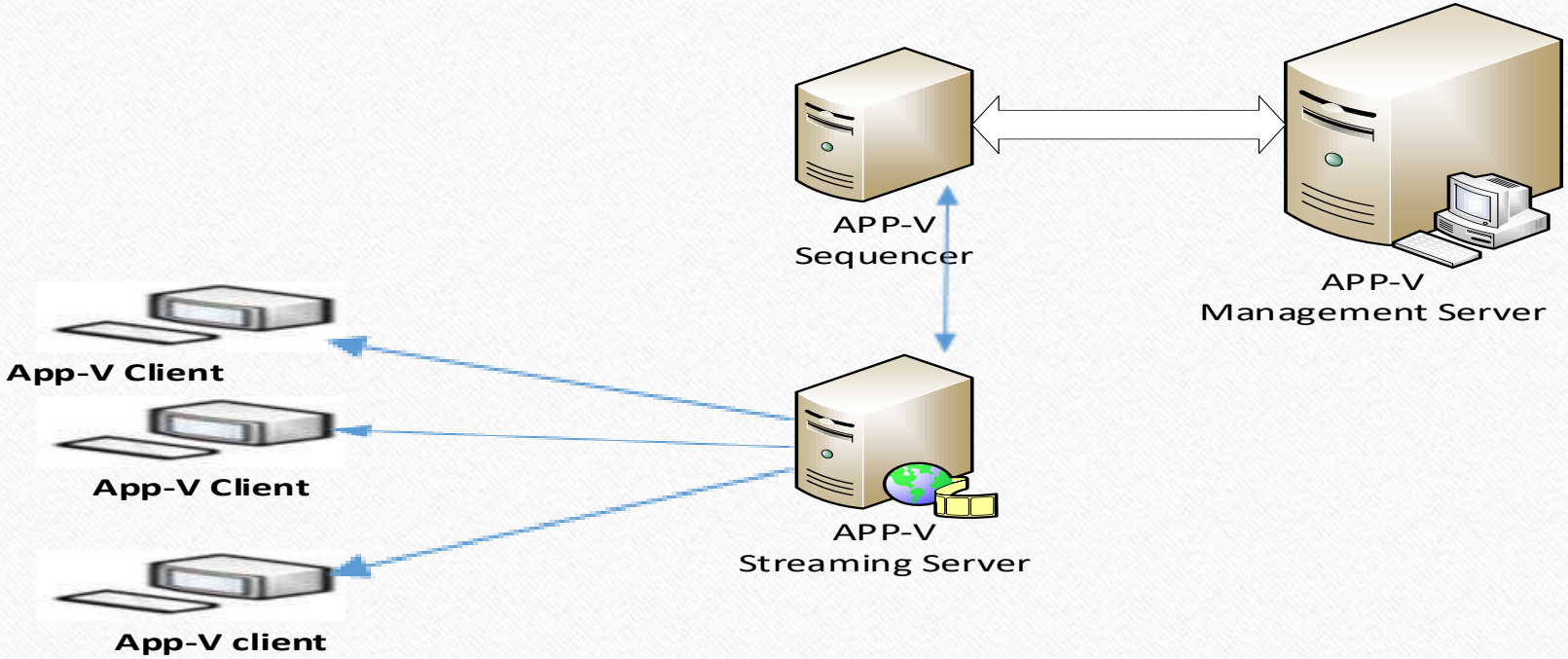
# Microsoft App-V

- **On Remote host,**
  - Virtualization App-V server components are required to be installed.
  - App-V server components include
    - App-V Management server,
    - App-V Sequencer
    - App-V Streaming Server
  - Virtual Applications are installed on centrally managed server
  - The App-V Sequencer converts application data into an App-V package for virtualized delivery
  - The package can be streamed down to the clients on demand and cached locally

# Microsoft App-V

- **On Client machine,**
  - App-V client is installed that runs virtualized applications on user devices
  - The App-V client presents a list of applications to the end-users to which that user has access

# App-V Architecture



# Comparison of Desktop and Application Virtualization



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# Comparison

Desktop Virtualization allows you to run your desktop and applications remotely, rather than on your local computer

## Desktop Virtualization

- It is the practice of hosting a Desktop Operating system and applications within a Virtual Machine on a centralized server or cloud
- In this technique, Desktop OS and Applications are decoupled from a hardware system that access it

## Application Virtualization

- Application Virtualization is a practice of hosting Applications only on remote host or server.
- In this technique, Applications are decoupled from a hardware system that access it

# Comparison

## Desktop Virtualization

- Desktop runs as Virtual Machines in a secure Data center
- Each VM is called as Virtual desktop
- Different vendors Virtual desktops can be hosted on virtual server to access.
- The users may be connected from anywhere, anytime to the central server and access Virtual desktop

## Application Virtualization

- Applications actually runs on secure server in Data center
- Each Application is called as Virtual application
- Different vendors applications can be hosted on remote server to run
- The users may be connected from anywhere, anytime to the central server and runs applications

# Comparison

## Desktop Virtualization

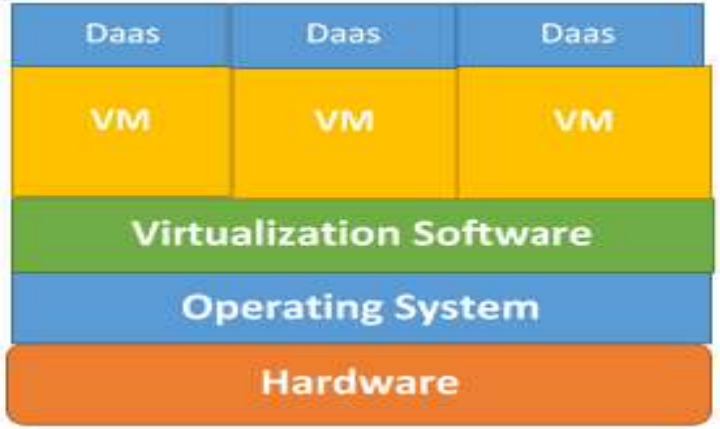
- The user may be a Desktop PC, laptop, thin client or mobile

## Application Virtualization

- The user may be a Desktop PC, laptop, thin client or mobile

# Comparison

## Desktop Virtualization




## Application Virtualization



# Software Define Data center SDDC



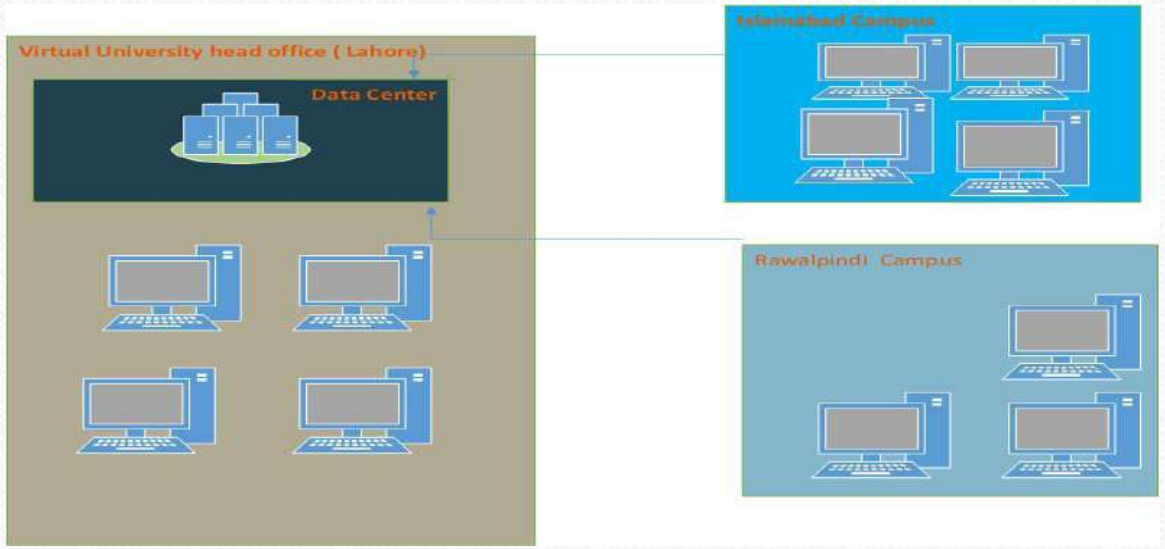
# Software Define Data center SDDC




- **Topics to be Covered**:
  - What is Data Center ?
  - What is Software Define Data center SDDC?
  - Component of SDDC?

# What is Data Center?

- A data center is a **place** that **gives** **people** access to **apps** and data **through** a **complicated** network **of computer** and storage **systems**



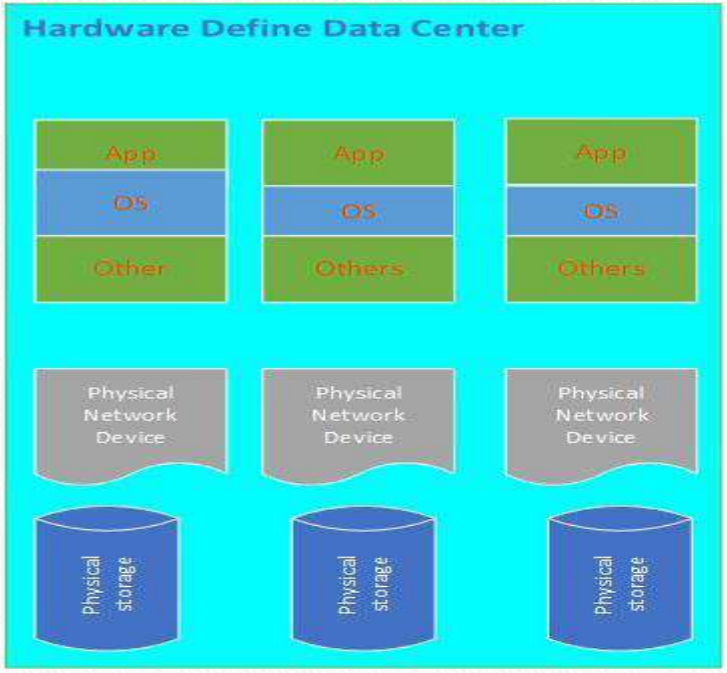
# Software define Data center( SDDC)?

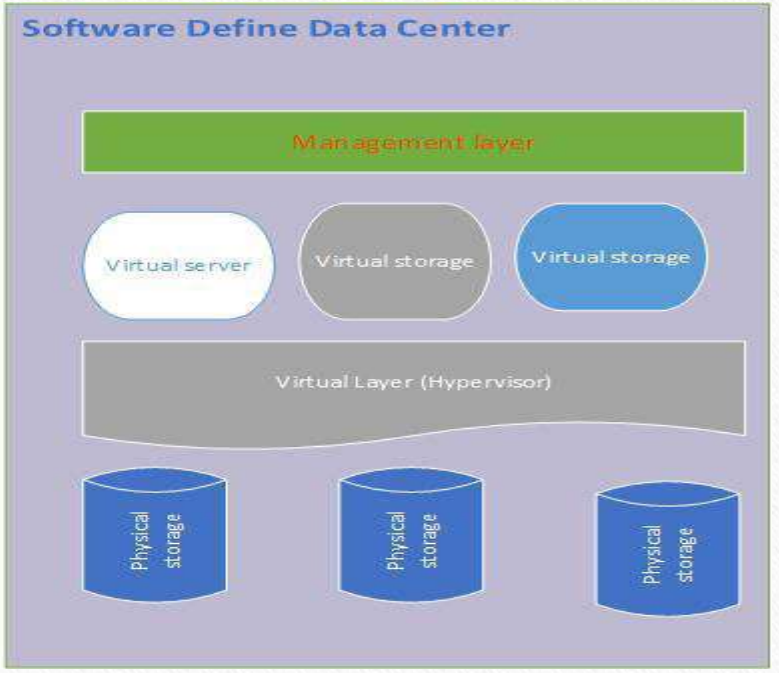
- **Keeping** your data center **up-to-date with the latest technology means using** software**-based components.**
- Provide different software to manage and automate the environment
- Better way of controlling the data Center
- Reduced it complexity
- Centralized management

# Software define Data center( SDDC)??

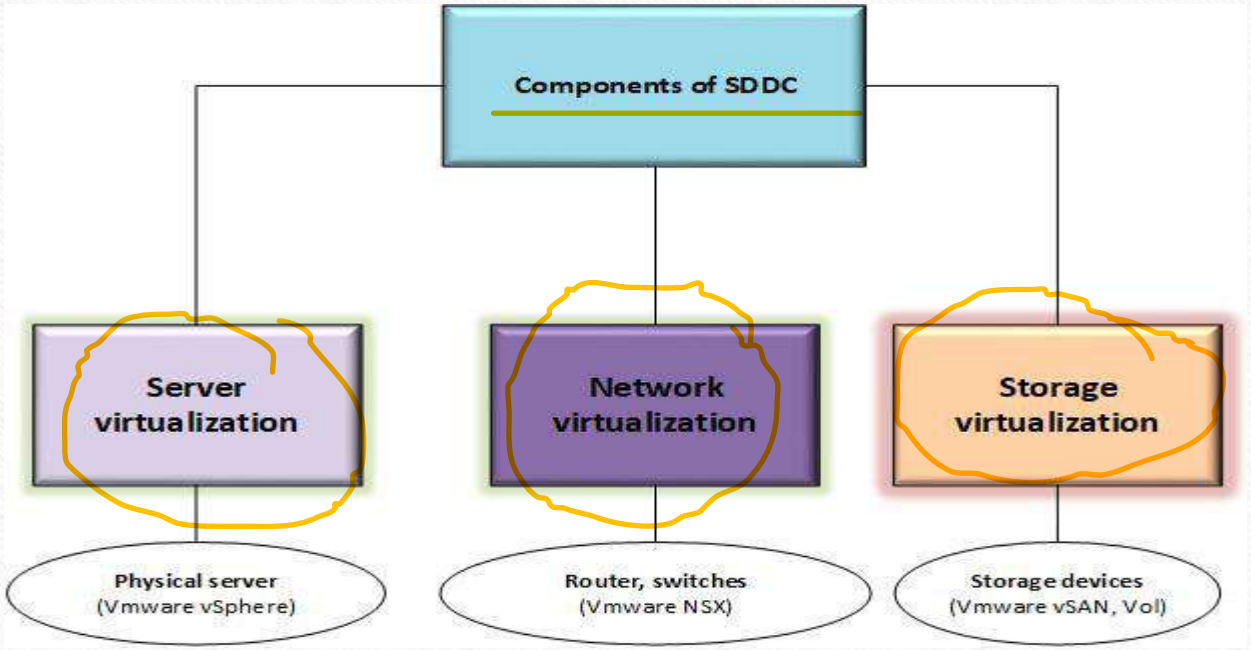
Hardware-Defined Data Center: Tied to physical hardware. Each application, OS, and resource is directly linked to the physical devices, making scaling and management more difficult.
Software-Defined Data Center: Virtualized environment, allowing resources to be managed and allocated through software. It provides better flexibility, scalability, and simplified management.

- ?

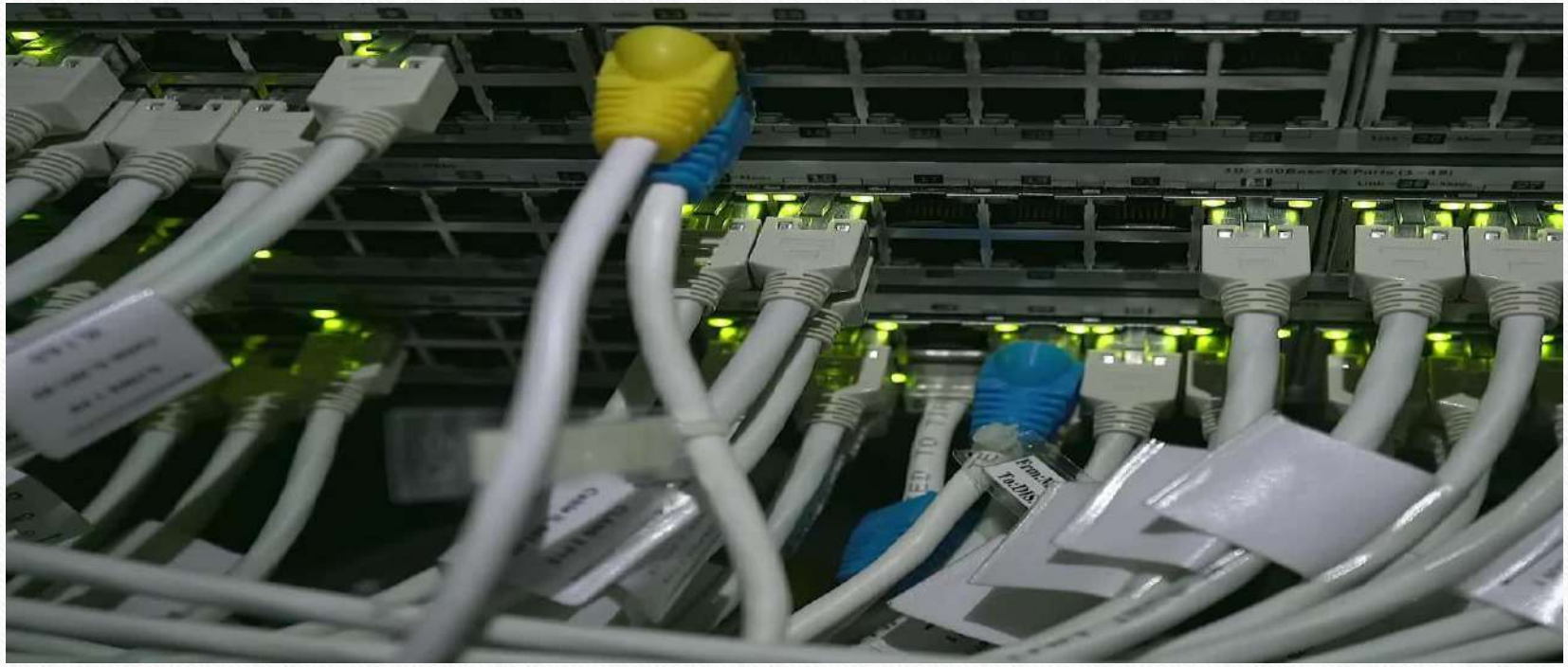


# Component of SDDC?



# Purpose of SDDC?



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# Purpose of SDDC?

| | |
|---|---|
| Abstraction | Abstraction of underling hardware (Physical CPU, RAM, Network) |
| **Pooling** | Pooling of resources |
| Automation | Automate the delivery of IT services |

# Abstraction




- In virtualization, abstraction refers to the process of hiding the underlying physical hardware from the user. This allows the user to view and interact with the resources as if they were dedicated to them, even though they are actually sharing resources with other users or virtual machines.

# Pooling of resources




- In virtualization, pooling refers to the grouping together of physical resources, such as CPU, memory, storage, and network bandwidth, so that they can be shared by multiple virtual machines. This allows organizations to improve the utilization of their IT infrastructure and reduce costs.

# Automation




- Automatically manage and control virtualized resources using software tools
- Involves tasks like provisioning, configuration, scaling, monitoring, and maintenance of virtualized environments
- Improve efficiency, reduce human errors, and enable rapid deployment and scaling

# Automation...




- Finding error in a server in datacenter with the help of vCenter Server will be lot easier if compared to Physical server
- Making copies of severs using virtual tools will be lot easier as compared to physical servers which will cost more and will take more time to do

# Week 4



# Architecture of Server Virtualization



# Architecture of Server Virtualization



# Architecture of Server Virtualization

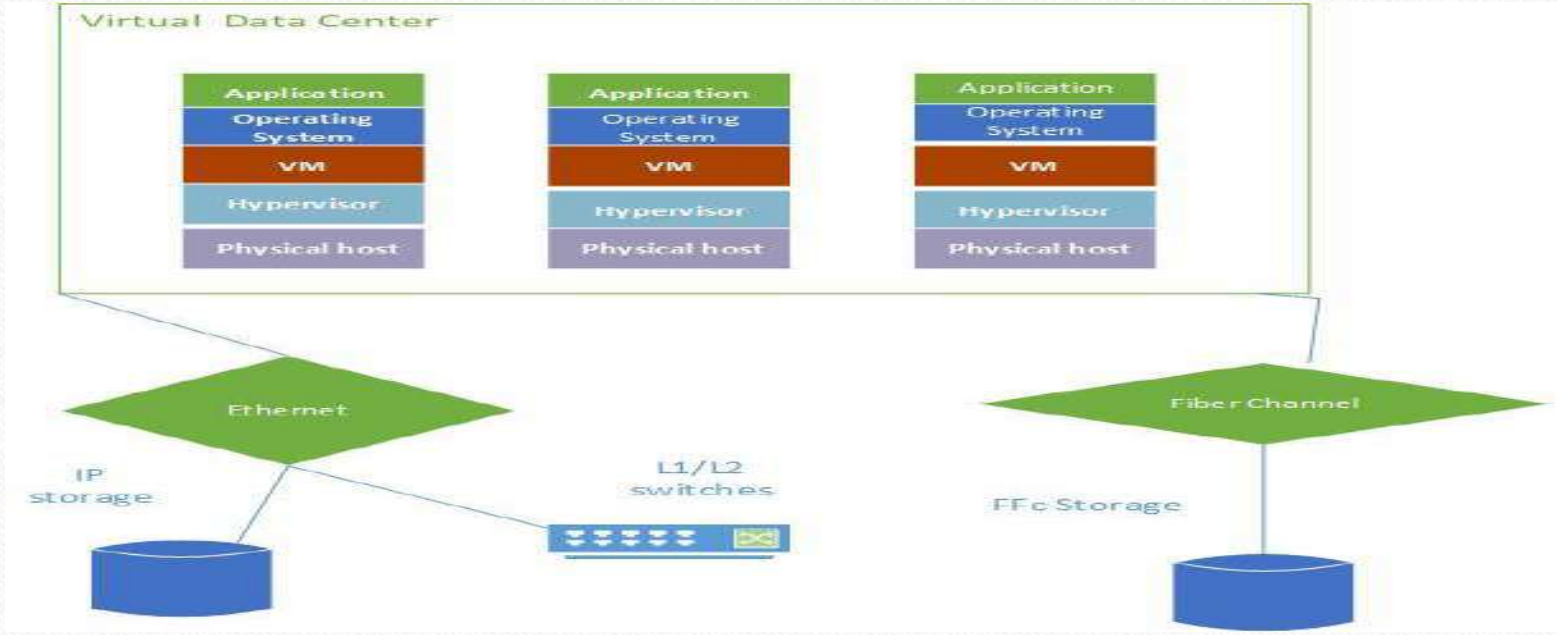
- **Topics to be Covered:**
  - The design of a physical data center. ?
  - **The design of a virtual data center.?**
  - Resource sharing ?

**Summary Table**

| Feature | Virtual Data Center | Physical Data Center |
|---|---|---|
| **Setup** | Software-based (Hypervisor & VMs) | Hardware-based (Dedicated servers) |
| **Resource Utilization** | High (Efficient sharing of resources) | Low (Dedicated hardware often underutilized) |
| **Scalability** | Easy and fast | Slow and expensive |
| **Reliability** | High (Easy failover & migration) | Low (Hardware failure impacts systems) |
| **Cost** | Cost-efficient in the long run | Expensive to maintain over time |

# The design of a Virtual datacenter?

- We can use a Virtual Data Center to get several virtual machines running on minimal hardware..



# Architecture of Server Virtualization



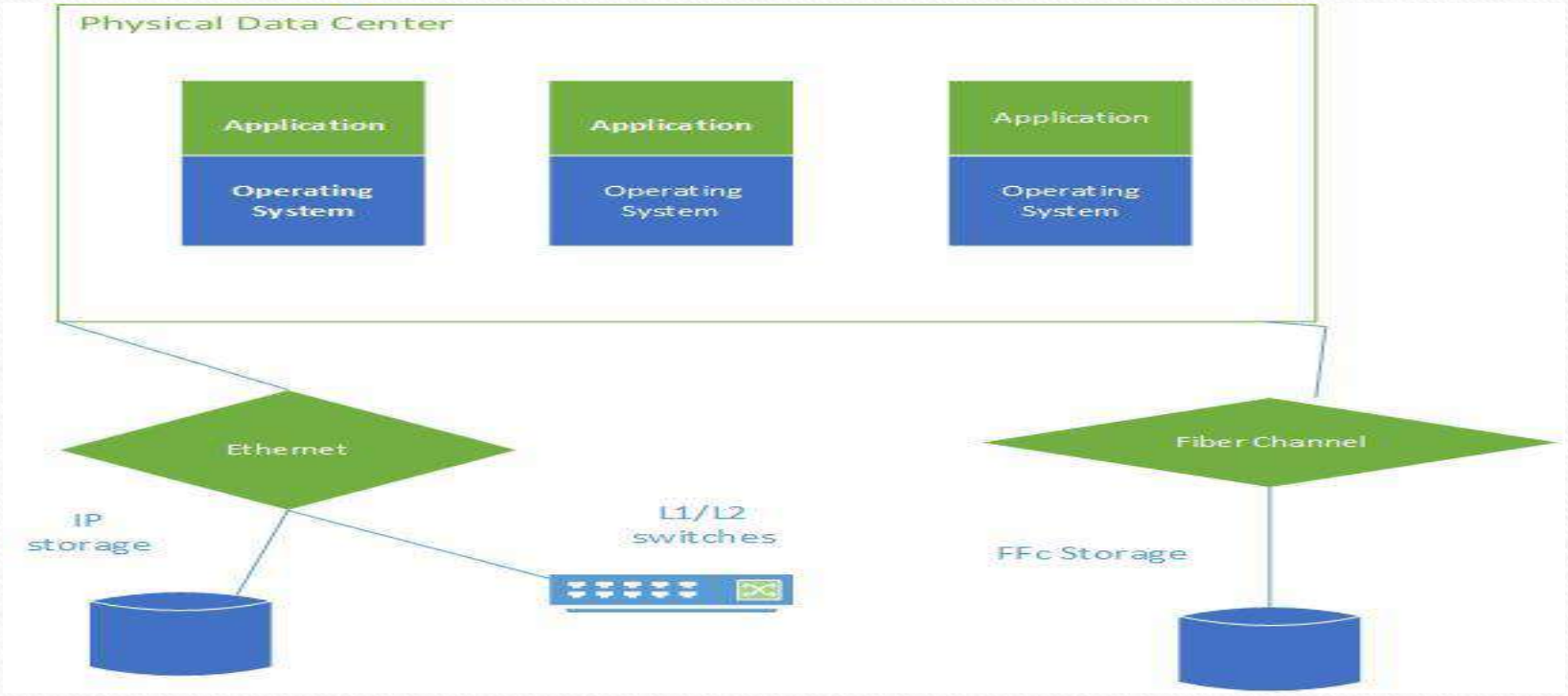

- **Topics to be Covered:**
  - **The design of a physical data center. ?**
  - The design of a virtual data center.?
  - Resource sharing ?

# The design of a physical data cent?




- In a physical data center, there are physical servers running.
- APP and Operating system are running on physical server
- **An** Ethernet switch is linked up with l2 and l3 switches**,** plus IP-**based** storage**.**

# The design of a physical data cent?



# Architecture of Server Virtualization

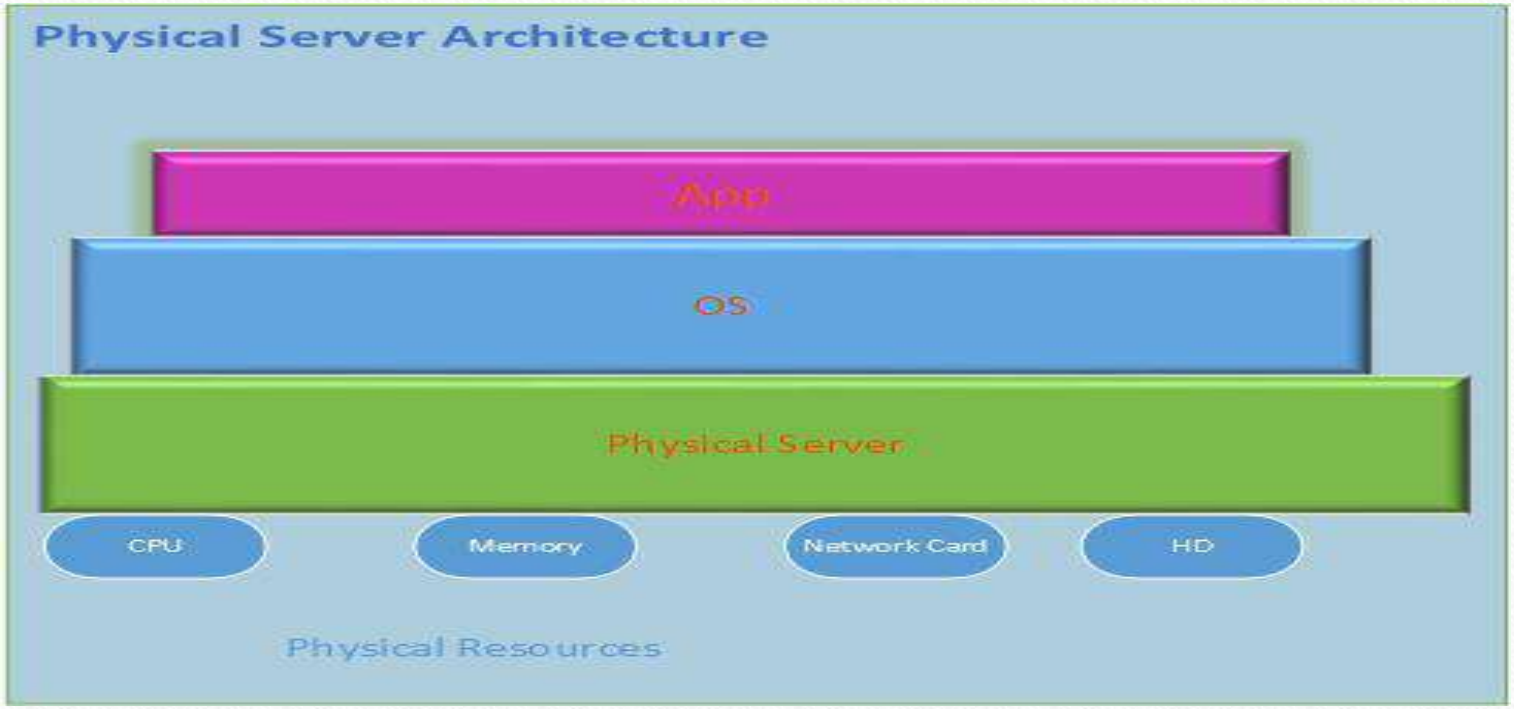


# Architecture of Server Virtualization

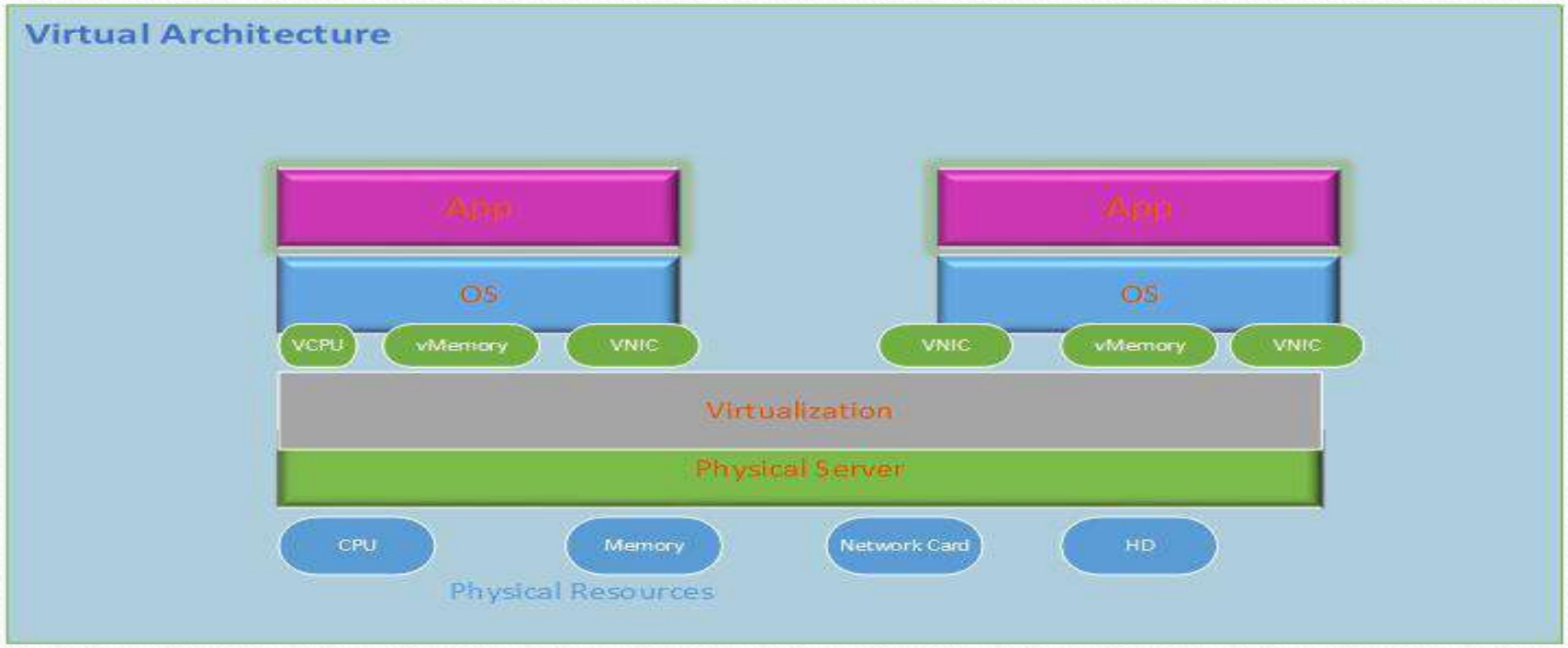



- **Topics to be Covered:**
  - **The design of a physical data center. ?**
  - **The design of a virtual data center.?**
  - Resource sharing

# Resource Sharing Physical Server?



# Resource Sharing Virtual  data center?

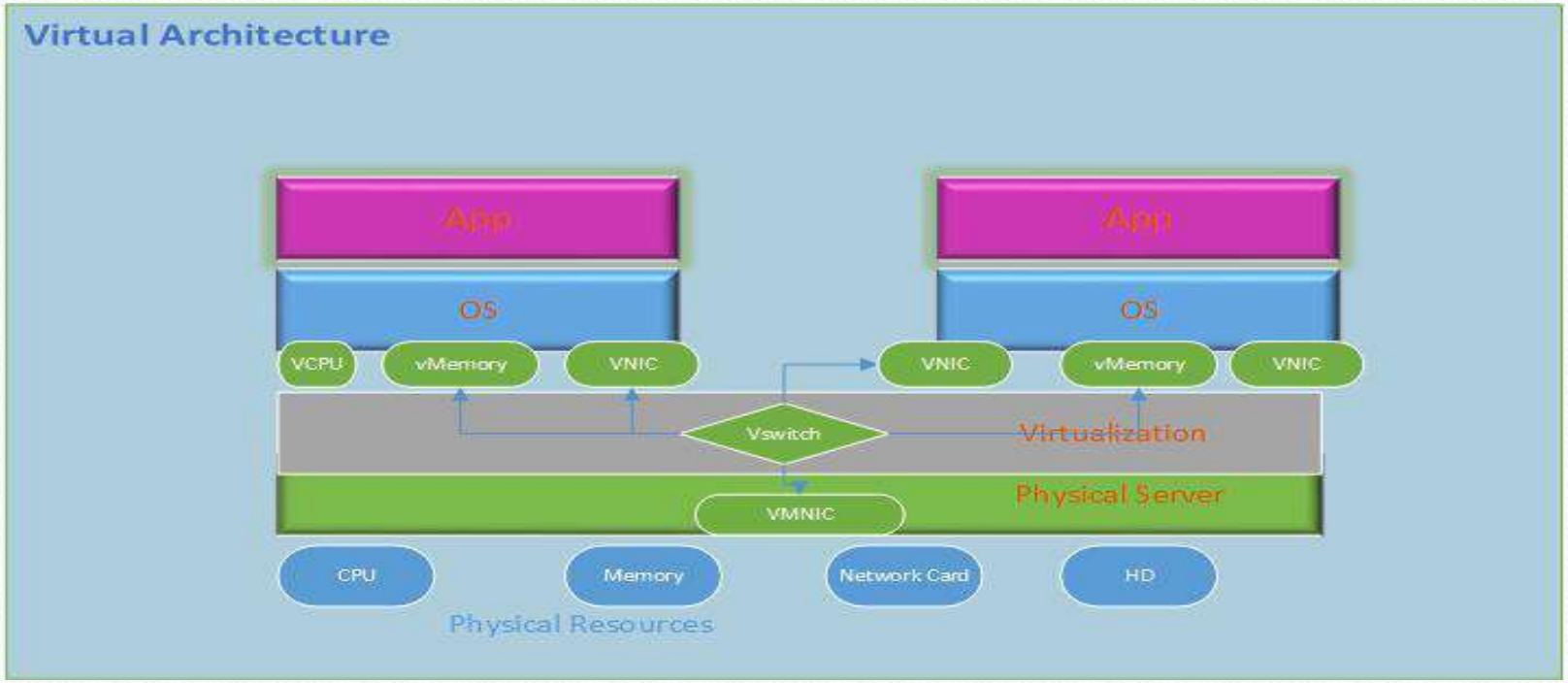


# Resource Sharing Virtual data center?




- **CPU**
- Maximum number of vCPU on VM depends on
  - number of logical CPU on the physical host
  - Type of OS on VM,
  - VMware supports up to 128 vCPU on one VM
- **Memory**
- Maximum number of virtual memory on VM depends on
  - Maximum memory available on Physical host
  - Type of OS on VM
- VMware supports up to 6TB Virtual memory on one VM

# Resource Sharing Virtual  data center?



# Resource Sharing Virtual  data center?




- VM support up to 10 VNIC

# Week 5



# Virtualization Technologies



# Architecture of Server Virtualization



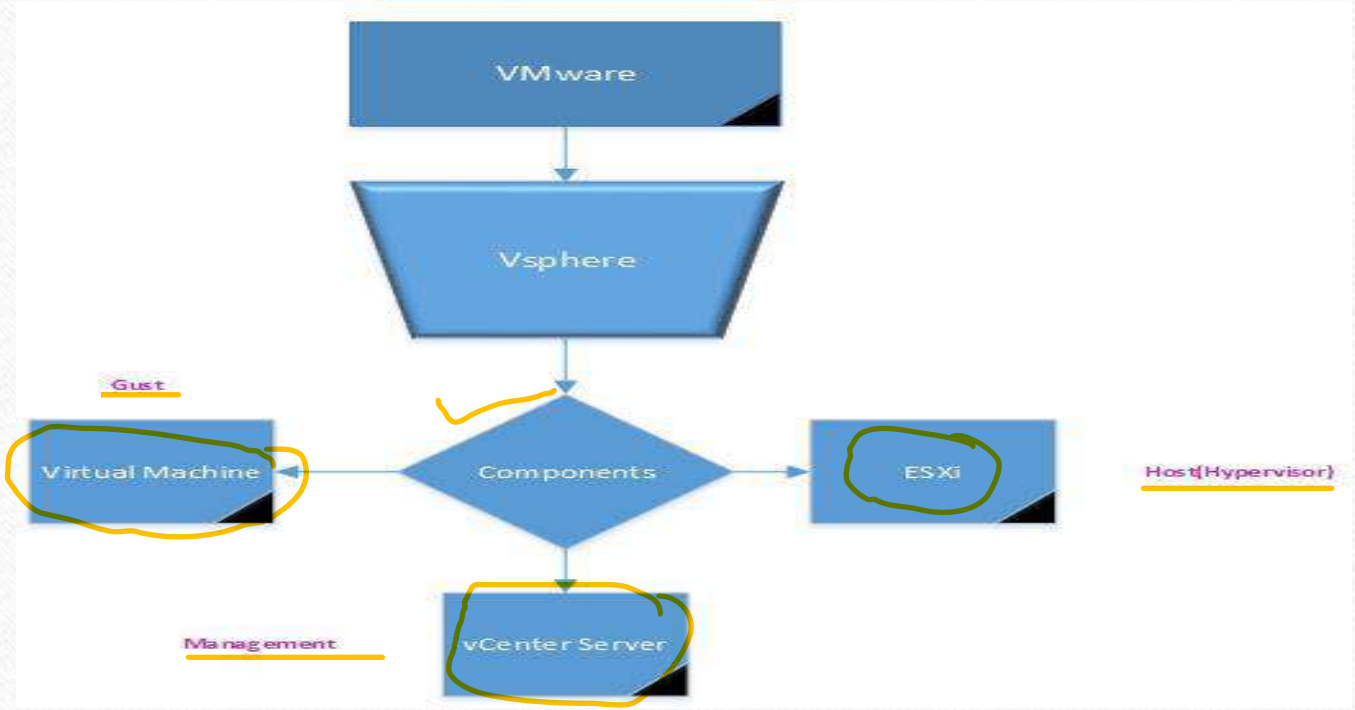

- **Topics to be Covered:**
  - **VMware vSphere ?**
  - **ESXI**

# VMware vSphere?




- VMware  is vendor who provide software's for server virtualization
- Version 6.7 vSphere (Current)

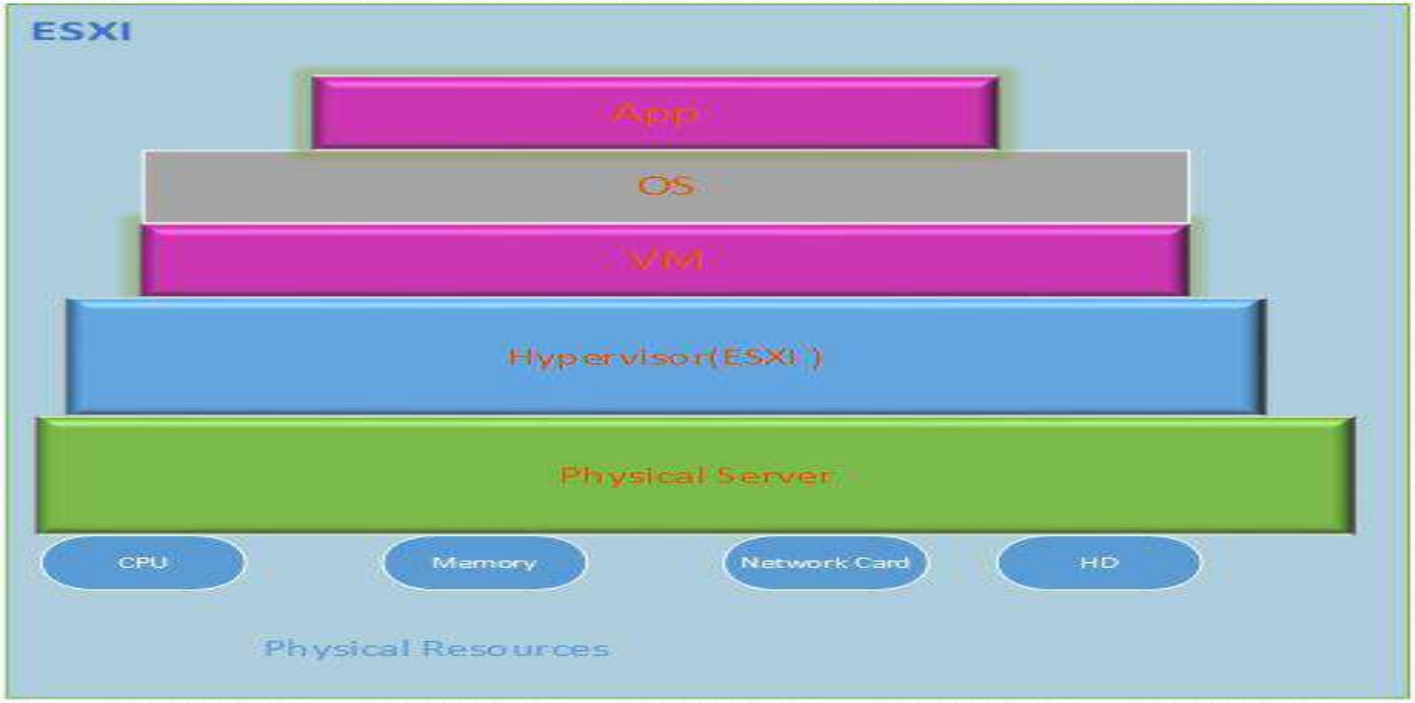
# Component of Vsphere



# ESXI

- ESXI is Hypervisor
- A hypervisor allows one host computer to support multiple guest VMs by virtually sharing its resources, such as memory and processing.
- Direct install on physical server as OS
- Assigned resources from physical server to VM

# ESXI


ESXI
App
OS
VM
Hypervisor(ESXI)
Physical Server
CPU
Memory
Network Card
HD
Physical Resources

# Virtualization Technologies

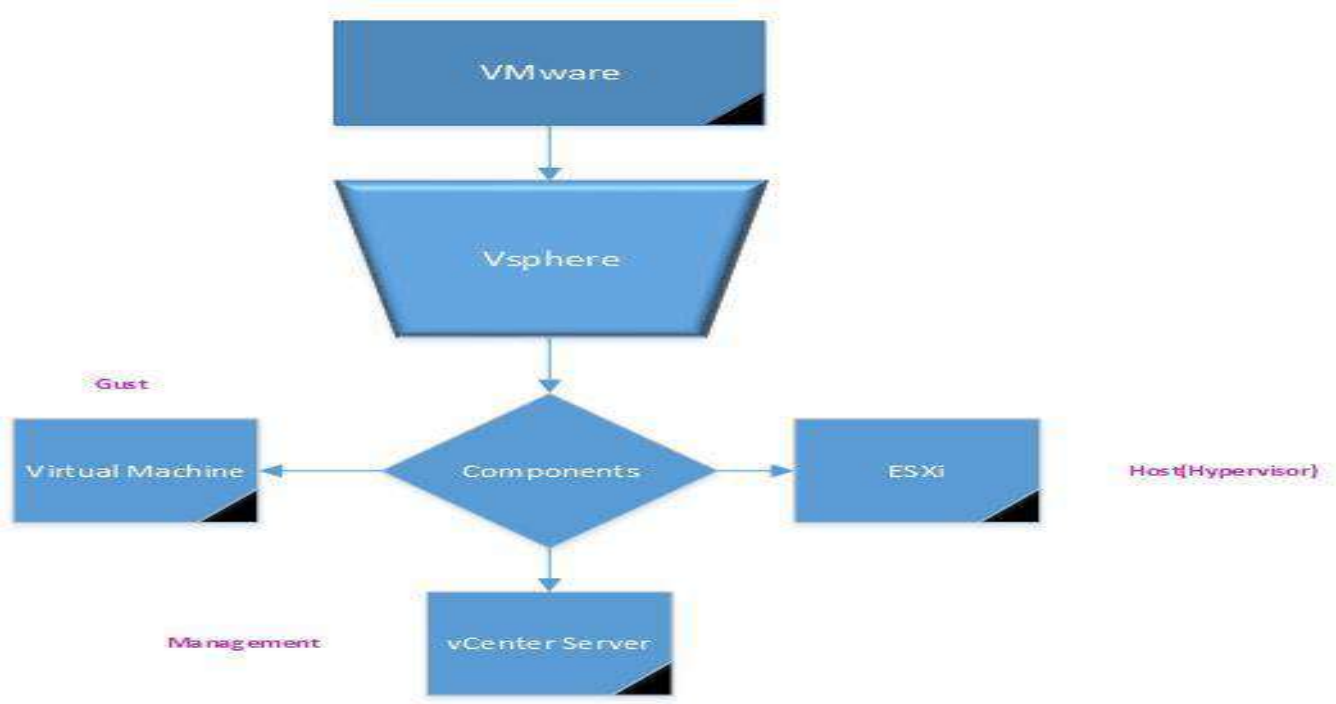


# Virtual Machine




- **Topics to be Covered:**
  - **VMware vSphere ?**
  - **ESXI**
  - **Virtual Machine**

# Virtual Machine

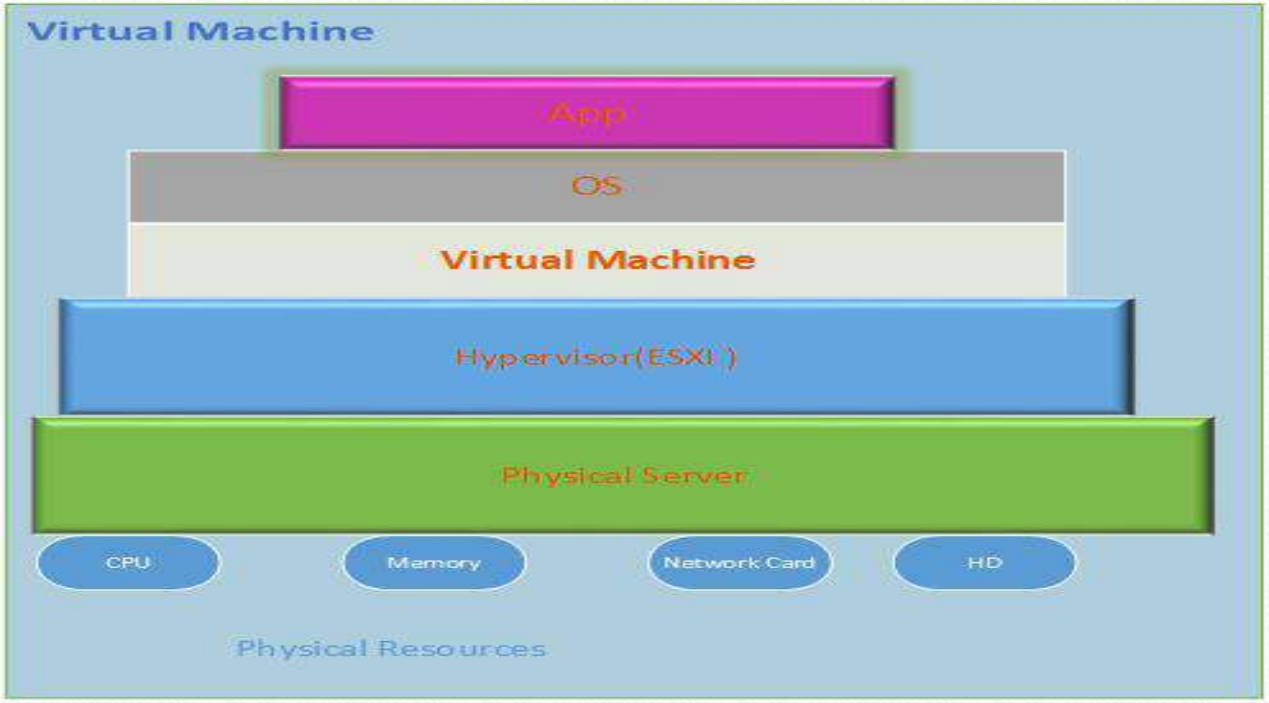


# What is Virtual Machine




- It is virtual Server
- Provide same function like Physical Server but you do not need hardware for VM
- Use Virtual resources (VCPU, vMem, vNic,vDisk)

# Virtual Machine



# Virtualization Technologies

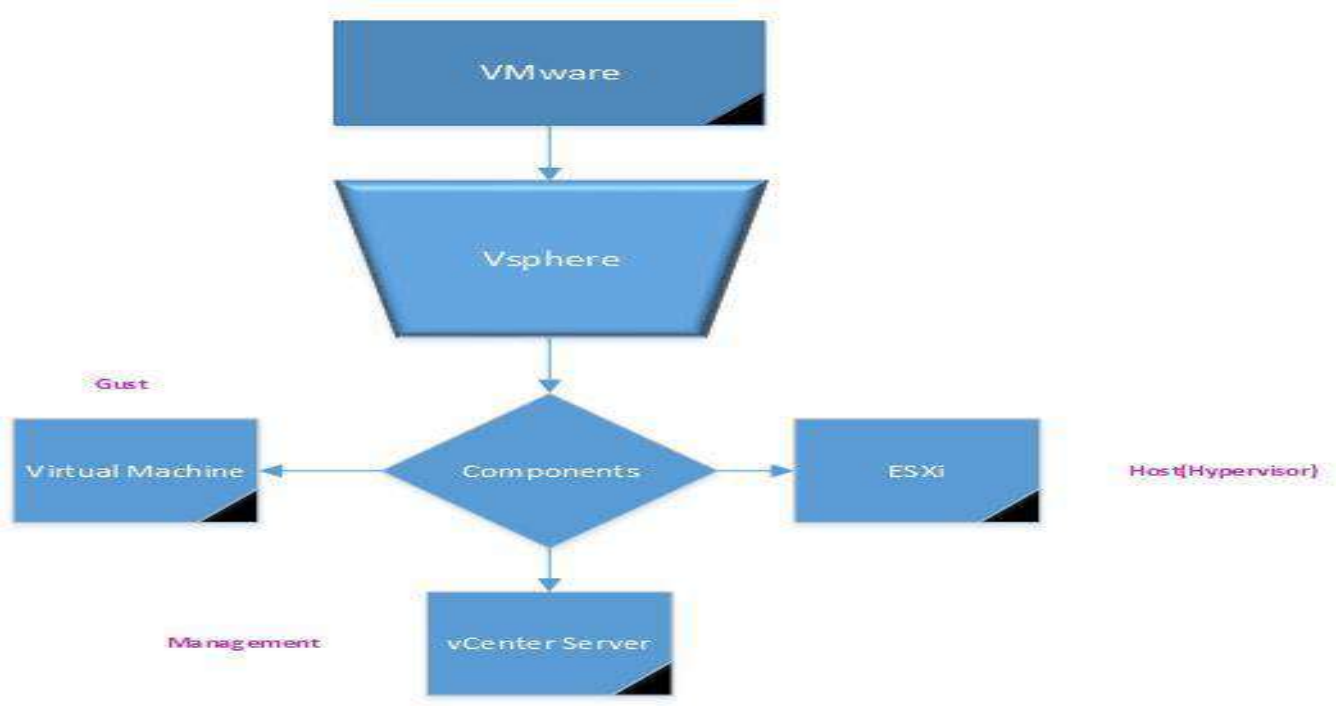


# vCenter




- **Topics to be Covered:**
  - **VMware vSphere ?**
  - **ESXI**
  - **Virtual Machine**
  - **VCenter Server**

# vCenter

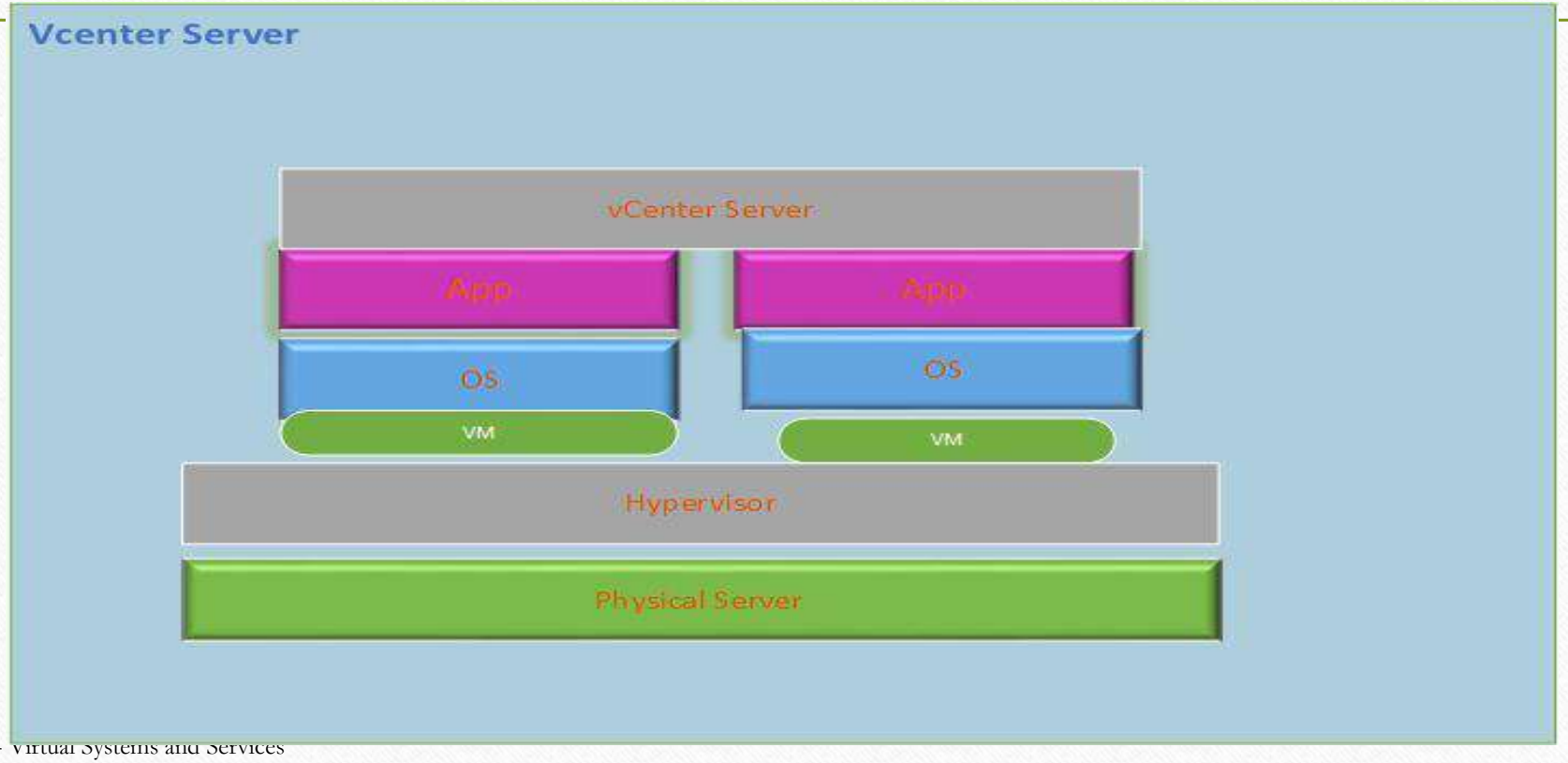

VMware
Vsphere
Gust
Virtual Machine
Components
ESXi
Host(Hypervisor)
Management
vCenter Server

# What is vCenter Server ?




- Manage entire Virtualization Data Center
- Before Virtualization (Manually visit Data Center )
- Management consoles
- It also perform day to day task for example take snap shot etc

# vCenter Server



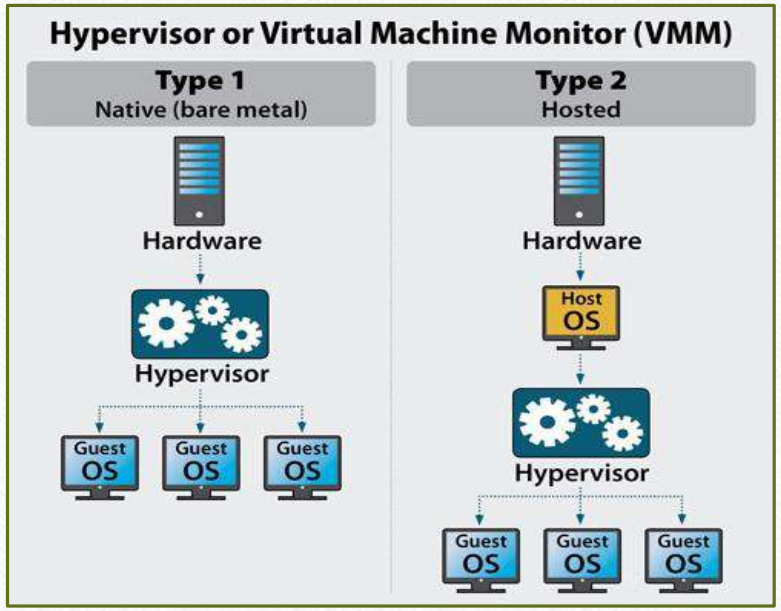
# Week 6



# Hypervisor

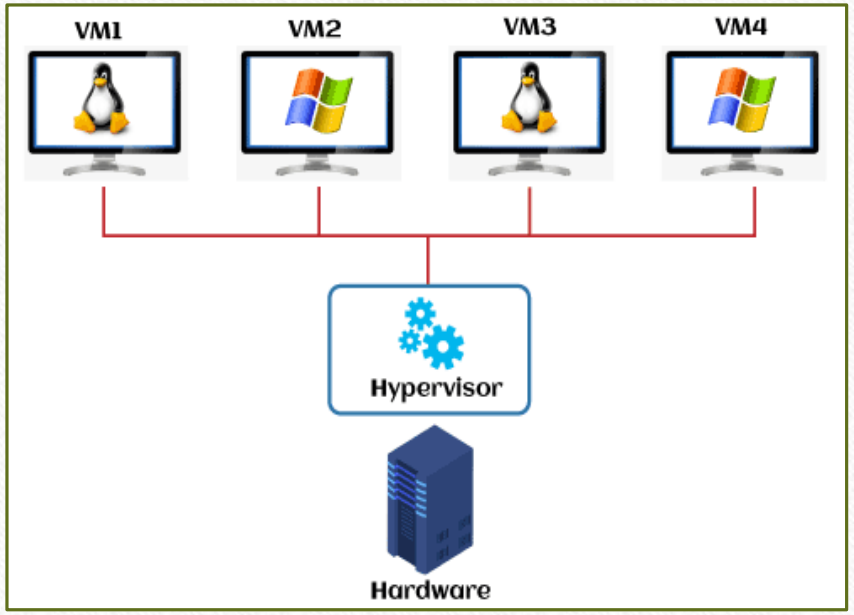
# Hypervisor

- **Topics to be Covered:**
  - A software layer
  - Creates and manages virtual machines (VMs)
  - Acts as a virtualization layer b/w the physical hardware and (VMs)
  - Allows multiple VMs to share the underlying hardware resources of Host



# Hypervisor

- Allows multiple operating systems to run on a single physical server
- Improving the utilization and flexibility of the underlying hardware resources



# Components of Hypervisor

- **Virtual Machine Monitor (VMM):** Yeh ek core system hai jo sari virtual machines ko control karta hai.

  Core component, manages Virtual Machines

- **Virtual Devices:** Yeh wo cheezein hain jo physical devices ka virtual version hoti hain, jaise virtual hard drive, virtual network card, ya virtual switch.

  NICs, virtual hard drives, and virtual switches used to provide virtual versions of physical devices

- **Virtual Machine Executables (VMEs):** Yeh wo operating systems hain jo virtual machines ke upar chal rahe hote hain.

  Virtual versions of operating systems that run on virtual machines

- **Drivers:** Yeh wo software hain jo virtual machines ko host ke physical hardware ke sath baat karne dete hain.

  Provide access to the physical hardware resources of a host device

# Components of Hypervisor

- **Management Interface:** Yeh ek interface hai jo administrators ko virtual machines (VMs) create, manage, aur monitor karne ka moka deta hai.

  Interface that allows administrators to create, manage, and monitor virtual machines (VMs)

- **Virtual Machine File System (VMFS):** Yeh ek storage system hai jo virtual machines ke data aur configuration files ko store karta hai.

  Stores virtual machines and their configuration files

It was a general overview of the Hypervisor structure it may vary from type to type

# Types of Hypervisors

# Types of Hypervisors

There are two main types of hypervisors:

- Type 1:

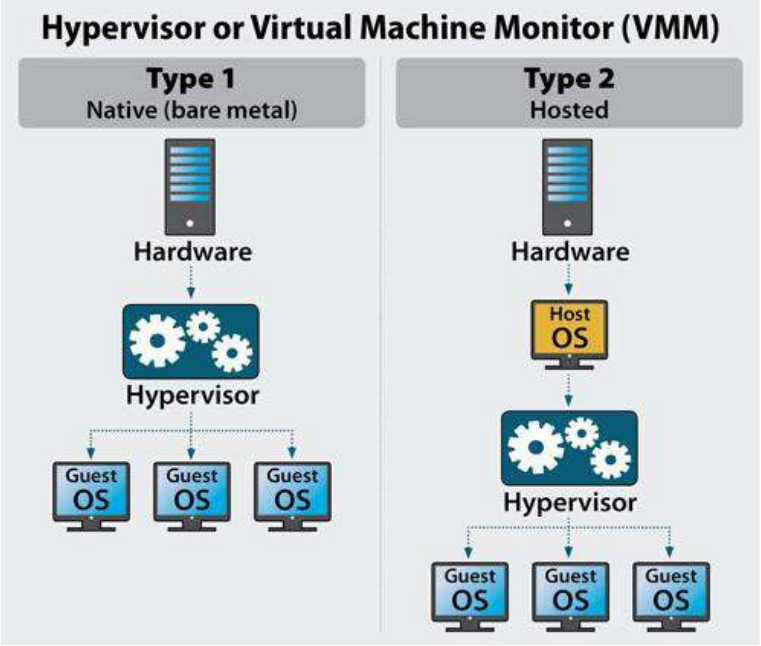
  -also known as bare-metal hypervisors

  -run directly on the host's physical hardware

- Type 2:

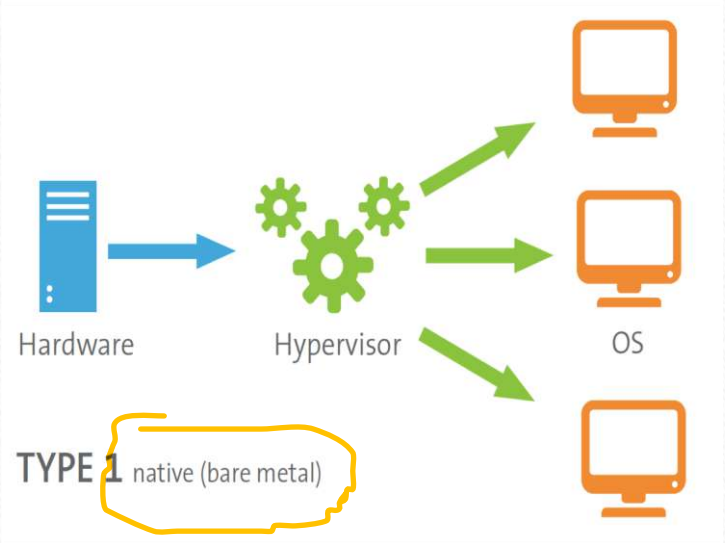
  -also known as Hosted Hypervisor

  -run as an application on a host operating system



# Type 1 Hypervisor

- Type 1 or bare-metal hypervisors, are virtualization software
- Runs on host's physical hardware.
- Not dependent on any host operating system
- Controls the underlying hardware resources of host
- Provides virtualized environments for guest operating systems



# Examples of Type 1 Hypervisor

- **VMware vSphere**

  -complete virtualization solution for data centers

  -used in large enterprise environments and cloud infrastructures

- **Microsoft Hyper-V**

  -provides virtualization support for Windows Server operating systems

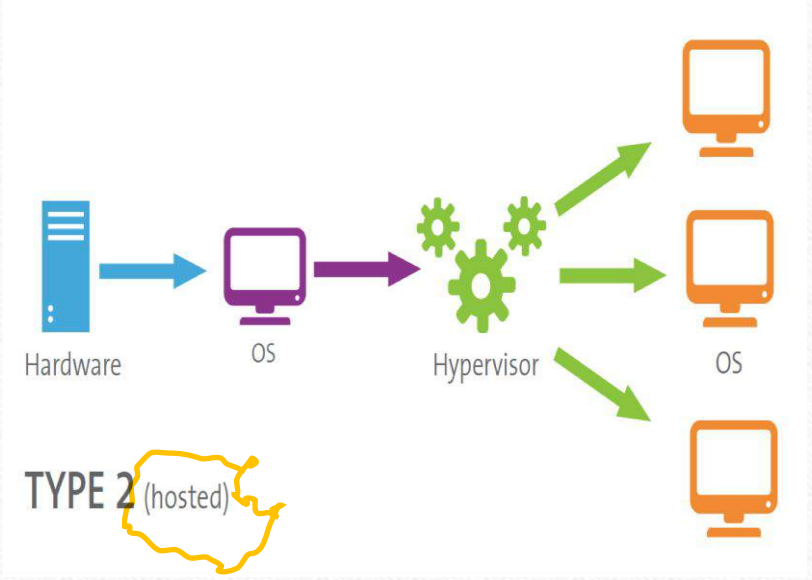
  -widely used in enterprise environments with a strong Microsoft technology presence.

- **Citrix XenServer**

  -has a strong presence in the financial services and healthcare industries

# Type 2 Hypervisor

- Commonly used for software development, testing, and personal desktop virtualization
- Used where a higher level of performance is not as critical
- Less secure than Type 1 hypervisors
- Vulnerable to attacks on the host operating system



# Examples of Type 2 Hypervisor

- Oracle VirtualBox

  -supports a wide range of host and guest operating systems

  -used for personal virtualization, software testing, and development

- VMware Workstation

  -commercial hypervisor

  -supports multiple operating systems, snapshots and cloning of virtual machines

  -support for virtualized hardware such as virtualized GPUs and virtualized USB devices.

# Week 7



# Type 2 hypervisor
# Wmware workstation

# Type 2 hypervisor
# Creating Virtual Machine




- **Topics to be Covered:**

  **What is Type 2 Hypervisor**

  **Type of Type 2 Hypervisor**

# Type 2 Hypervisor




- It run at top of OS
- First you install OS and top of OS you install Wmware workstation / Wmw are fusion as application
- Using that application you create multiple VM

# Type 2 Hypervisor




- **VMware Fusion**
- **Workstation**
- **VirtualBox**

# VMware Fusion:




- This is the company's MacOS-focused offering, which lets Mac users run a large range of guest operating systems.
- Run window application in Mac OS

# Workstation:

- VMware's Linux- and Windows-focused platform comes in two flavors
  - **Pro:**which is a paid version, The Pro version allows users to run multiple operating systems on a single PC
  - **Player**, which is free for personal use, Player only supports a single guest OS

# VirtualBox




- A Type 2 hypervisor running on Linux, Mac OS, and Windows operating systems.

# Type 2 hypervisor
# Wmware workstation

# Lab:
# Creating Virtual Machine on Vmware Workstation

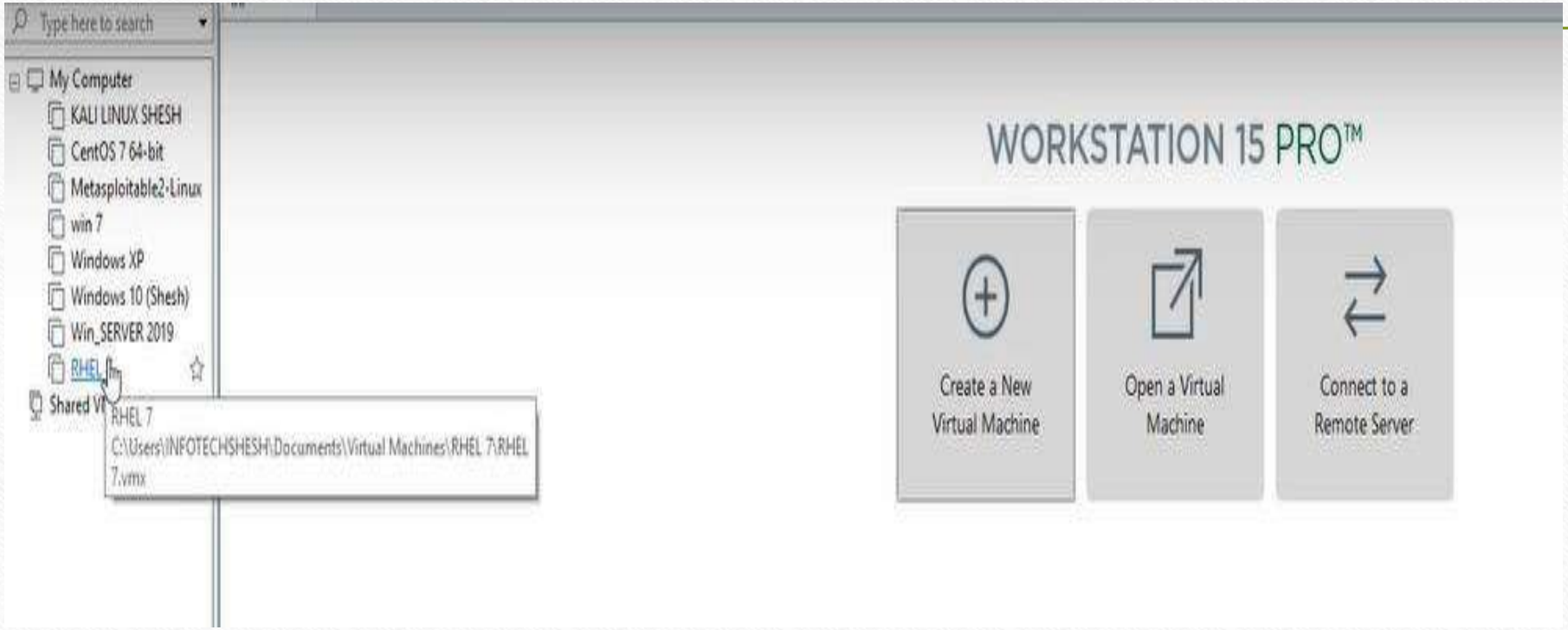



- **Topics to be Covered:**

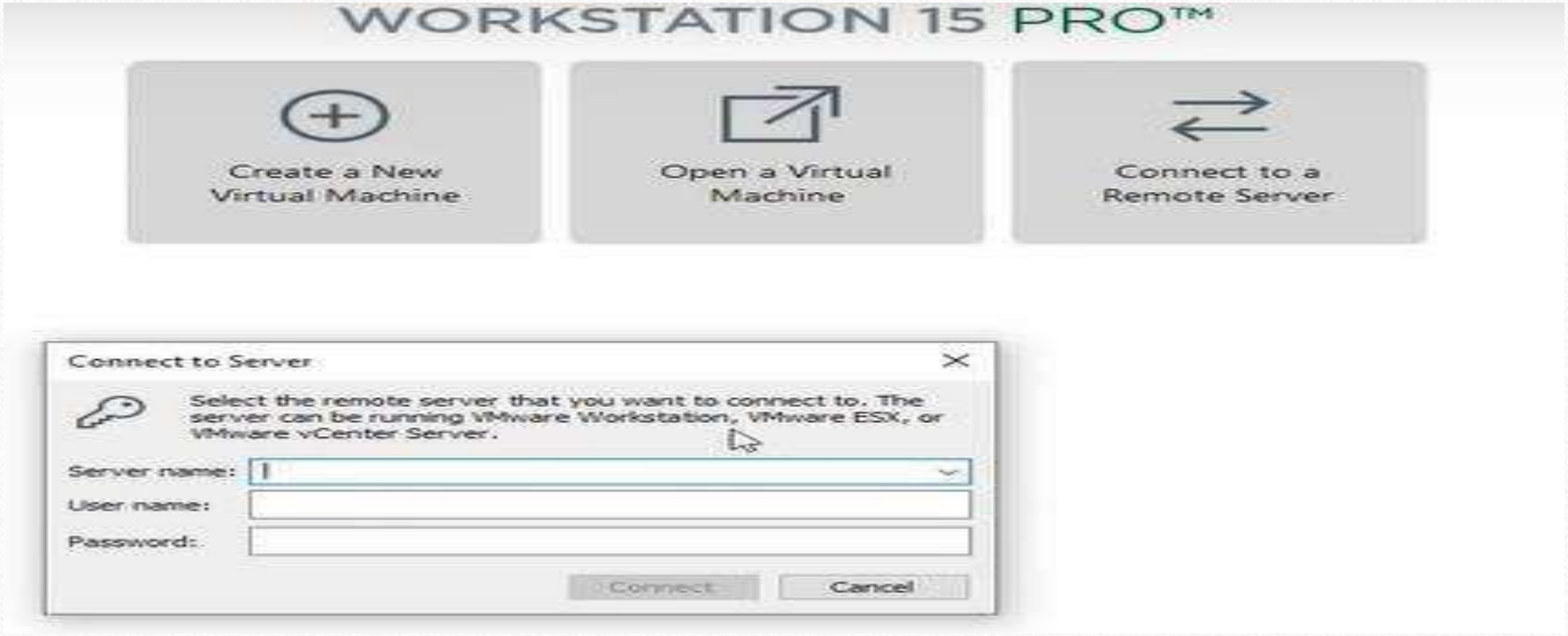
  **Interface**

  **Create VM**

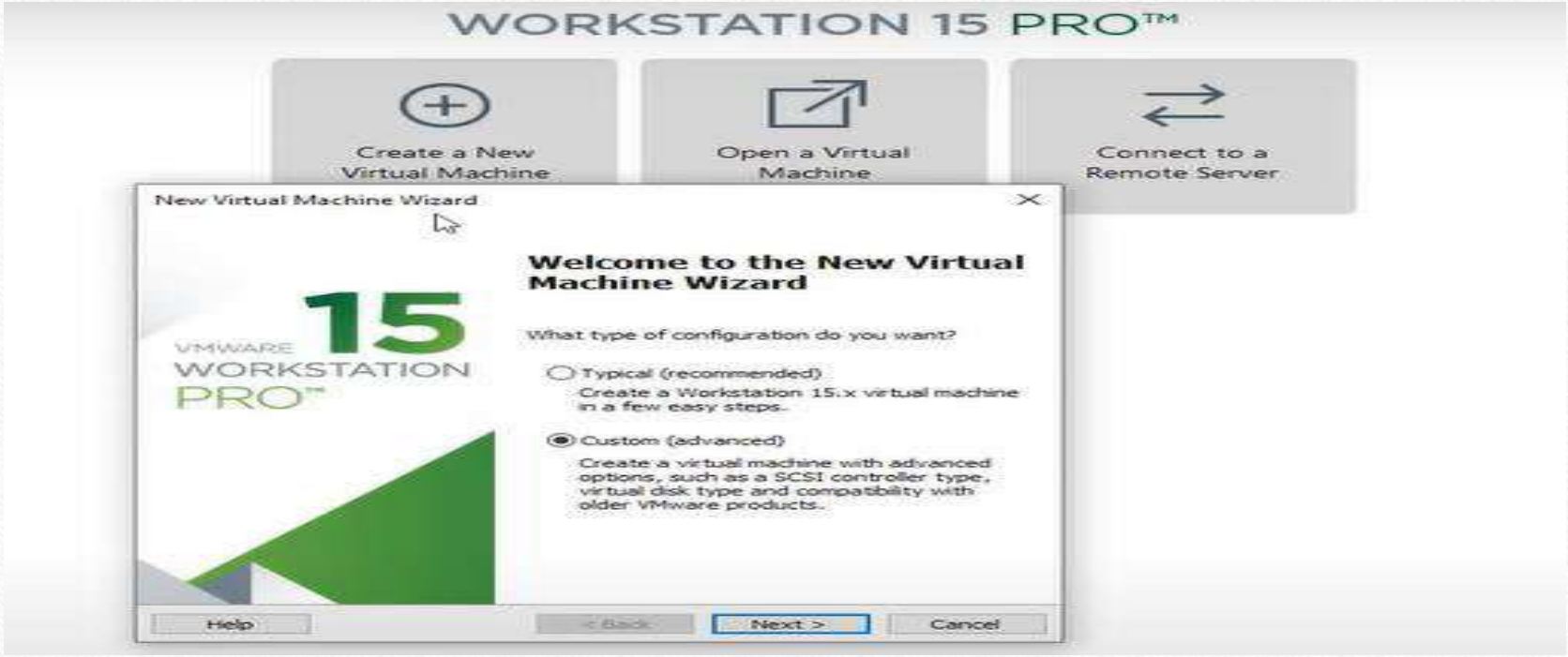
# Interface


Type here to search
My Computer
KALI LINUX SHESH
CentOS 7 64-bit
Metasploitable2-Linux
win 7
Windows XP
Windows 10 (Shesh)
Win_SERVER 2019
RHEL 7
Shared VMs
RHEL 7
C:\Users\INFOTECHSHESH\Documents\Virtual Machines\RHEL 7\RHEL 7.vmx
WORKSTATION 15 PRO™
Create a New Virtual Machine
Open a Virtual Machine
Connect to a Remote Server

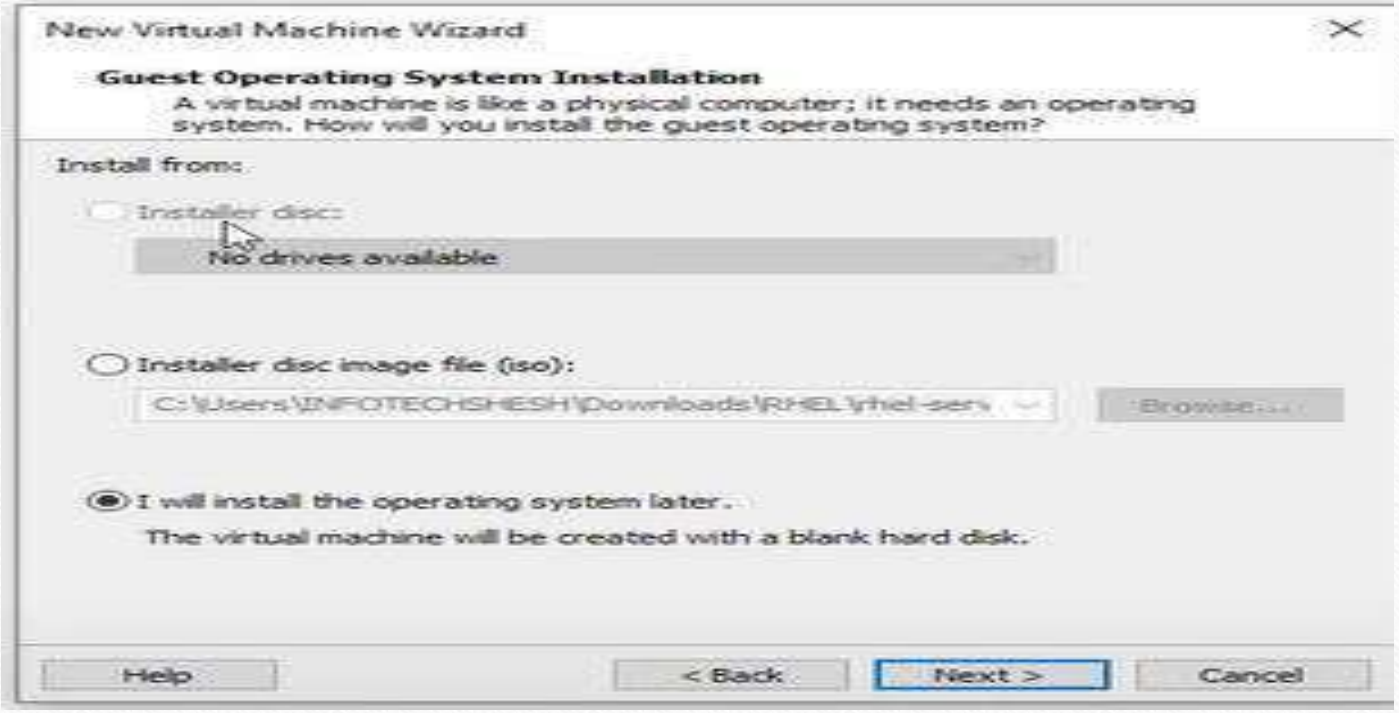
# Interface


WORKSTATION 15 PRO™
Create a New Virtual Machine
Open a Virtual Machine
Connect to a Remote Server
Connect to Server
Select the remote server that you want to connect to. The server can be running VMware Workstation, VMware ESX, or VMware vCenter Server.
Server name:
User name:
Password:
Connect
Cancel

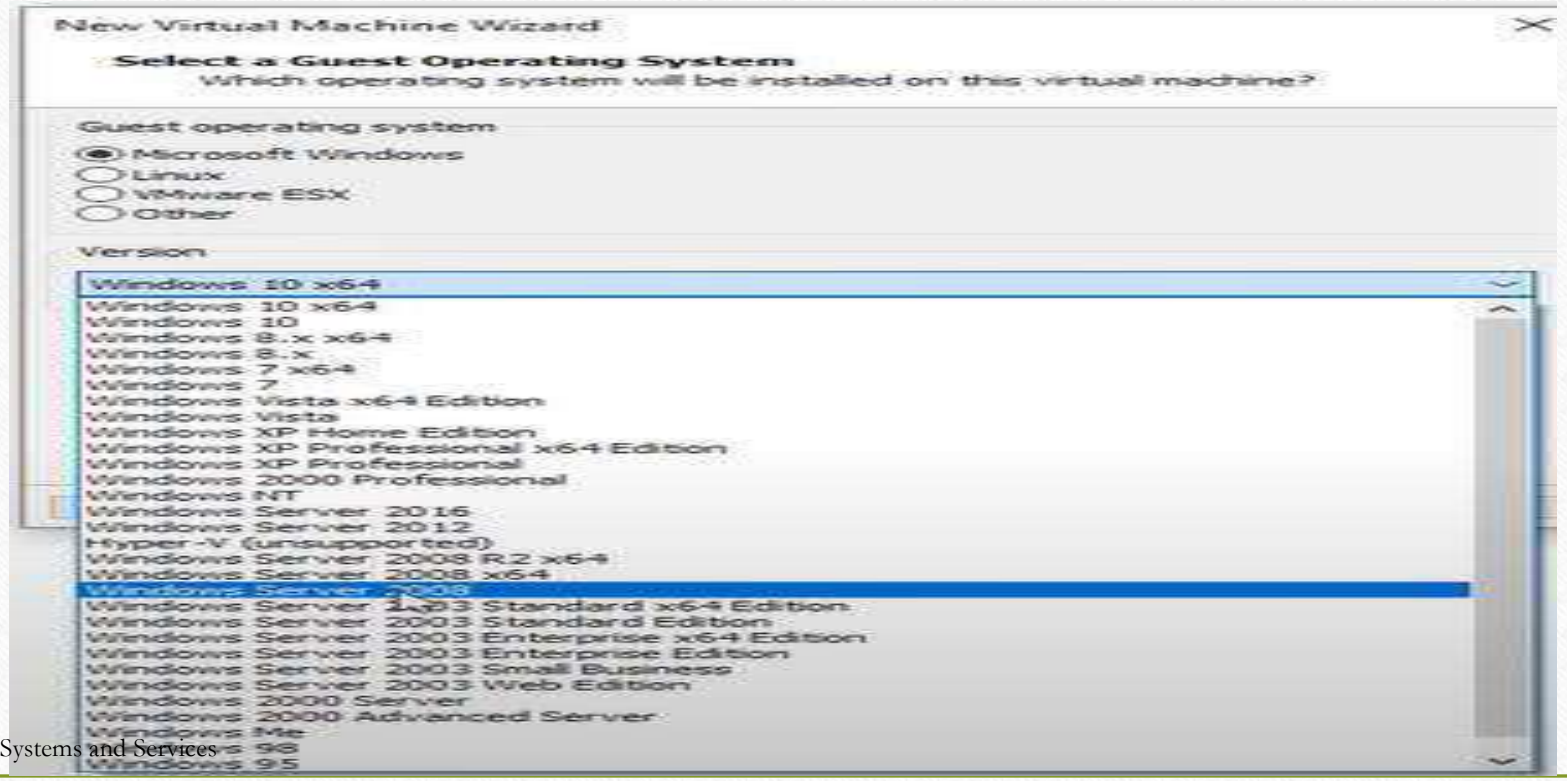
# Create VM


WORKSTATION 15 PRO™
Create a New Virtual Machine
Open a Virtual Machine
Connect to a Remote Server
New Virtual Machine Wizard
Welcome to the New Virtual Machine Wizard
What type of configuration do you want?
Typical (recommended)
Create a Workstation 15.x virtual machine in a few easy steps.
Custom (advanced)
Create a virtual machine with advanced options, such as a SCSI controller type, virtual disk type and compatibility with older VMware products.
VMWARE WORKSTATION PRO™ 15
Help
< Back
Next >
Cancel

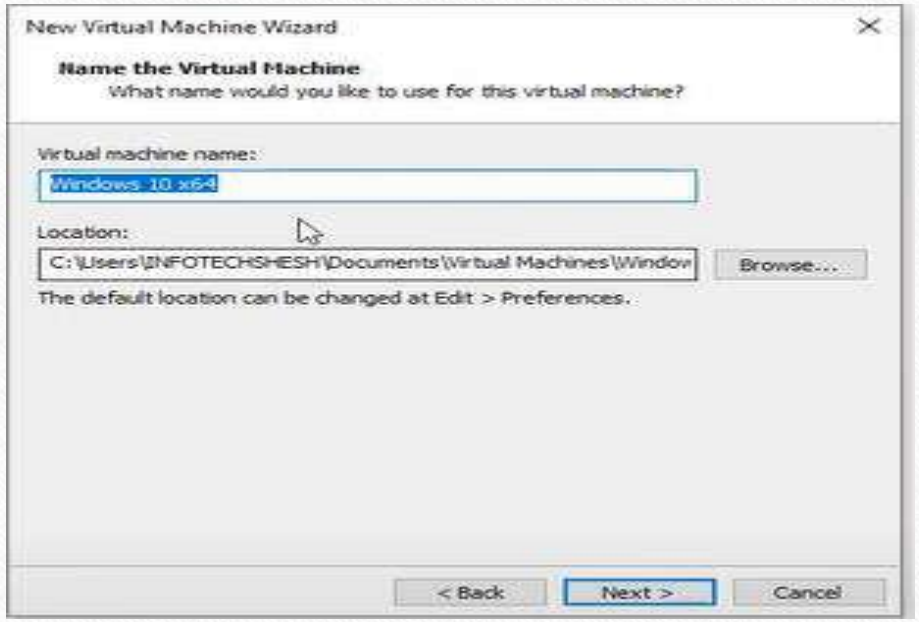
# Create VM


New Virtual Machine Wizard
Guest Operating System Installation
A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?
Install from:
Installer disc:
No drives available
Installer disc image file (iso):
C:\Users\INFOTECHSHESH\Downloads\RHEL\rhel-serv
Browse...
I will install the operating system later.
The virtual machine will be created with a blank hard disk.
Help
< Back
Next >
Cancel

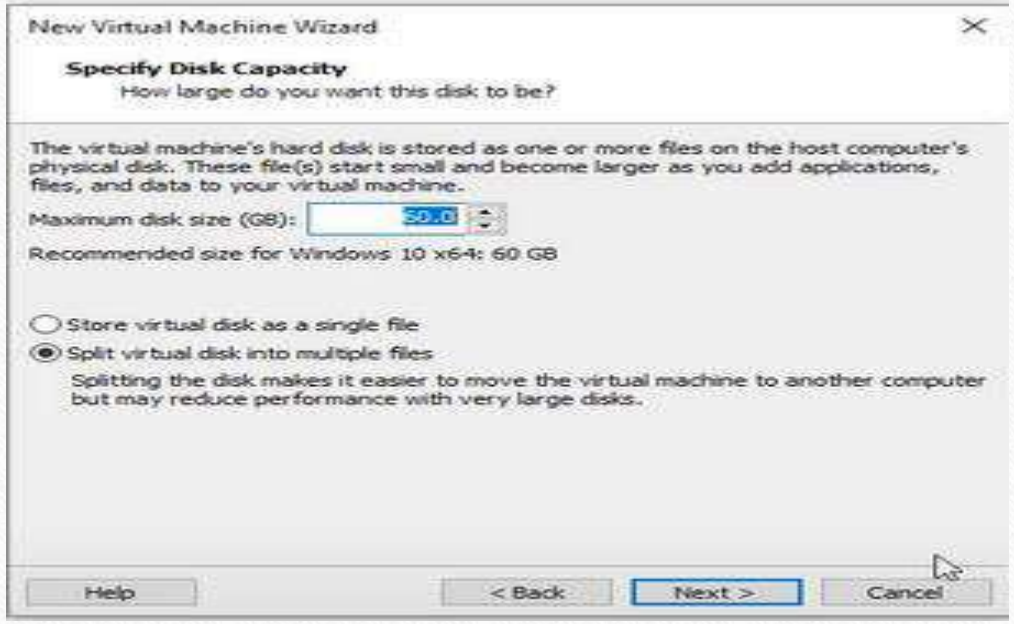
# Create VM


New Virtual Machine Wizard
Select a Guest Operating System
Which operating system will be installed on this virtual machine?
Guest operating system
Microsoft Windows
Linux
VMware ESX
Other
Version
Windows 10 x64
Windows 10 x64
Windows 10
Windows 8.x x64
Windows 8.x
Windows 7 x64
Windows 7
Windows Vista x64 Edition
Windows Vista
Windows XP Home Edition
Windows XP Professional x64 Edition
Windows XP Professional
Windows 2000 Professional
Windows NT
Windows Server 2016
Windows Server 2012
Hyper-V (unsupported)
Windows Server 2008 R2 x64
Windows Server 2008 x64
Windows Server 2008
Windows Server 2003 Standard x64 Edition
Windows Server 2003 Standard Edition
Windows Server 2003 Enterprise x64 Edition
Windows Server 2003 Enterprise Edition
Windows Server 2003 Small Business
Windows Server 2003 Web Edition
Windows 2000 Server
Windows 2000 Advanced Server
Windows Me
Windows 98
Windows 95

# Create VM


New Virtual Machine Wizard
Name the Virtual Machine
What name would you like to use for this virtual machine?
Virtual machine name:
Windows 10 x64
Location:
C:\Users\INFOTECHSHESH\Documents\Virtual Machines\Windov
Browse...
The default location can be changed at Edit > Preferences.
< Back
Next >
Cancel

# Create VM


New Virtual Machine Wizard
Specify Disk Capacity
How large do you want this disk to be?
The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.
Maximum disk size (GB):
50.0
Recommended size for Windows 10 x64: 60 GB
Store virtual disk as a single file
Split virtual disk into multiple files
Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.
Help
< Back
Next >
Cancel

week 8




# ESXi (Hypervisor)

# ESXi (Hypervisor)




- **Topics to be Covered:**

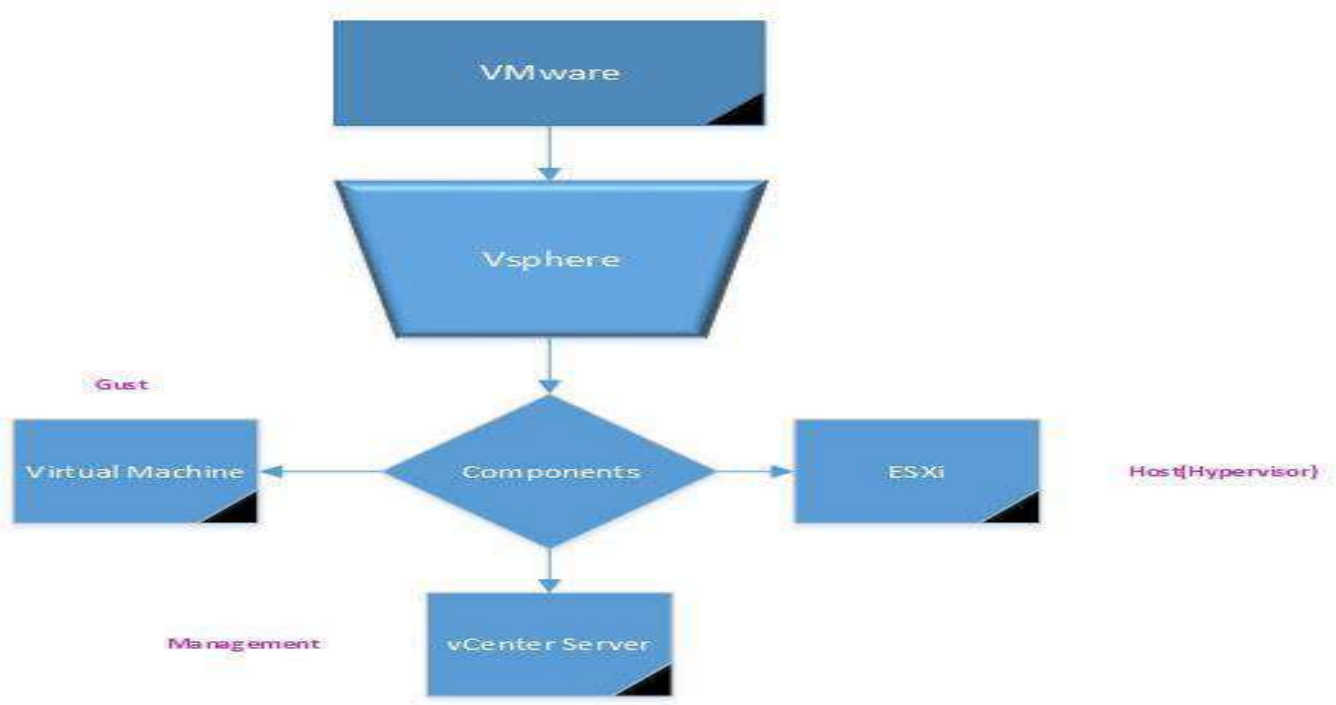
  What is ESXI

# What is ESXI




- It is Type 1 Hypervisor
- Version 6.7 vSphere (Current)

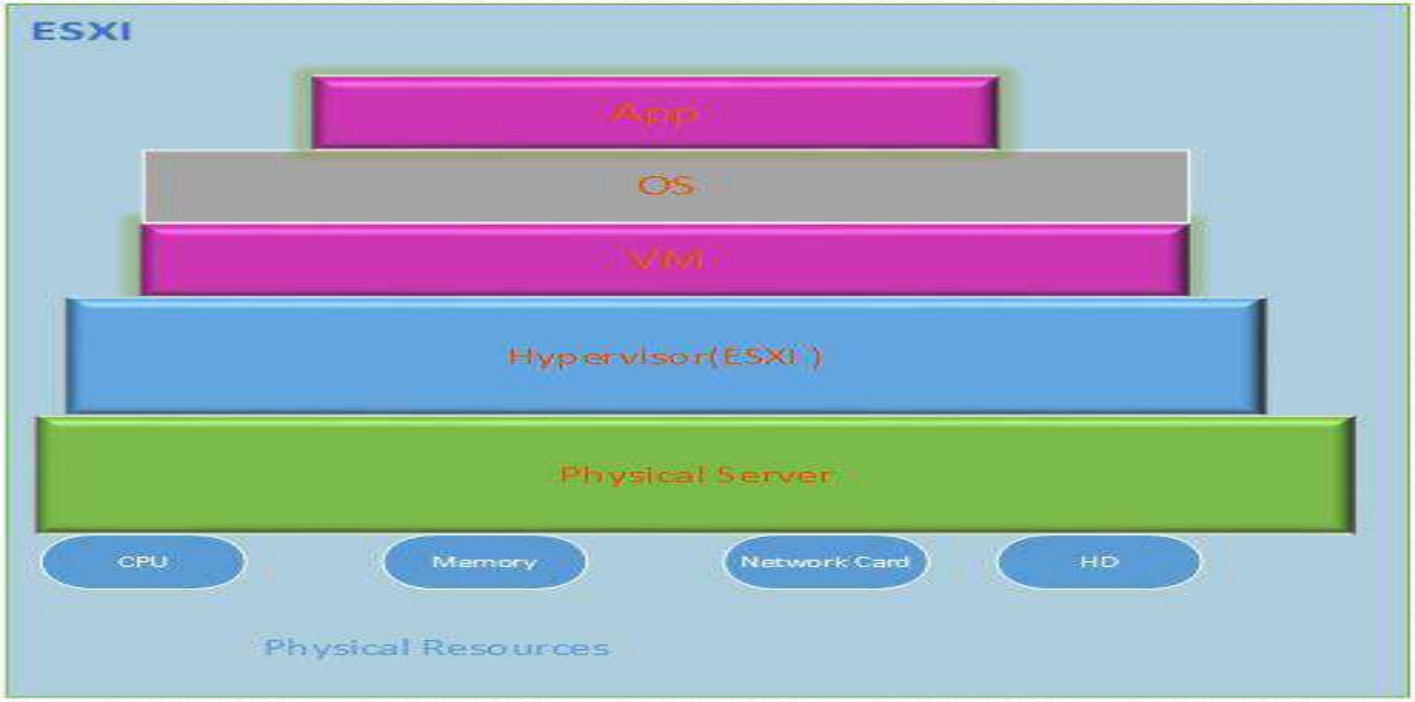
# Component of Vsphere



# ESXI

- ESXI is Hypervisor
- ESXI take physical resource  from physical server and assigned to VM
- Assigned resources from physical server to VM
- Reduced data center space , power , IT administration services

# ESXI


ESXI
App
OS
VM
Hypervisor(ESXI)
Physical Server
CPU
Memory
Network Card
HD
Physical Resources

# ESXi
# Installation Steps



# ESXi
# Installation Steps



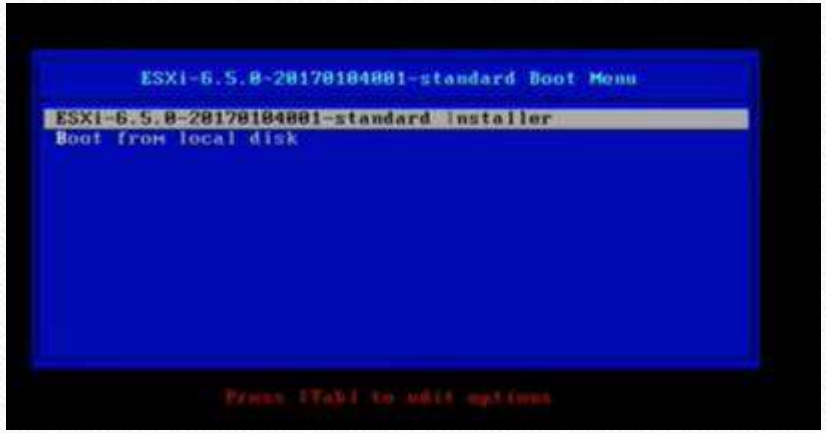

- **Topics to be Covered**:

  What is ESXI

  Lab: Installation steps

# Installation Steps

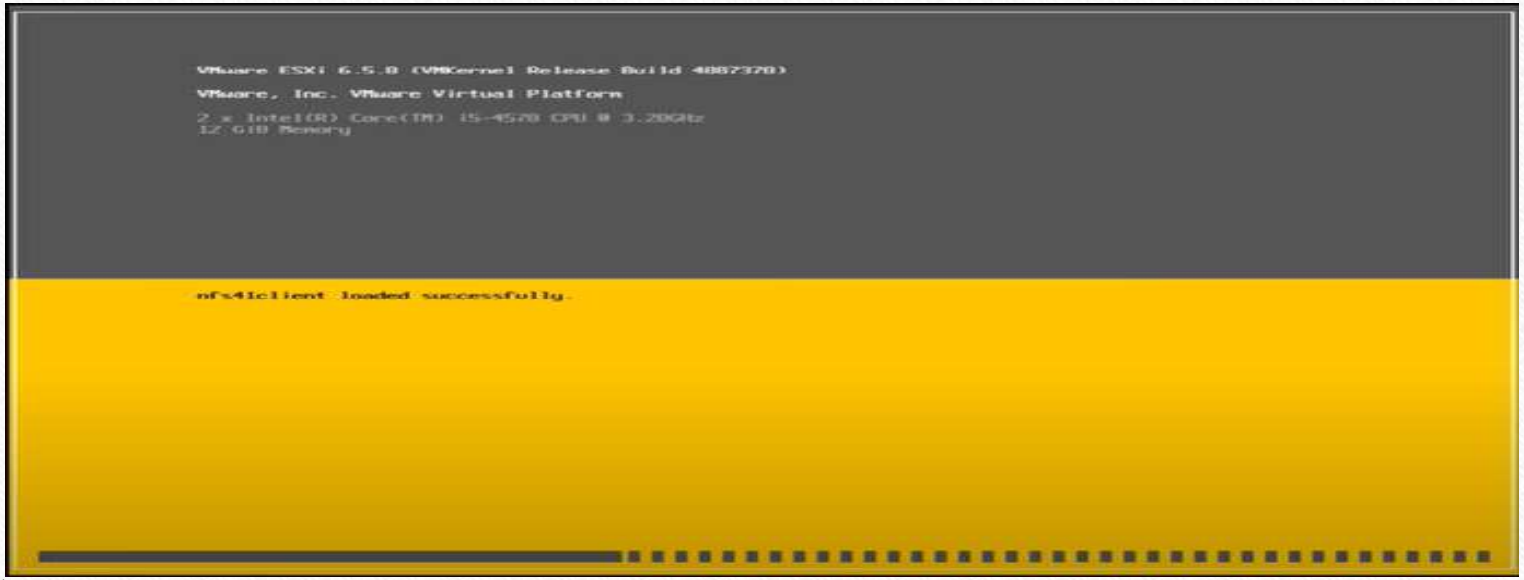
- Download **ESXi 6.5 Image from VM ware website**
- Boot System with Bootable ESXI Image



# Installation Steps

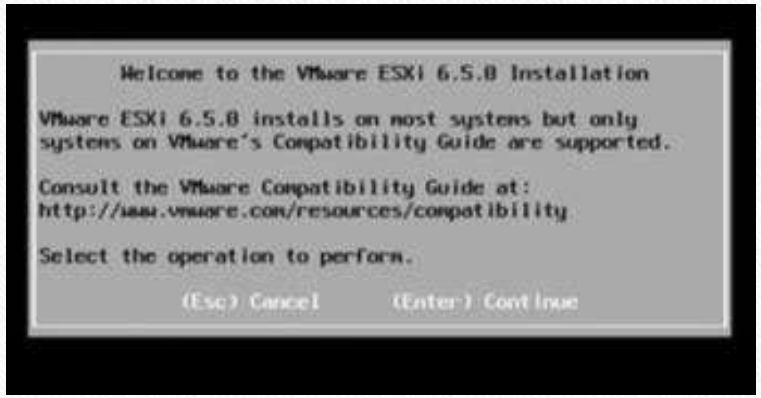


# Installation Steps



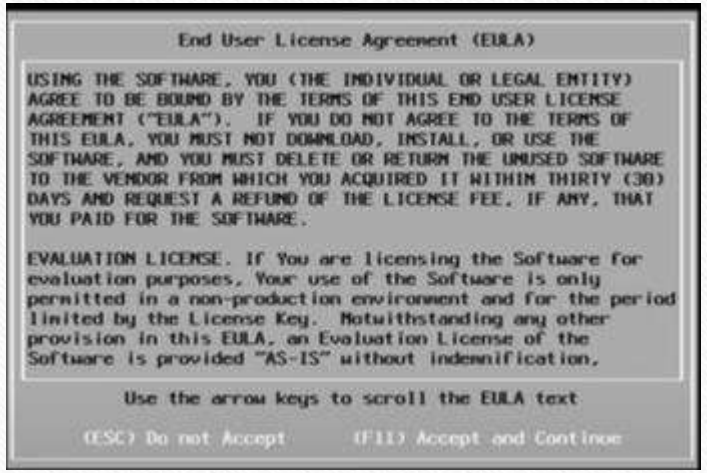

- This is not new one

# Installation Steps

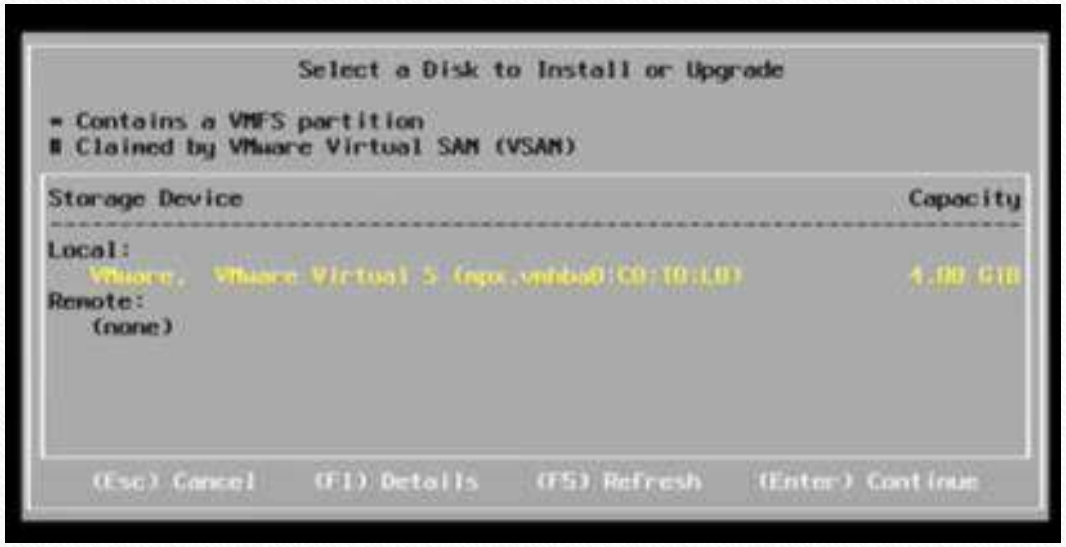



- Find HW and then used that compatible image
- Enter continue

# Installation Steps



# Installation Steps




- Scan disk to install ESXI
- Single disk used by ESXI

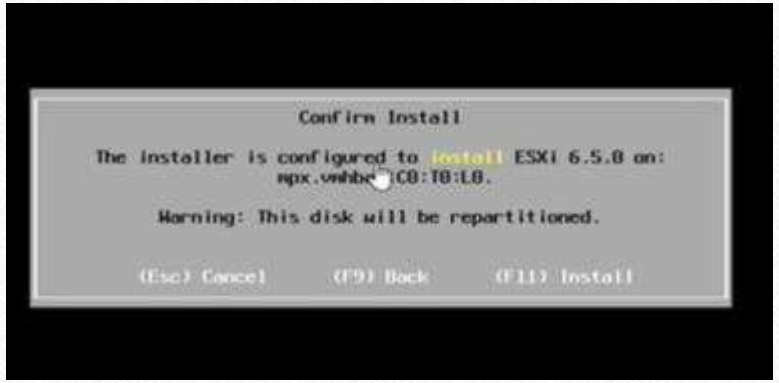
# Installation Steps



# Installation Steps



# Installation Steps



# Installation Steps



# Installation Steps



# Installation Steps

- Remove IOS image



# ESXi
# Configuration Steps



# ESXi
# Installation Steps




- **Topics to be Covered**:

  What is ESXI

  Lab: Installation steps

  **Lab: 2 Configuration Steps**

# Configuration Steps



# Configuration Steps




- As authenticate ESXI Host
- Enter user name and pass

# Configuration Steps (System Customization)



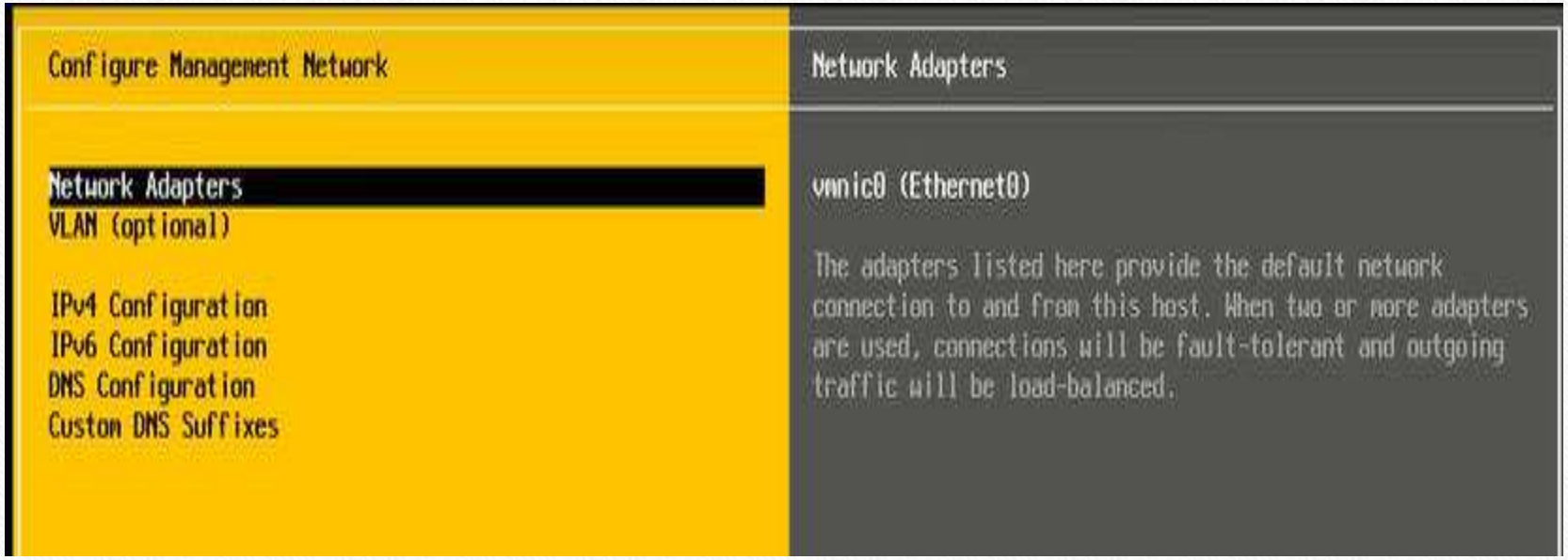
# Configuration Steps
# Password



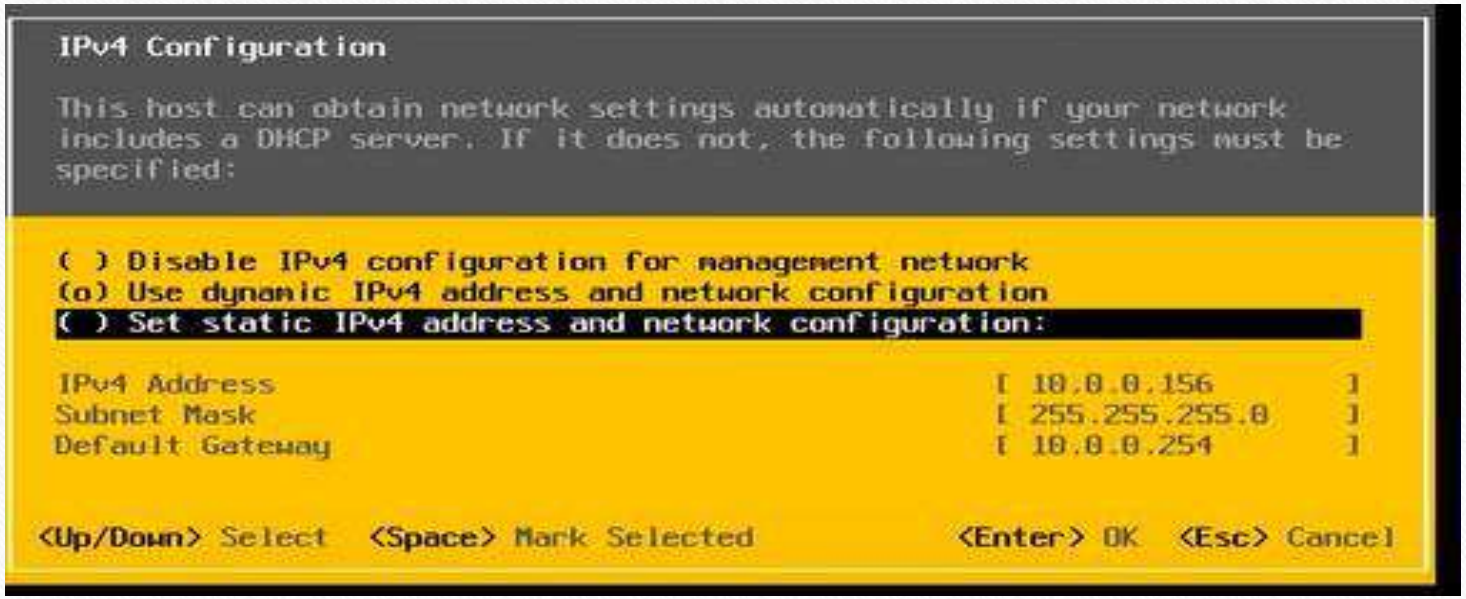
# Configuration Steps
# Configure lockdown Mode



# Configure Management Network



# Configure Management Network



# Restart Management Network

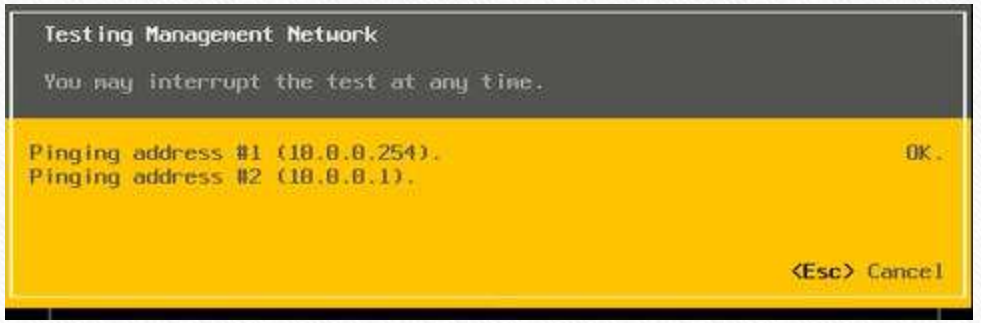



- If you have problem to connecting ESXI Host you can restart the management network

# Test Management Network



# Test Management Network

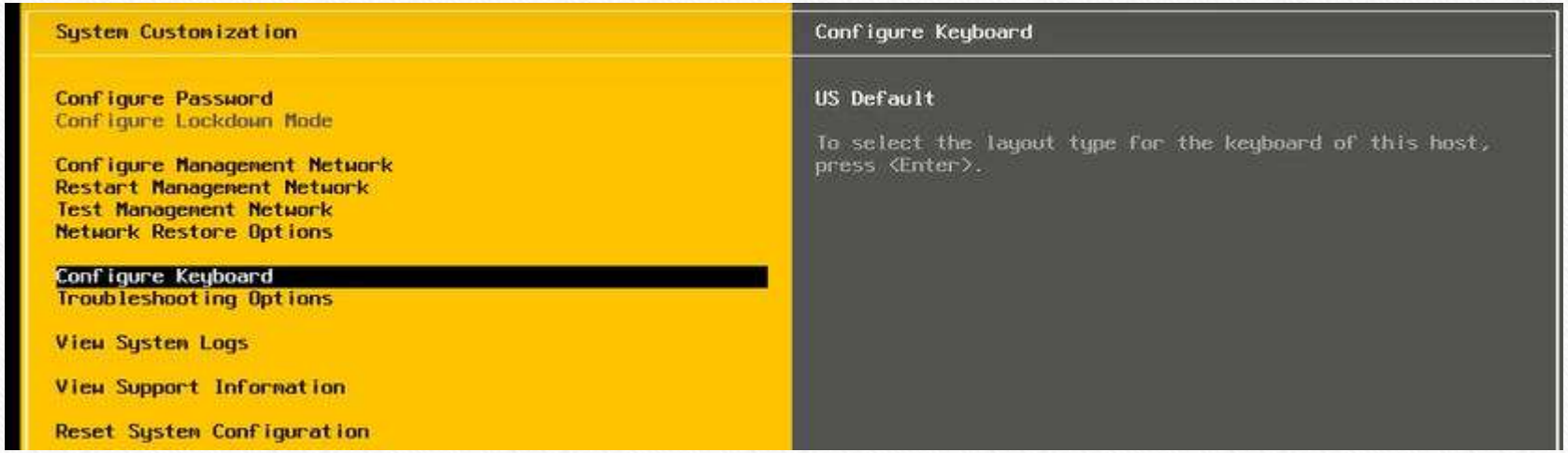



- Ping getaway

# Network Restore Options



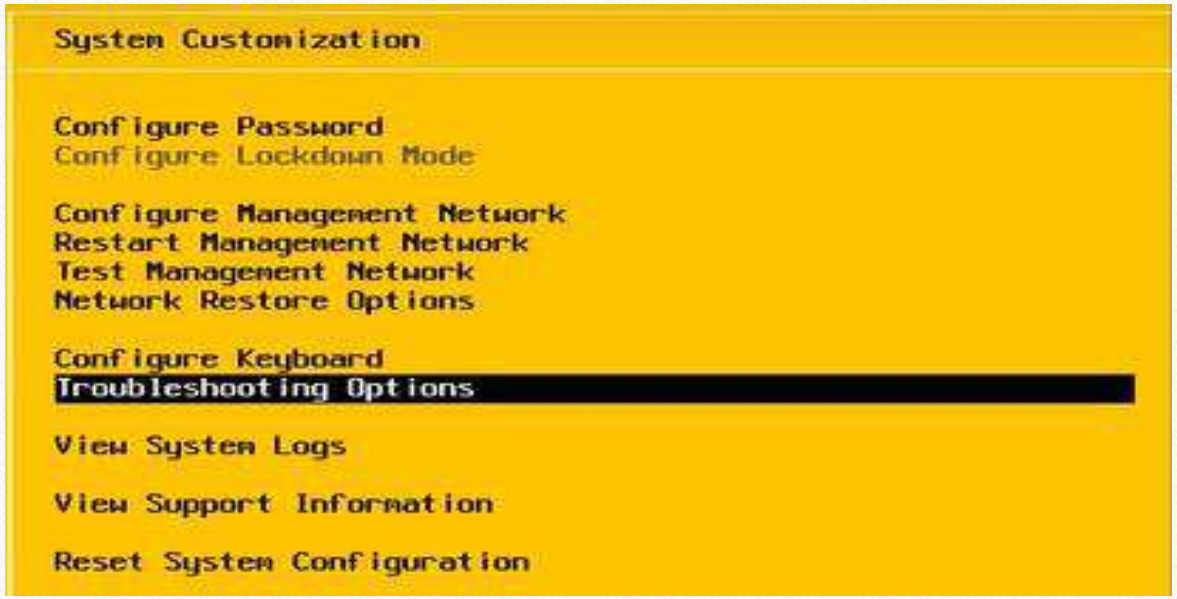
# Configure Keyboard



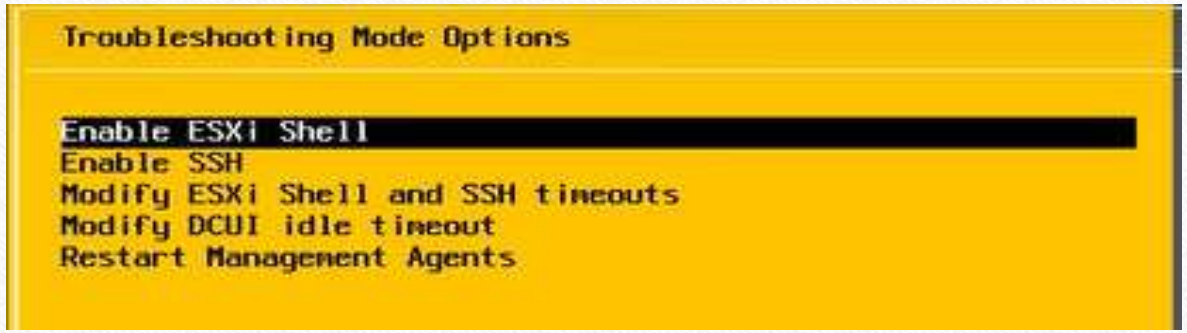
# Configure Keyboard



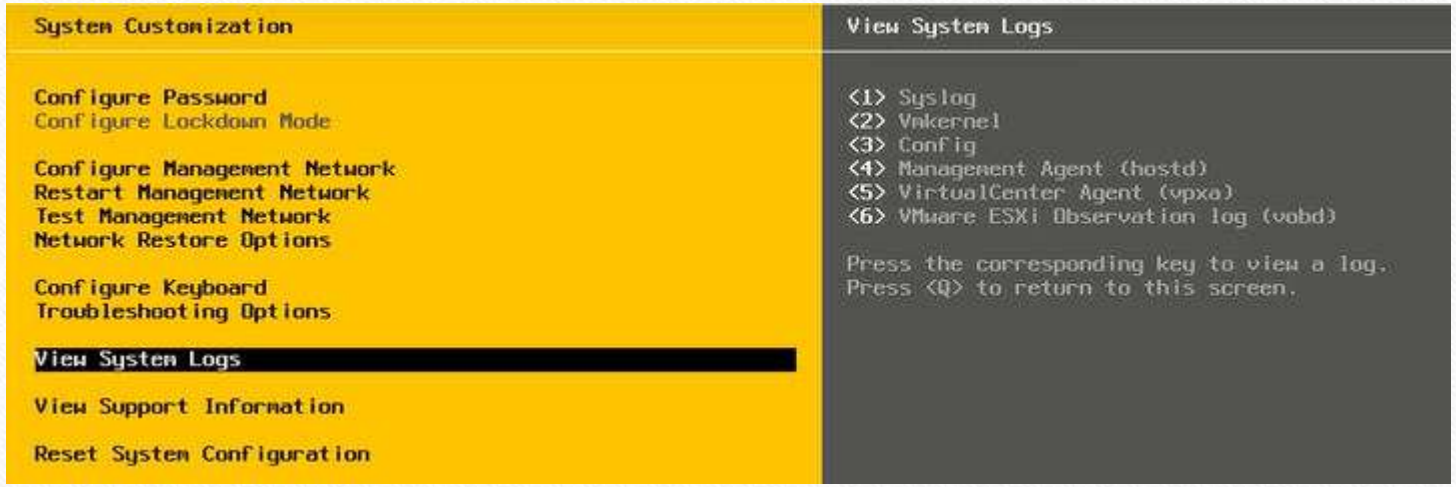
# Trouble shooting options



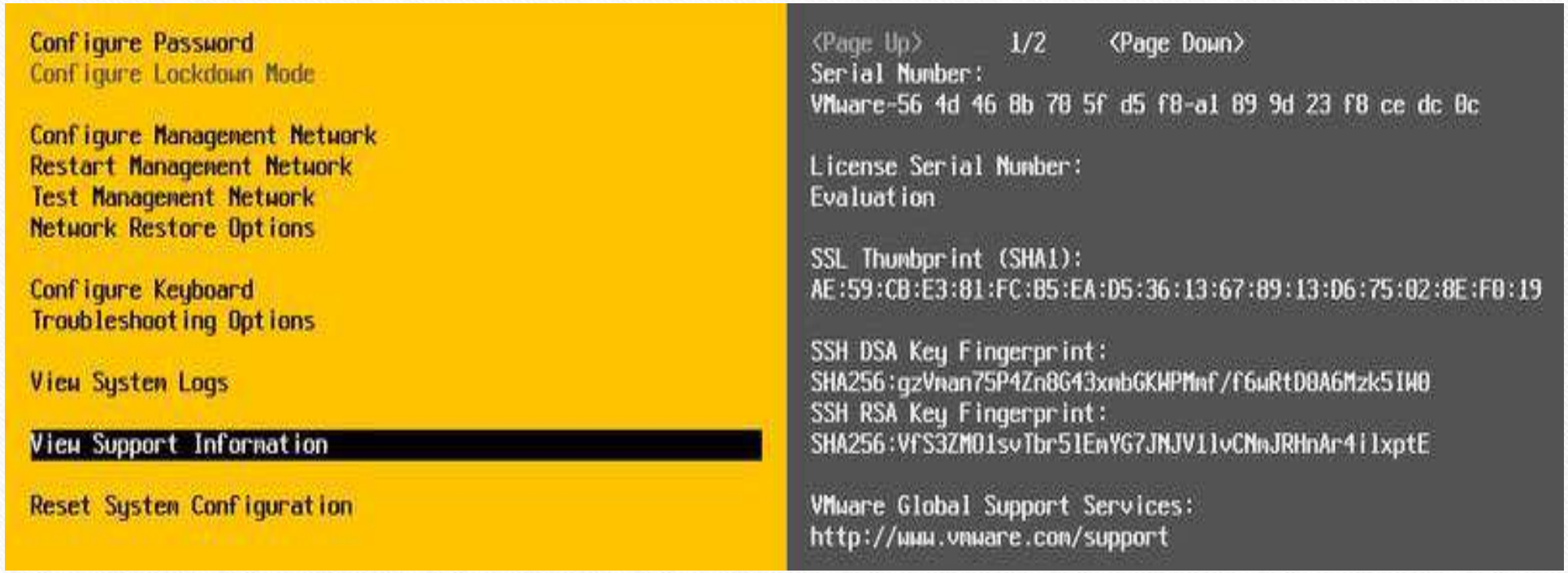
# Trouble shooting options



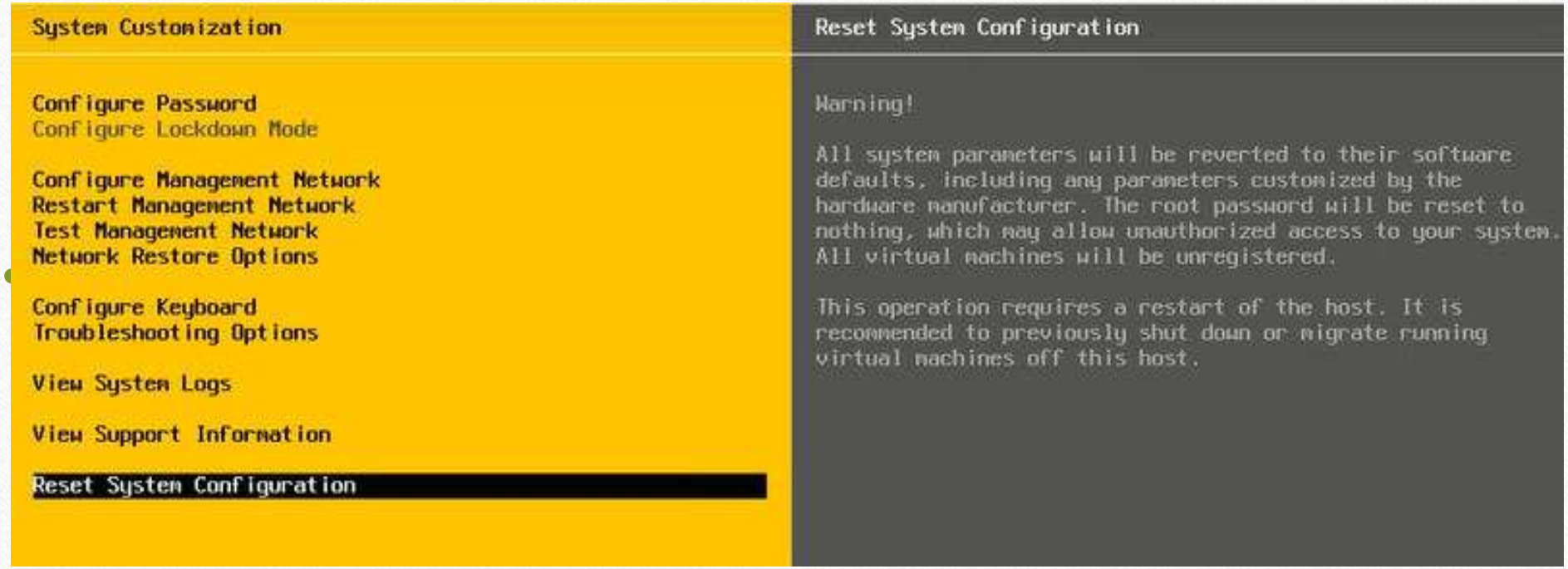
# View System Log



# View Support Information



# Reset System Configuration



# ESXi
# Creating Virtual Machine



# ESXi
# Creating Virtual Machine




- **Topics to be Covered**:

  What is ESXI

  Lab: Installation steps

  Lab: 2 Configuration Steps

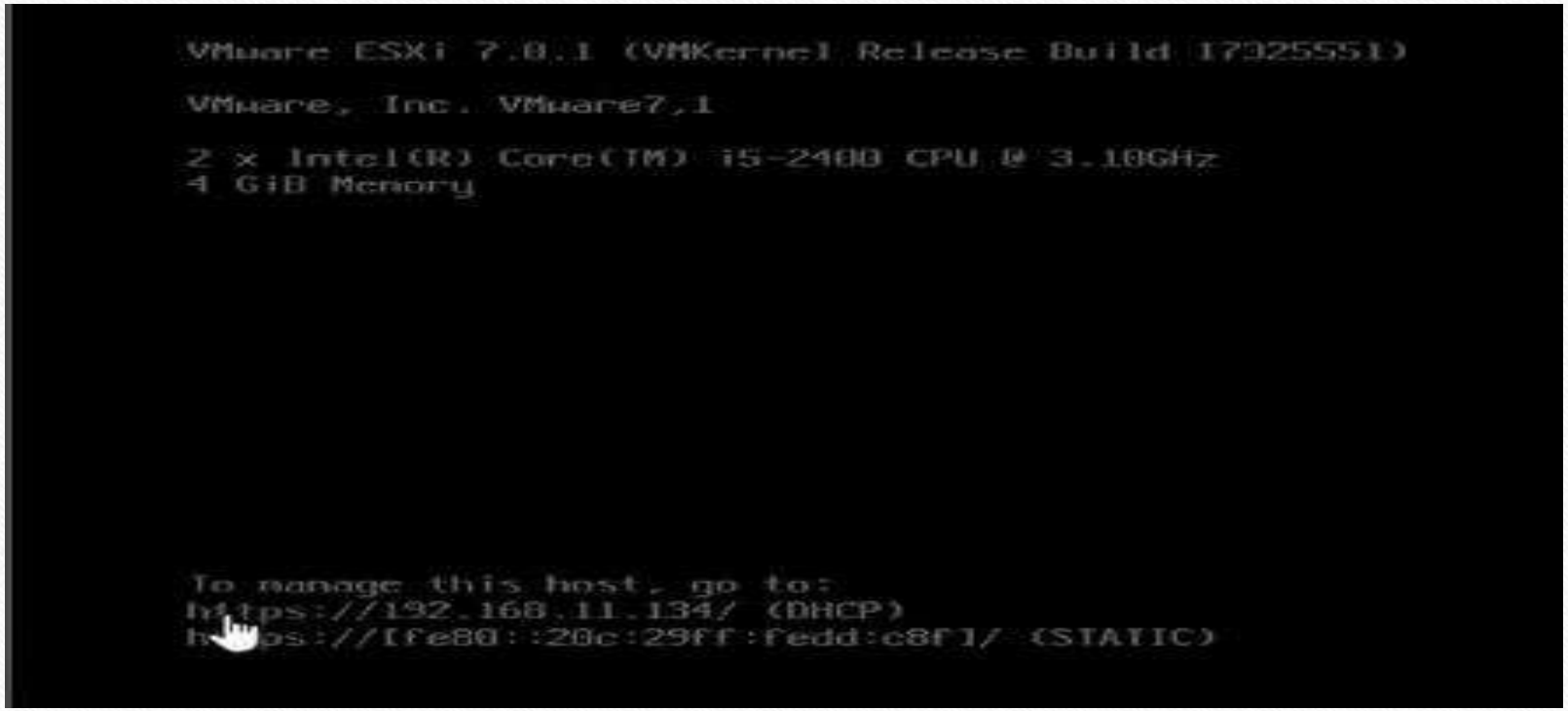
  **Lab 3: Creating Virtual Machine**

# Recommended Steps

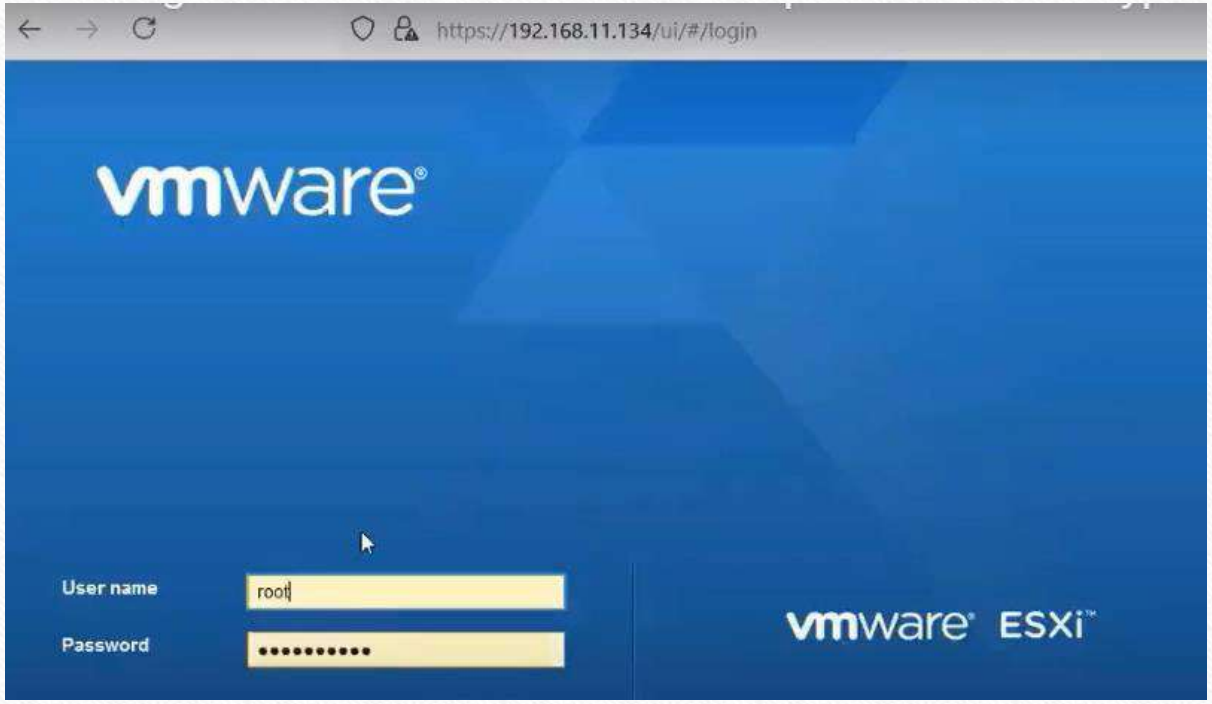



- Creating Data Store
- Upload image of OS
- Installing VM

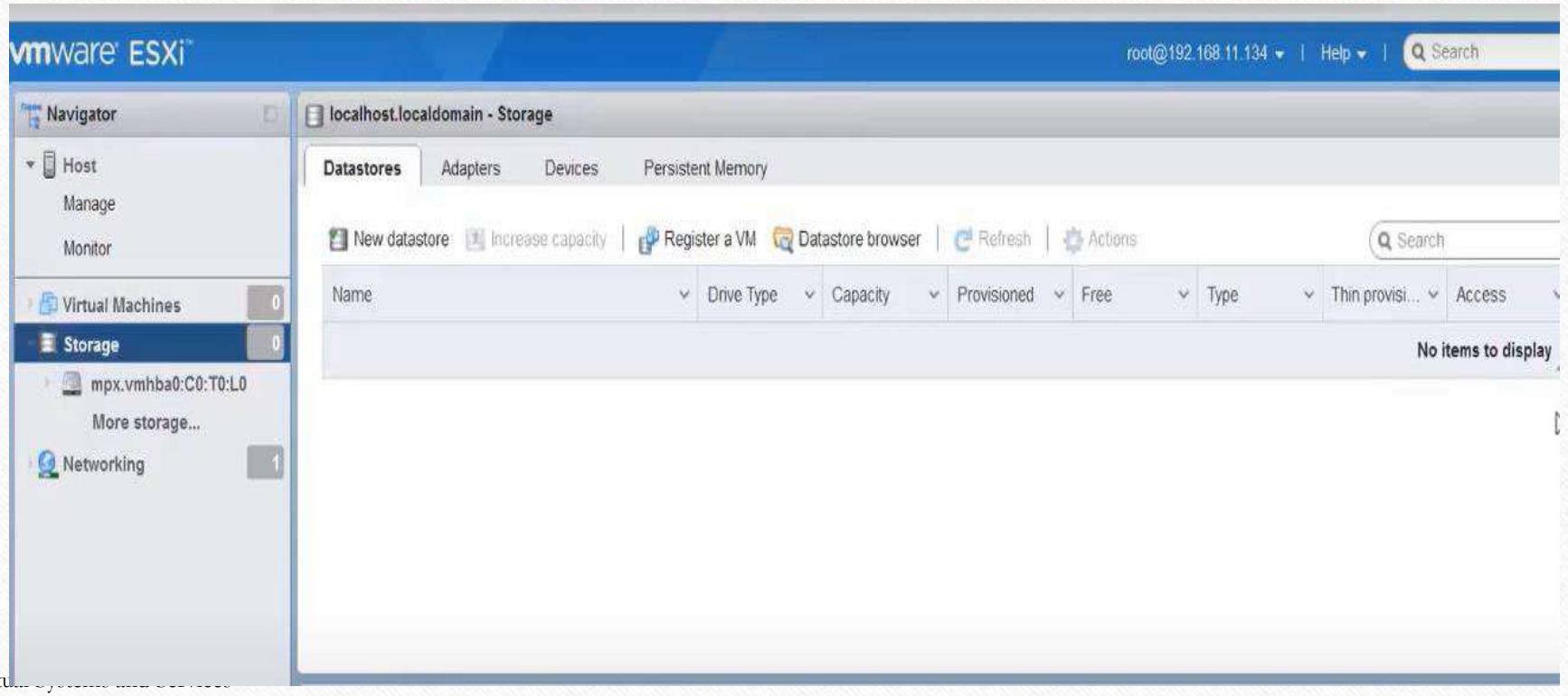
# Creating Data Store



# Creating Data Store



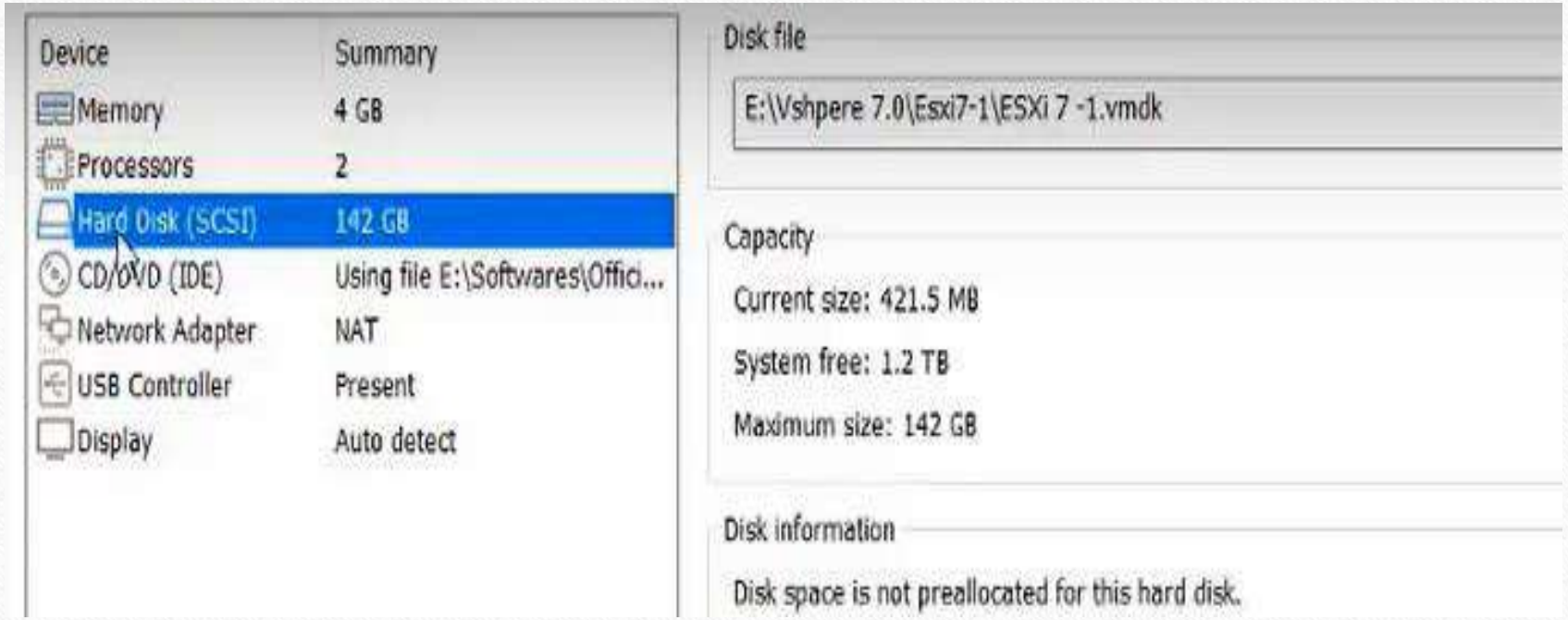
# Creating Data Store



# Creating Data Store



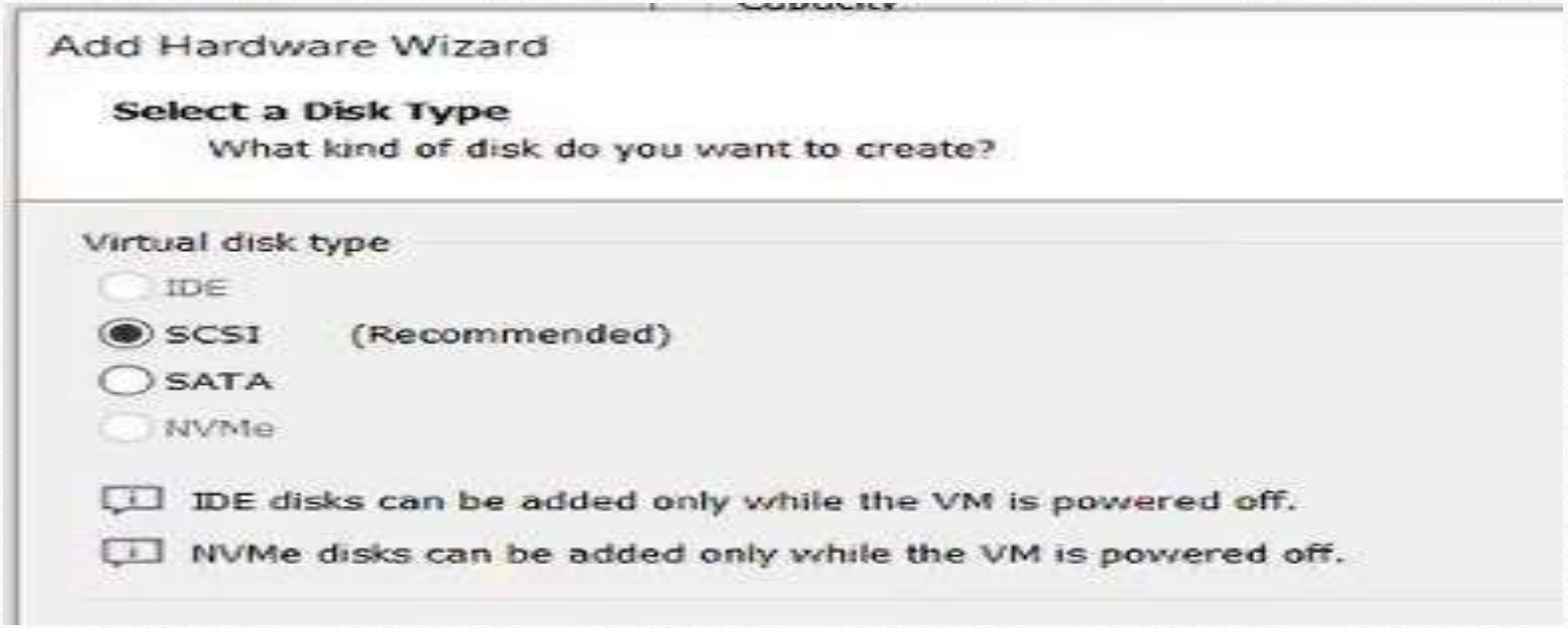
# Creating Data Store



# Creating Data Store



# Creating Data Store



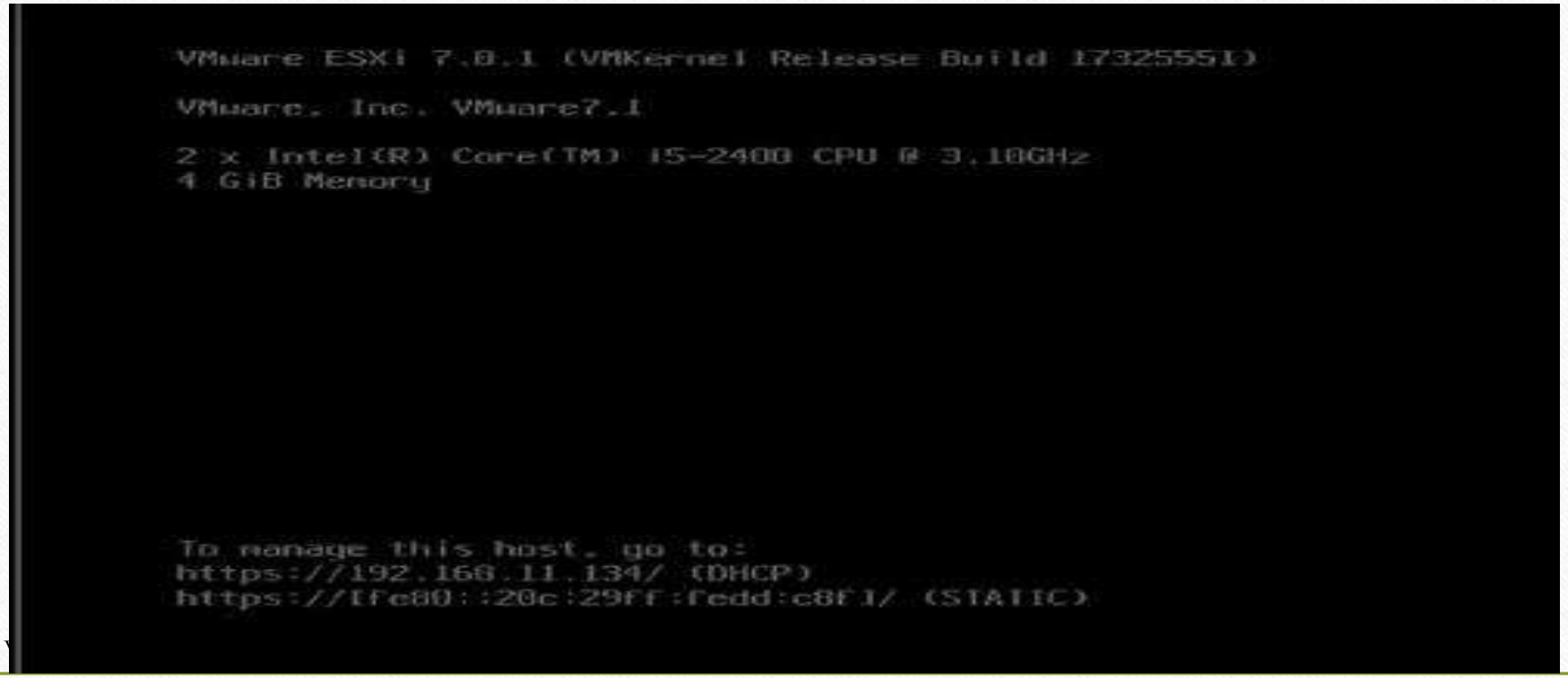
# Creating Data Store



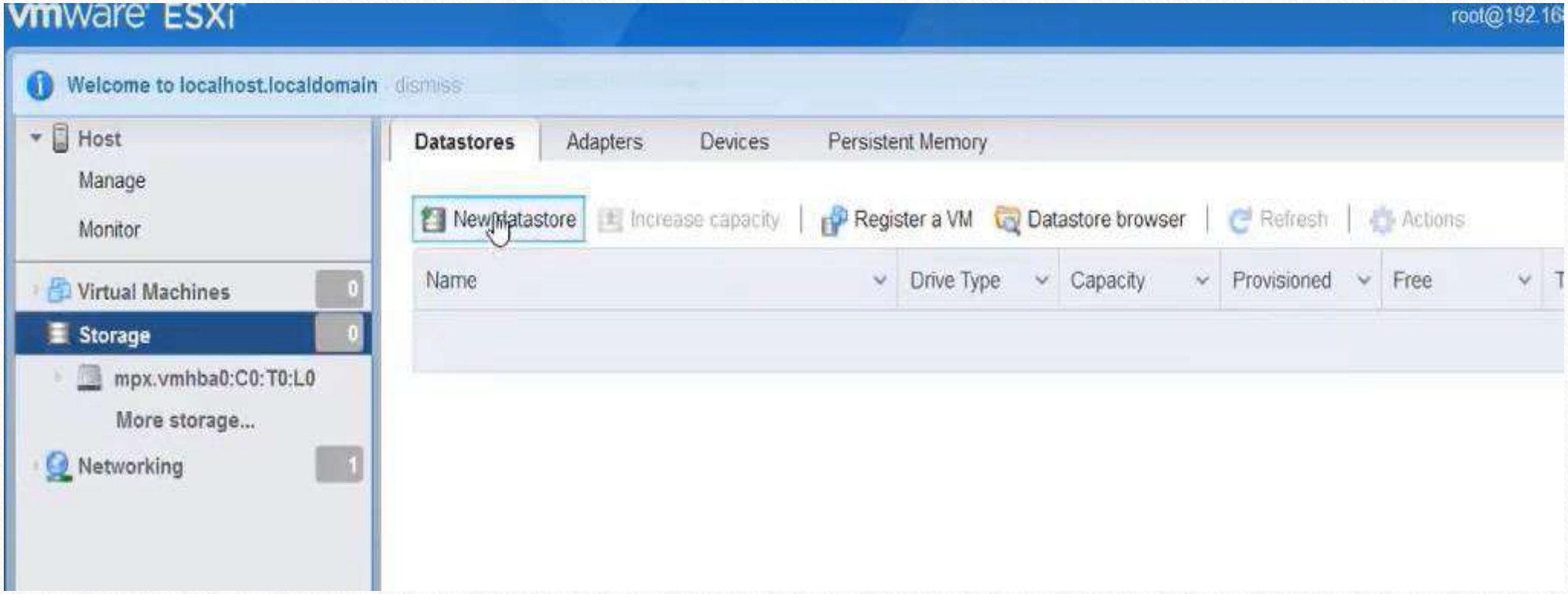
# Creating Data Store



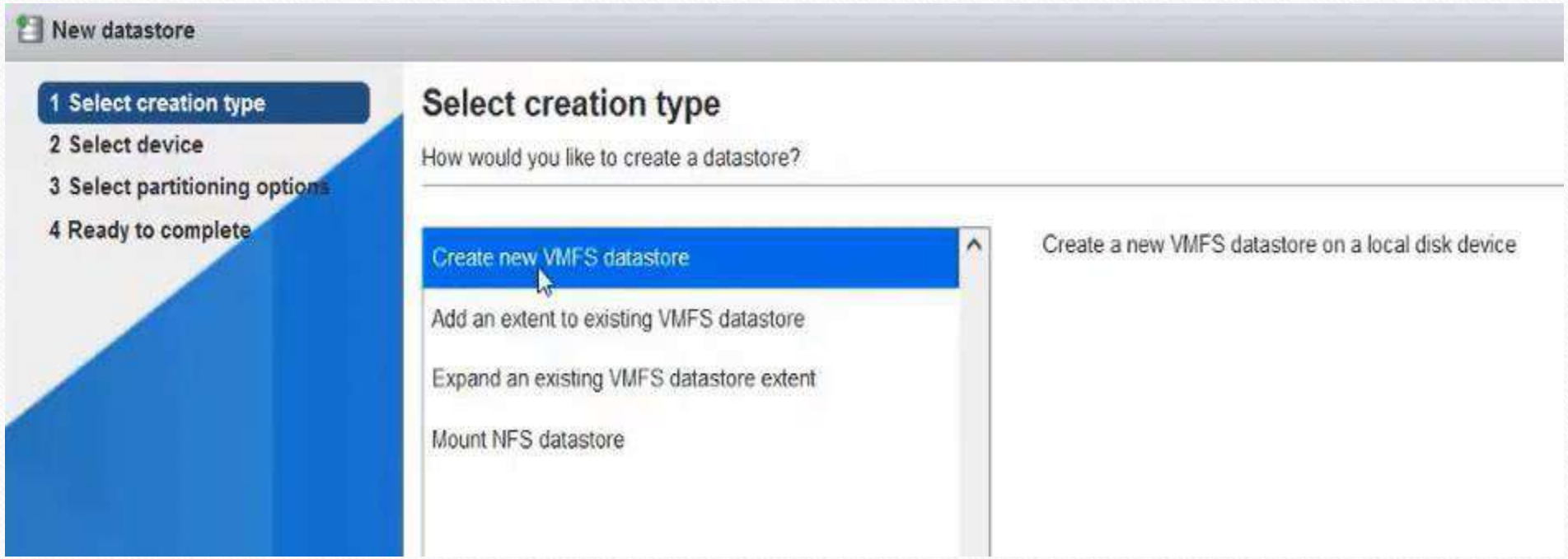
# Creating Data Store
# Restart



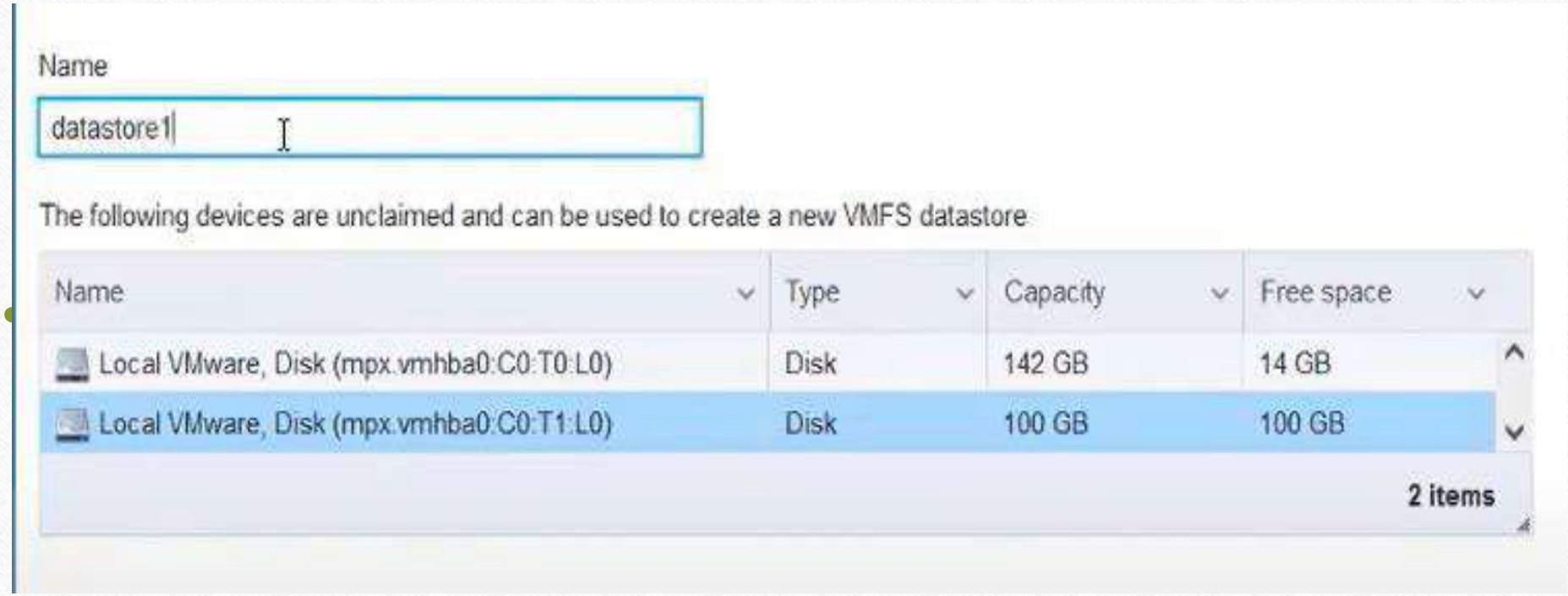
# Creating Data Store



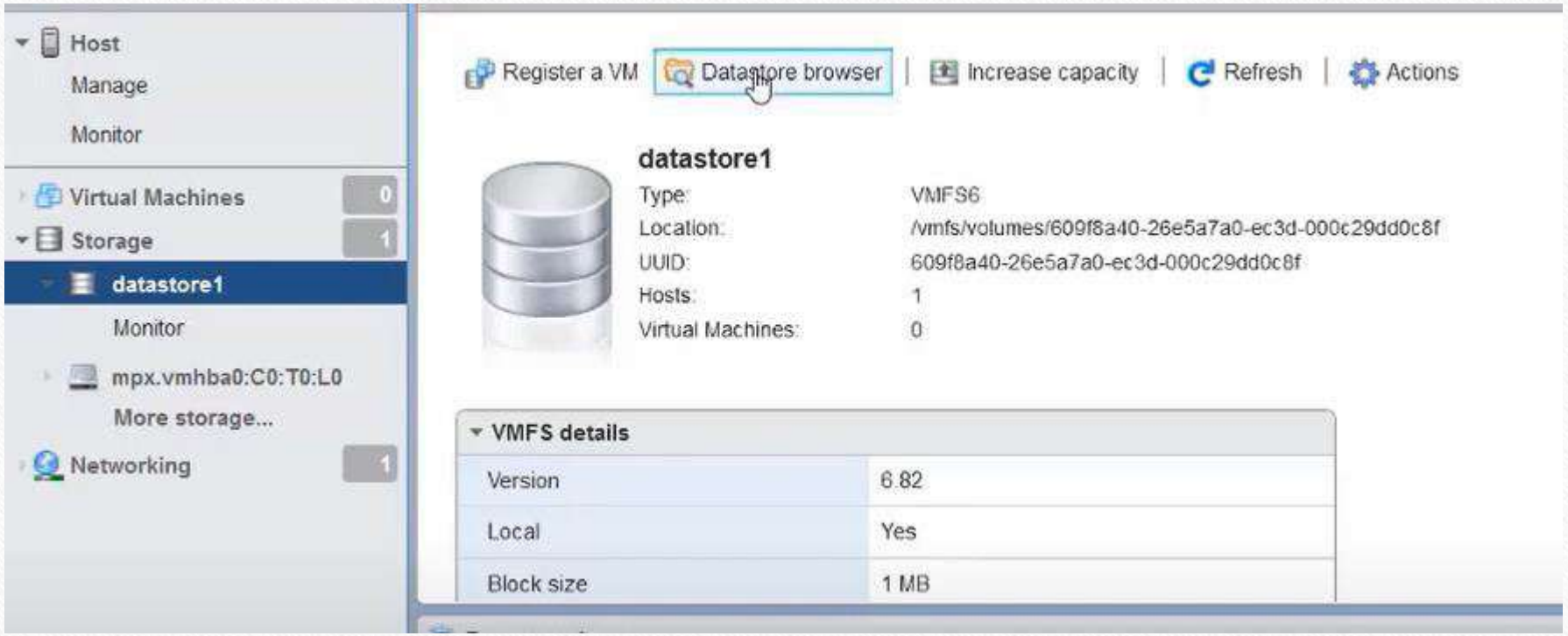
# Creating Data Store



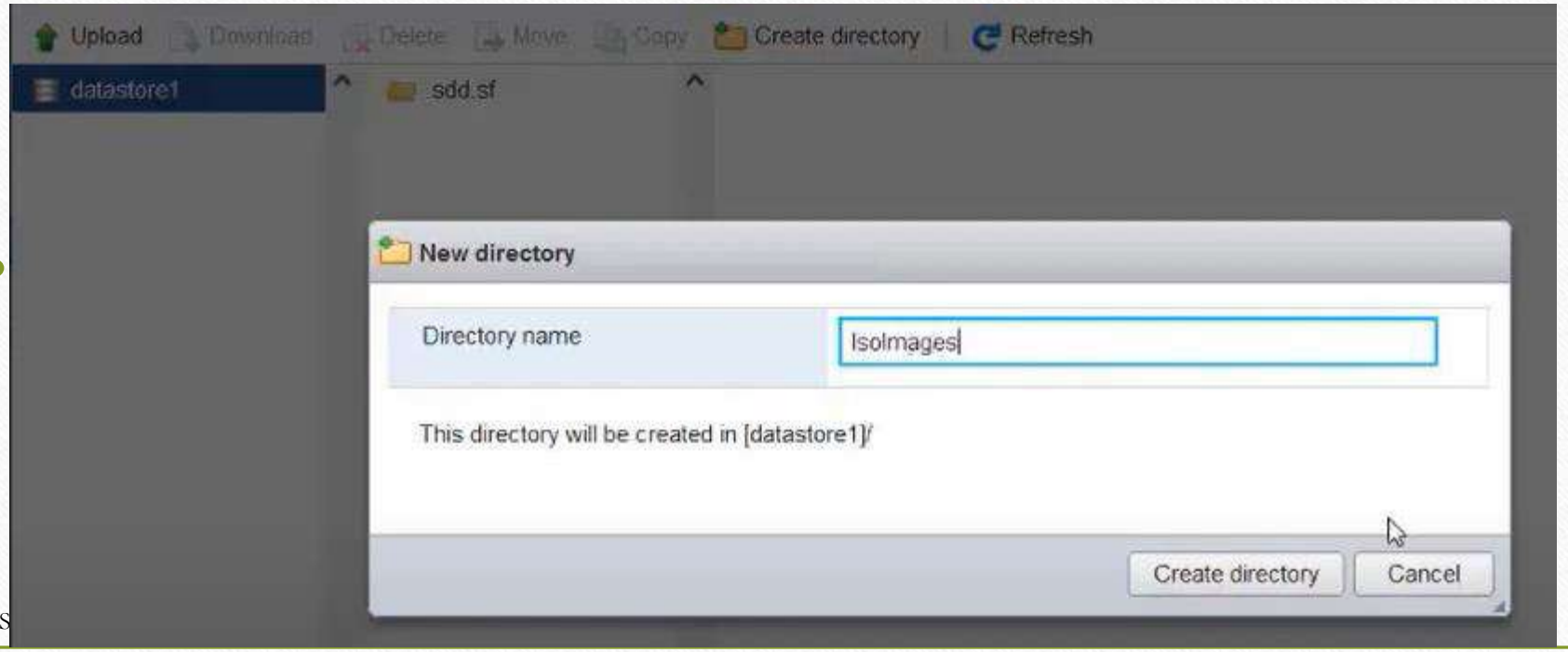
# Creating Data Store



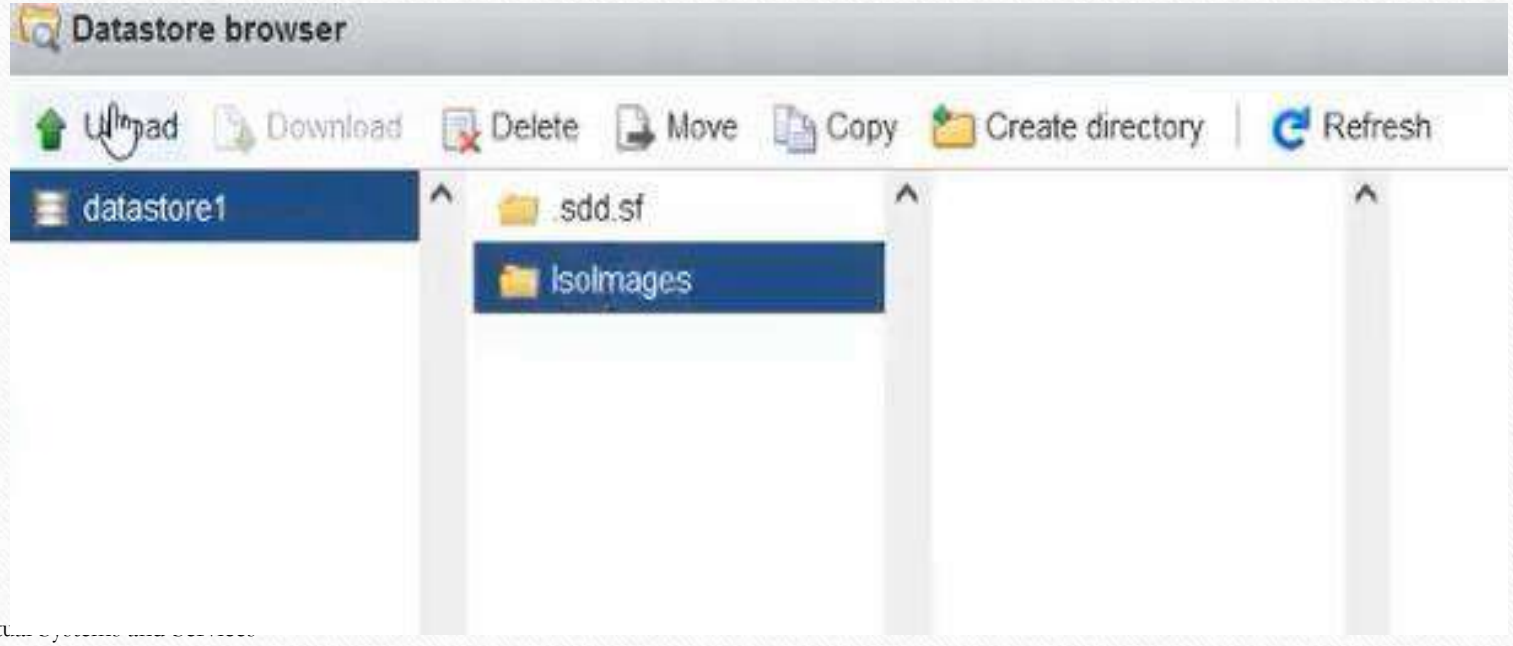
# Create Directory on Data Store



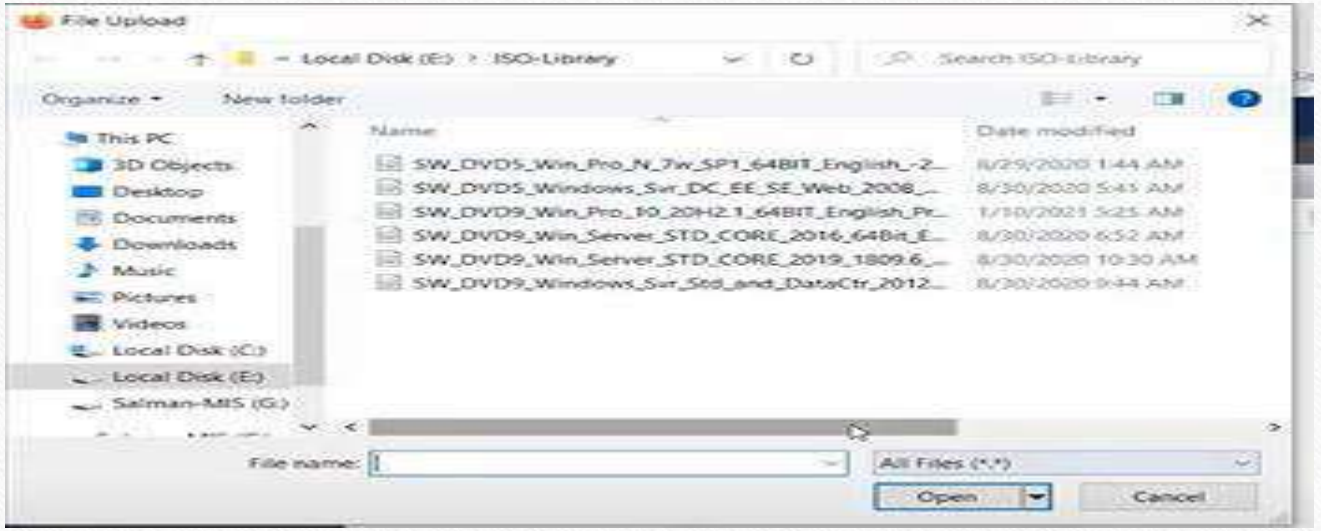
# Create Directory on Data Store



# Upload OS image



# Upload OS image



# Upload OS image



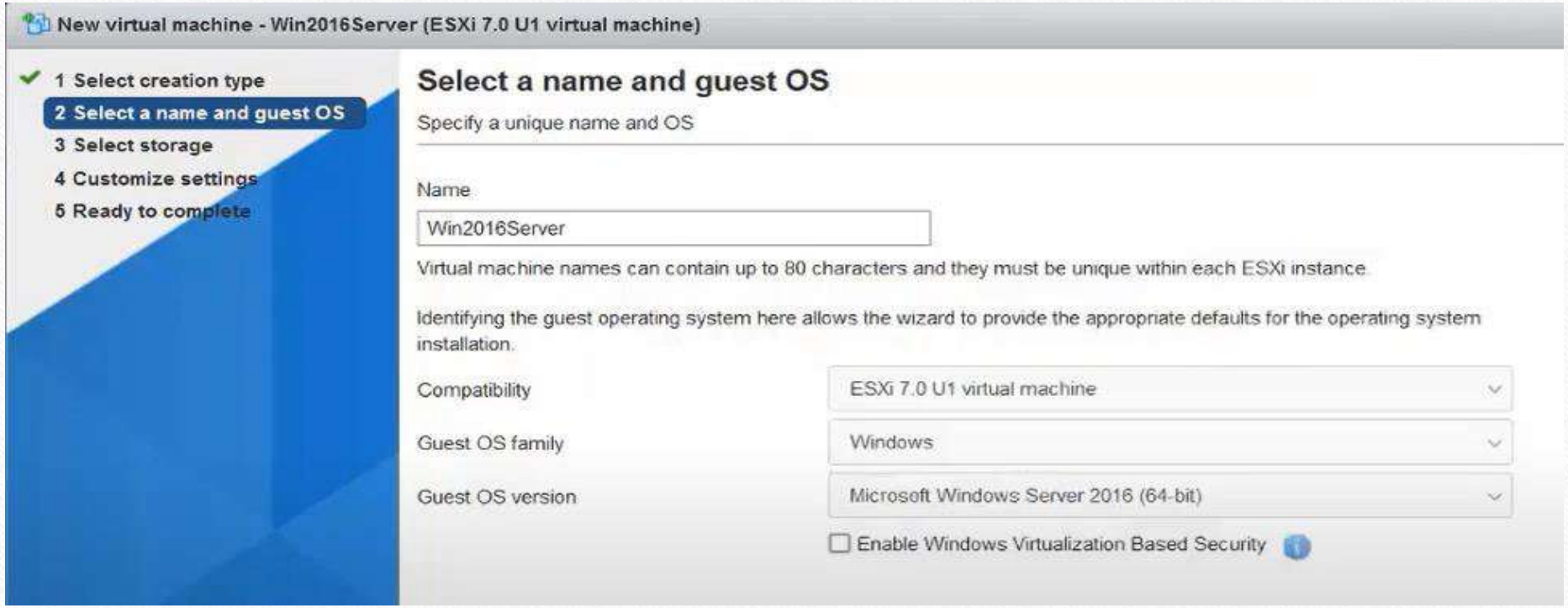
# Installing VM



# Installing VM



# Installing VM



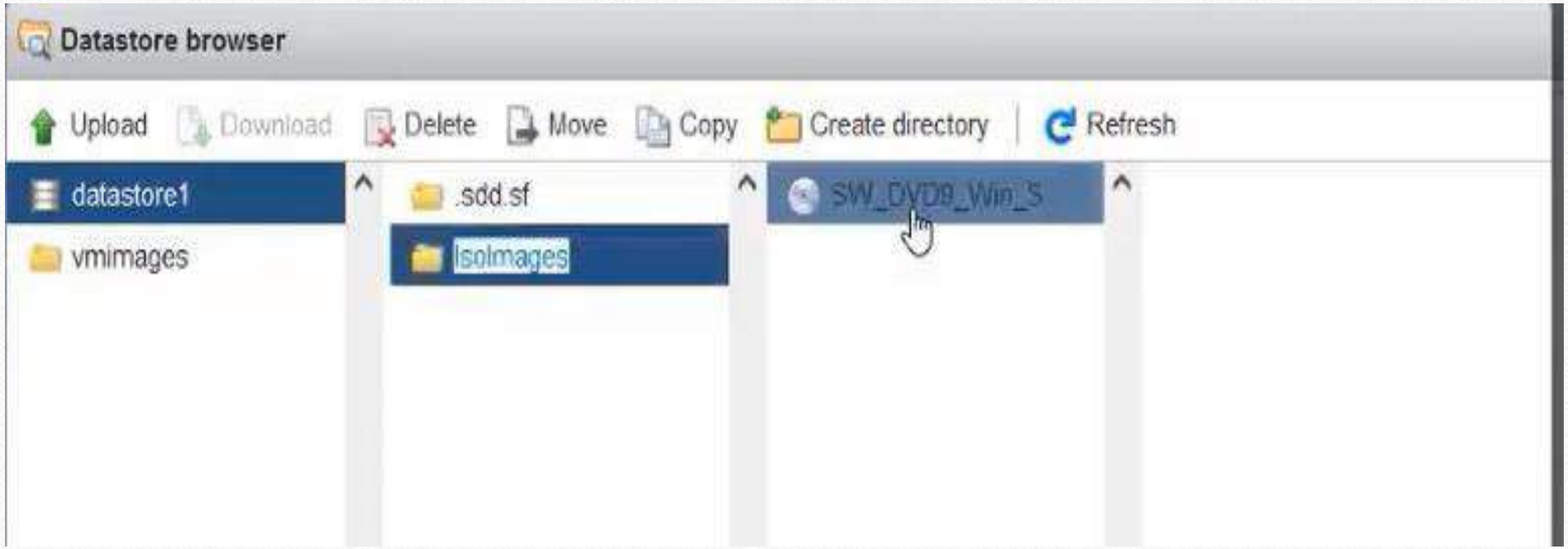
# Installing VM



# Installing VM



# Installing VM



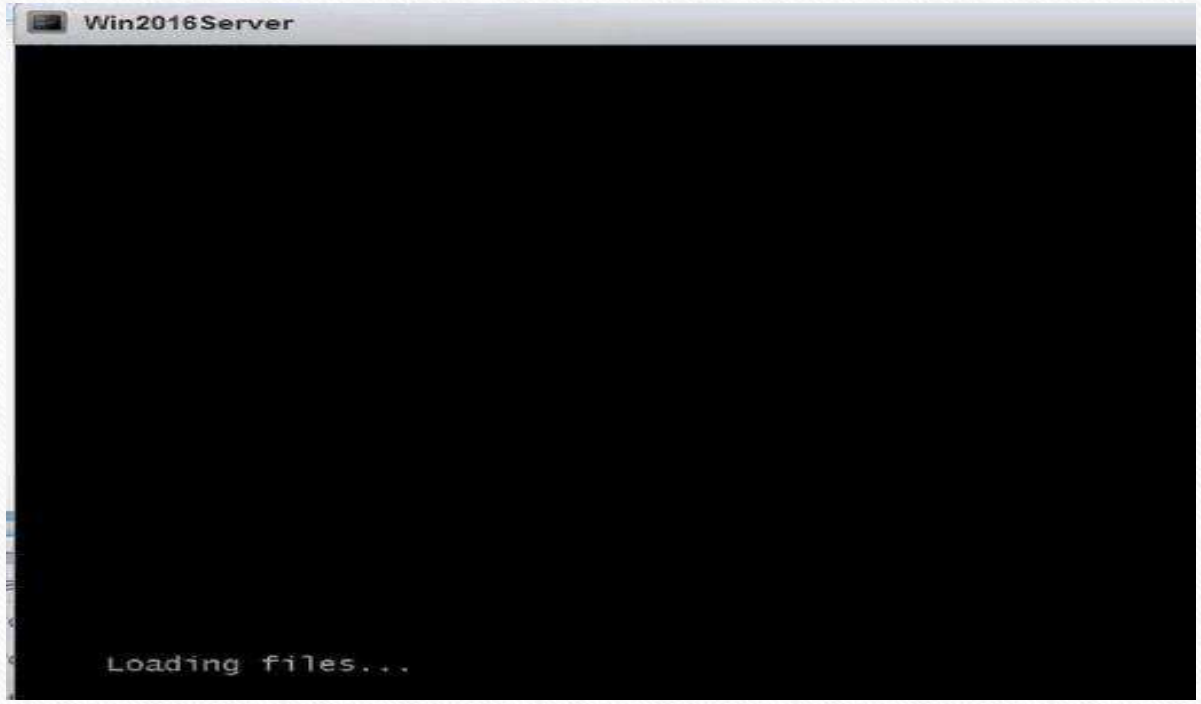
# Installing VM



# Install Window



# Install Window



# week9






# VirtualBox



# VirtualBox




- Oracle VM VirtualBox is a cross-platform virtualization product
- It is used for virtualizing x86 and AMD64/Intel64 computing architecture
- It is freely available as open source software under the terms of GNU General public license (GPL)
- It runs on all major operating systems including Windows, Linux, Mac and Solaris

# VirtualBox

- Different Versions of Oracle VM VirtualBox exists to install ( for Windows, Linux and Solaris etc.)
- VM VirtualBox is first downloaded and installed on any host OS
- It has both the graphical and Command Line interface (GUI and CLI)
- Using VirtualBox, multiple guest operating systems can be installed as VMs on host OS
- We can install Windows 7 or Linux as guest OS on hosted OS Windows 10

# VirtualBox




- VirtualBox supports both the hardware and software virtualization
- It is recommended to use VirtualBox for small and mid-range projects

# VMware Vs VirtualBox




Virtual Systems and Services



# VMware Vs VirtualBox




## VMware

- It is a virtualization software platform developed by vendor, VMware
- It is a type-1 hypervisor
- VMware supports hardware virtualization
- Its workstation player is free while other VMware products require a paid license.

## VirtualBox

- It is a virtualization software platform developed by Oracle Corporation
- It is a type-2 hypervisor
- VirtualBox supports both software & hardware virtualization
- It is an open source virtualization product

# VMware Vs VirtualBox

## VMware

- VMware workstation works on Windows and Linux. For macOS, we need to install VMWare Fusion/Fusion Pro
- It supports VMDK disk format

## VirtualBox

- VirtualBox can install on Windows, Linux, Solaris, MacOS, FreeBSD
- It supports VDI, VHD, VMDK disk formats

# VMware Vs VirtualBox




## VMware

- VMware supports snapshots on its paid products
- VMware Virtual Machines run faster
- For enterprise projects, VMware is recommended

## VirtualBox

- VirtualBox supports snapshots free of cost
- VirtualBox Virtual Machines run slower comparatively.
- For small and mid-level projects, VirtualBox is recommended

# week10






# Hyper-V




Ijaz Ullah

Lecturer in Computer Science Department

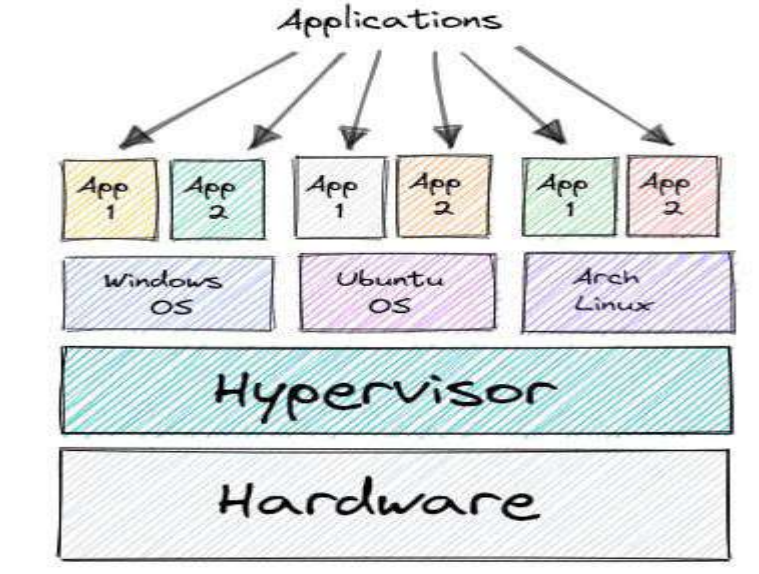
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# Hyper-V

- Hyper-V is a virtualization platform through which multiple isolated environments can be created on single physical server.
- Hyper-V is the product of Microsoft
- Hyper-V can be enabled from the Windows operating system
- Hyper-V is type 1 hypervisor

# Type 1 Hypervisor

- Hyper-V is type 1 hypervisor






# Advantages of Hyper-V

- Cost can be reduced
- Hardware can be utilized with full capacity
- Provide flexibility
- Different environments like testing and production environment can be created
- Power consumption can be reduced too

# Compare

- Hyper-V
  - Microsoft dependent
  - Checkpoints
  - Hyper-V live migration tool for migrating workload
  - Hyper-V provide support for Linux
- VMware
  - Complete echo system for virtualization
  - Snapshots
  - VMware vMotion for migration of work load
  - Fault tolerance and disaster recovery

# Conclusion

- Hyper-V is Virtualization platform
- All advantages of virtualization can be achieved with the help of Hyper-V
- Both VMware and Hyper-V are the market players of Virtualization
- Hyper-V is type 1 hypervisor

# Image sources




- https://www.youtube.com/watch?v=tlL1W9MLobo
- https://dev.to/kcdchennai/virtual-machines-101-2gmg
- http://www.greenit-monaco.com/en/software/hyper-v.html

# Hyper-V Architecture




Ijaz Ullah

Lecturer in Computer Science Department

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# Hyper-V Architecture

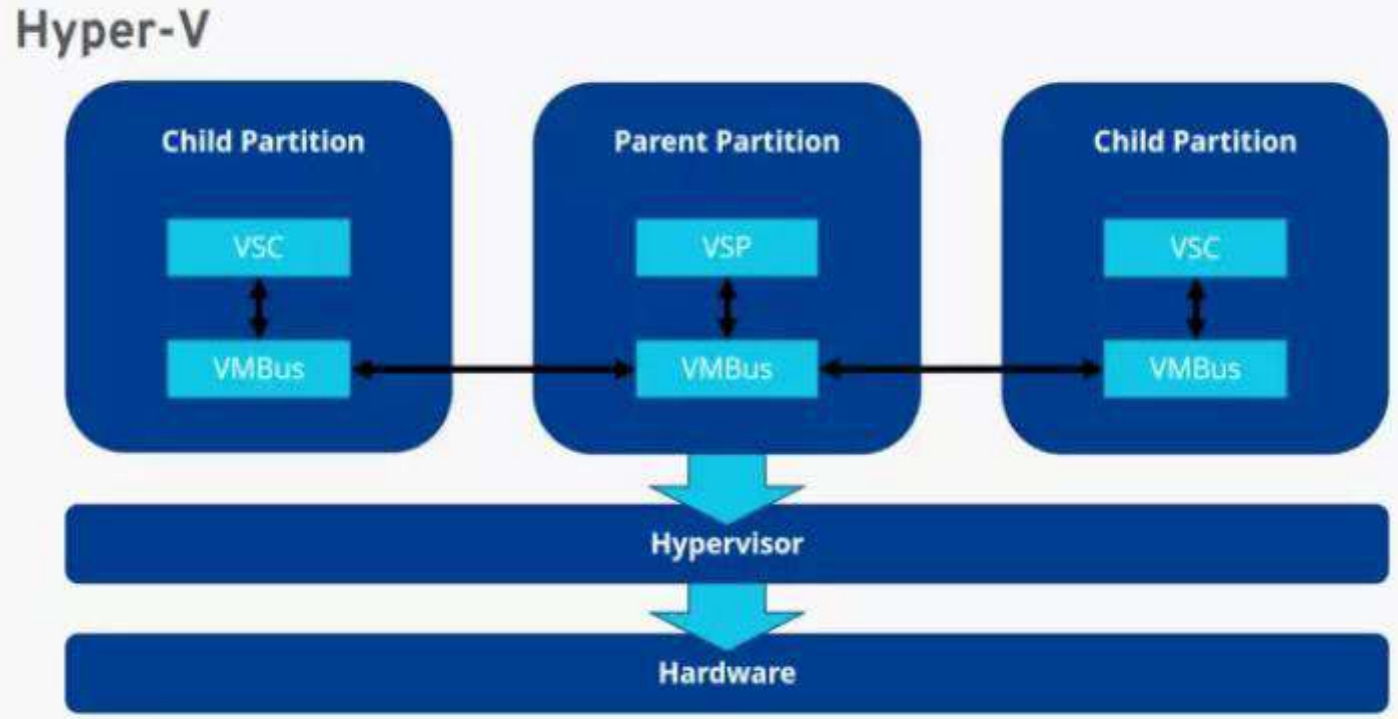



Image Source:https://www.ionos.com/digitalguide/server/know-how/what-is-hyper-v/

# Hyper-V Lab

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Lecturer in Computer Science Department

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# Hyper-V Lab




- Hyper-V  is a Microsoft Product for Virtualization
- Hardware Virtualization
- Flexibility
- Modify your infrastructure according to your workload

# Step 1

- Enable Hyper-V in Windows 10 pro or Windows Server
- Go to Apps & features
  - Then Programs & Features
  - Turn Windows features on or off
  - Check Hyper-V option from the list so that it needs to be selected .
- Now Hyper-V will be accessible from the search bar in windows

# Step 2

- Create a Virtual Machine
  - Search Hyper-V Manager open it
  - From Hyper-V Manager Dashboard in Actions Click on  New
  - Click on Create Virtual Machine
  - Configure it according to your needs

# Step 3

- Install Operating system in the Virtual Machine
- Select Virtual Machine you just created in step 2 which will be shown in the center
- From the menu bar Click on Actions and then on Connect then proceed with Start option on Screen
- Now your Operating System installation step will start

# Step 4




- Play with Virtual Machine state
- Either turn it off if its running
- Or turn it on if its off
- Also check Shutdown status
- Right click on the Virtual Machine and Look for above Options

# Step 5




- Explore the setting option of the Virtual Machine
- Increase the memory or CPU according to your workload from the setting option of VM

# Try it Yourself




- Now its your turn explore and apply all options covered in this Lab

# Week 11

# Virtual Machine

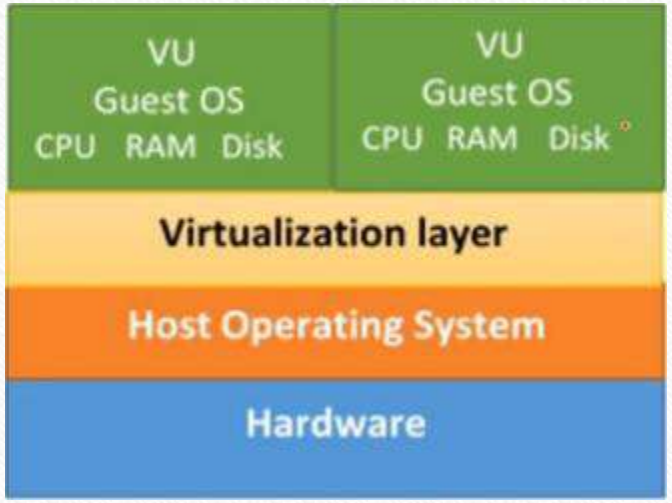
# Topics to be covered

- What is Virtual Machine?
- Uses of Virtual Machine
- Virtual Machine's Files

# Virtual Machine

- Virtual Machine is a virtual representation of a Physical machine
- It is called as Guest machine or Virtual Server
- It provides same functionalities as Physical machine or Server
- It uses the resources of physical machine through virtualized software
- Each Virtual Machine has its own operating system called as Guest Operating System

# Virtual Machine



# Uses of Virtual Machine

- VM uses as a substitute of Physical Server in a Data Center
- Reduce Data Center Cost
- VM provides different types of Virtualization like Hardware virtualization, desktop virtualization etc.
- Provides Scalability (add new VM or upgrade existing one)Migration is quite easier

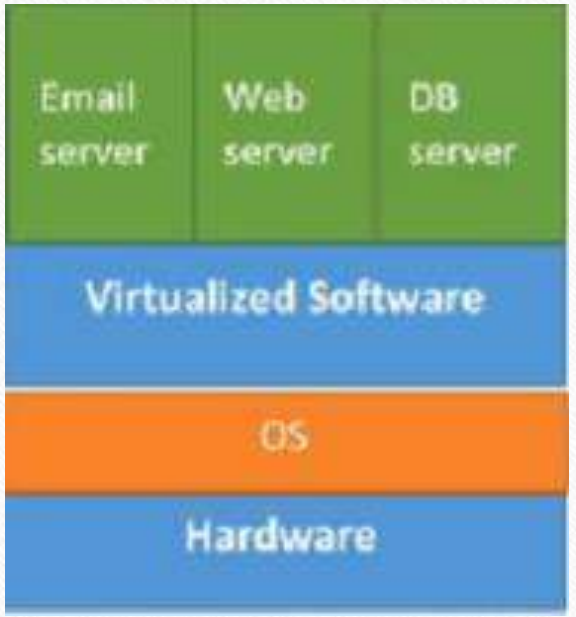
# Virtual Machine Files

| File | Brief Description |
|---|---|
| vmname.nvram | VM BIOS Configuration file |
| . vmx | Configuration file of VM. It contains information of hardware setting |
| . vmsn | VM snapshot data files |
| . vmem | Virtual Machine Paging backup file |
| . vmdk | Virtual disk characteristics |
| vmware.log | Stores the log activities of VM |

# Merits and Demerits of Virtual Machine

# Merits of Virtual Machine

- Virtualization
  - Virtual Machine transforms traditional hardware into virtual environment
- Reducing cost:
  - Several VMs can host on a single physical machine, reducing
  - the cost of data center
- Run Multiple Operating Systems
  - VMs allow us to run multiple operating systems on a single host



# Merits

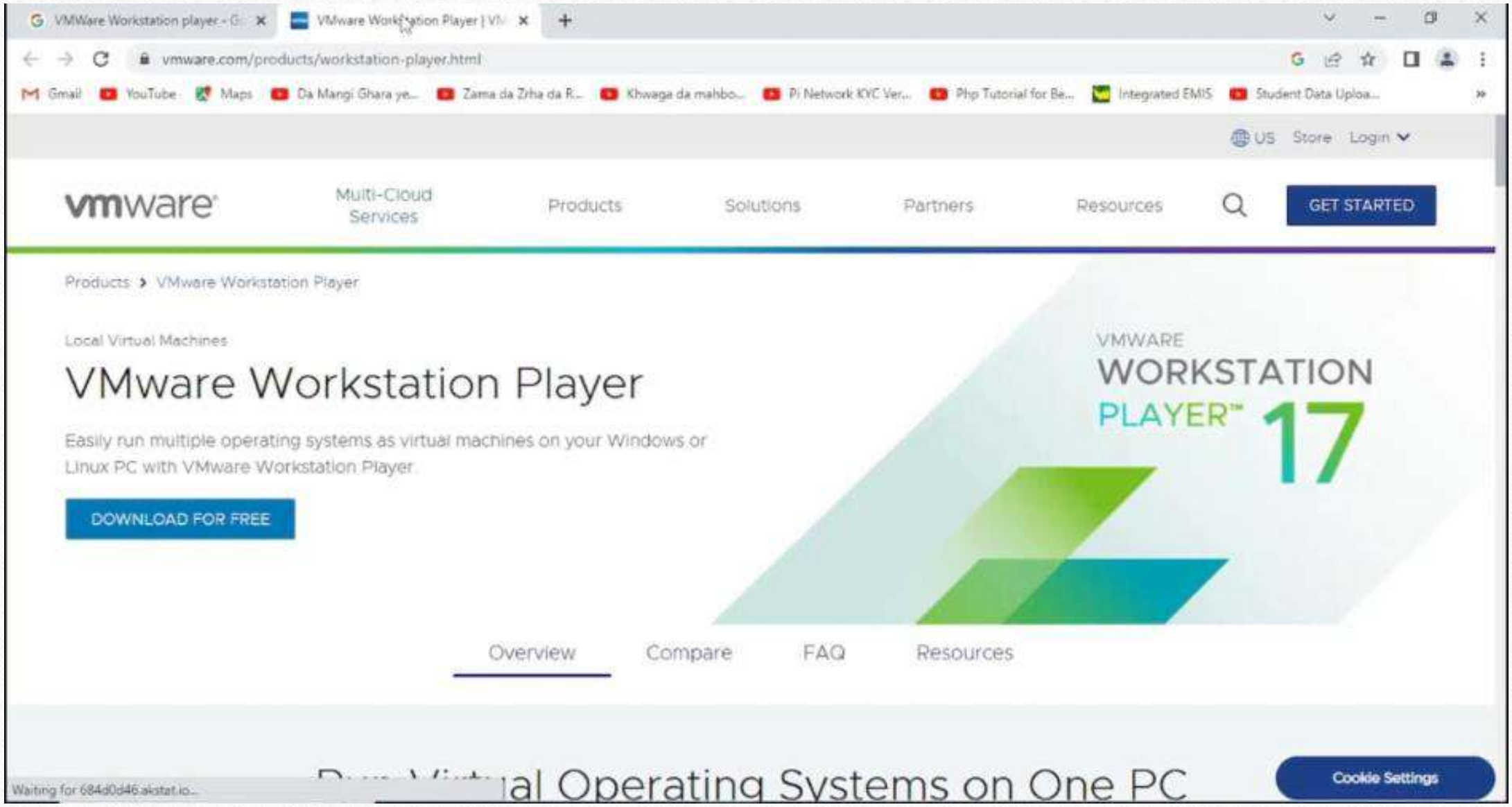
- Portability
  - VM can be easily migrated across the Virtual Environment
- Security:  VMs run in isolation. If security is compromised in one VM, then no effect on other VMs
- Runs incompatible Software
- If one VM fails, then other machines will not be affected
- Virtual Machine enables administrators to simplify backup, disaster recovery, new deployment

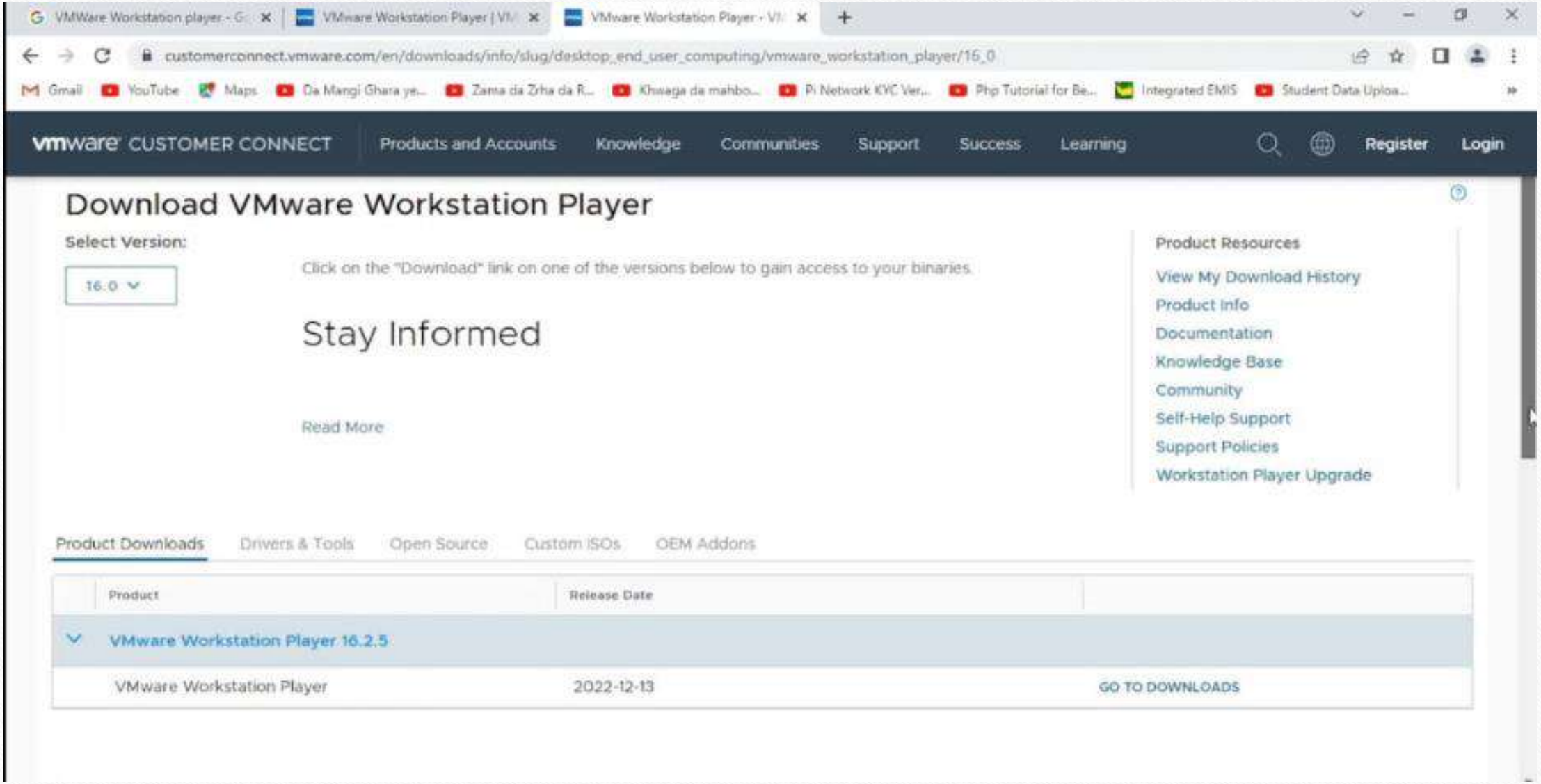
# Demerits of Virtual Machine

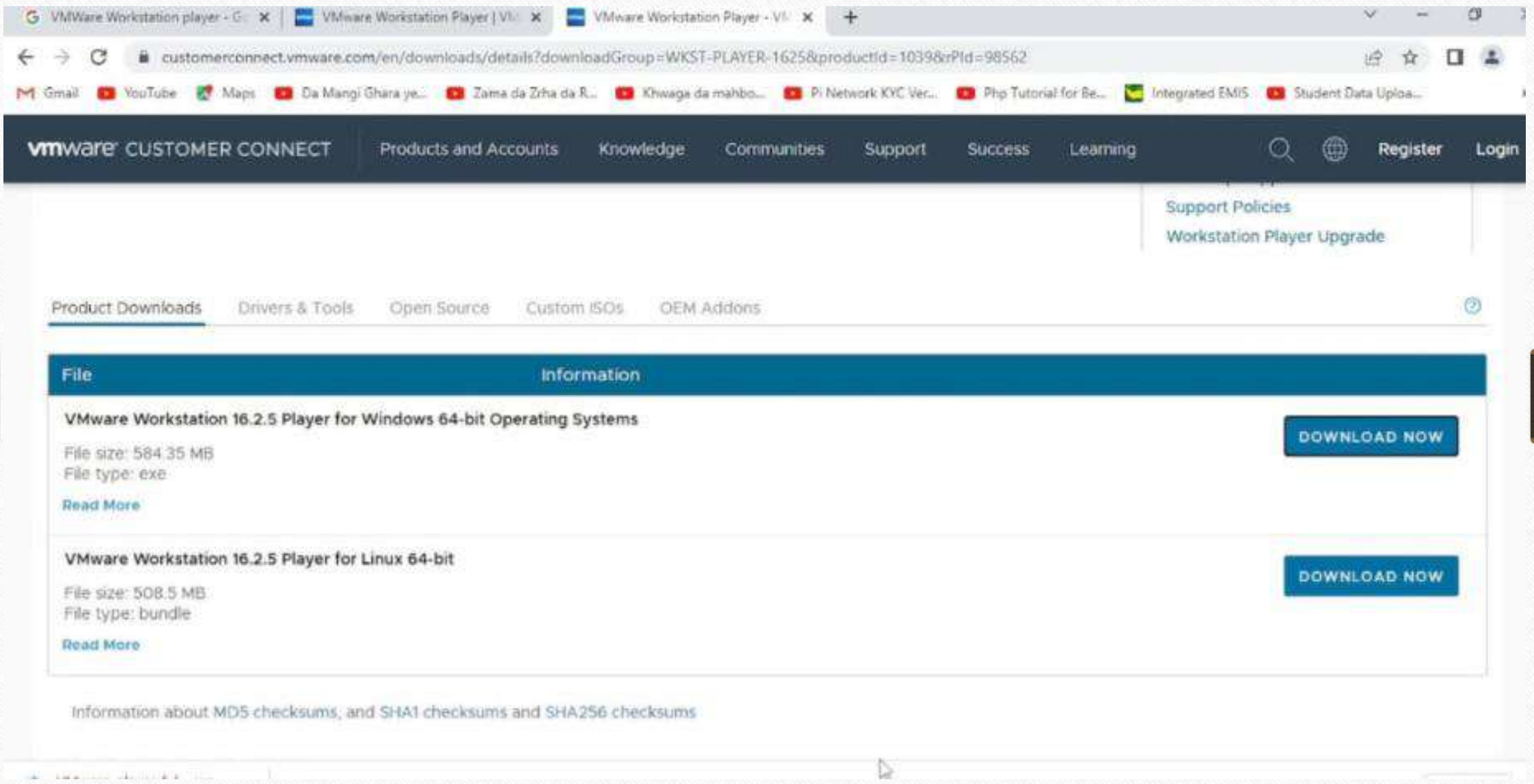
- Cost
  - Running VMs with cloud computing is expensive
- Low Performance
  - VM performance is less than the real machine
- Less Efficiency
  - Less efficient than real machine
- Single point dependency
  - If physical host down, then all the hosted VMs becomes useless

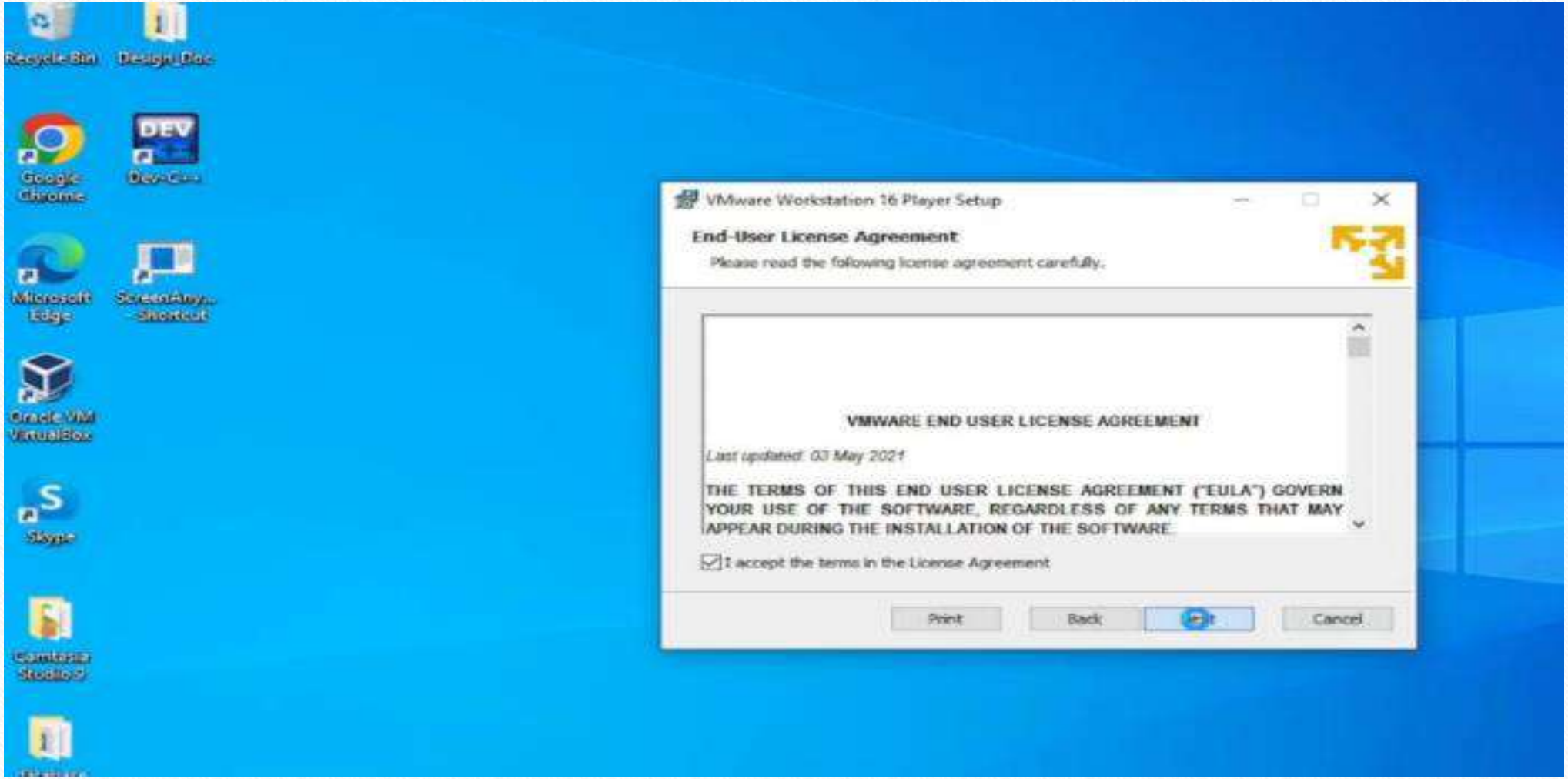
# Practical Lab

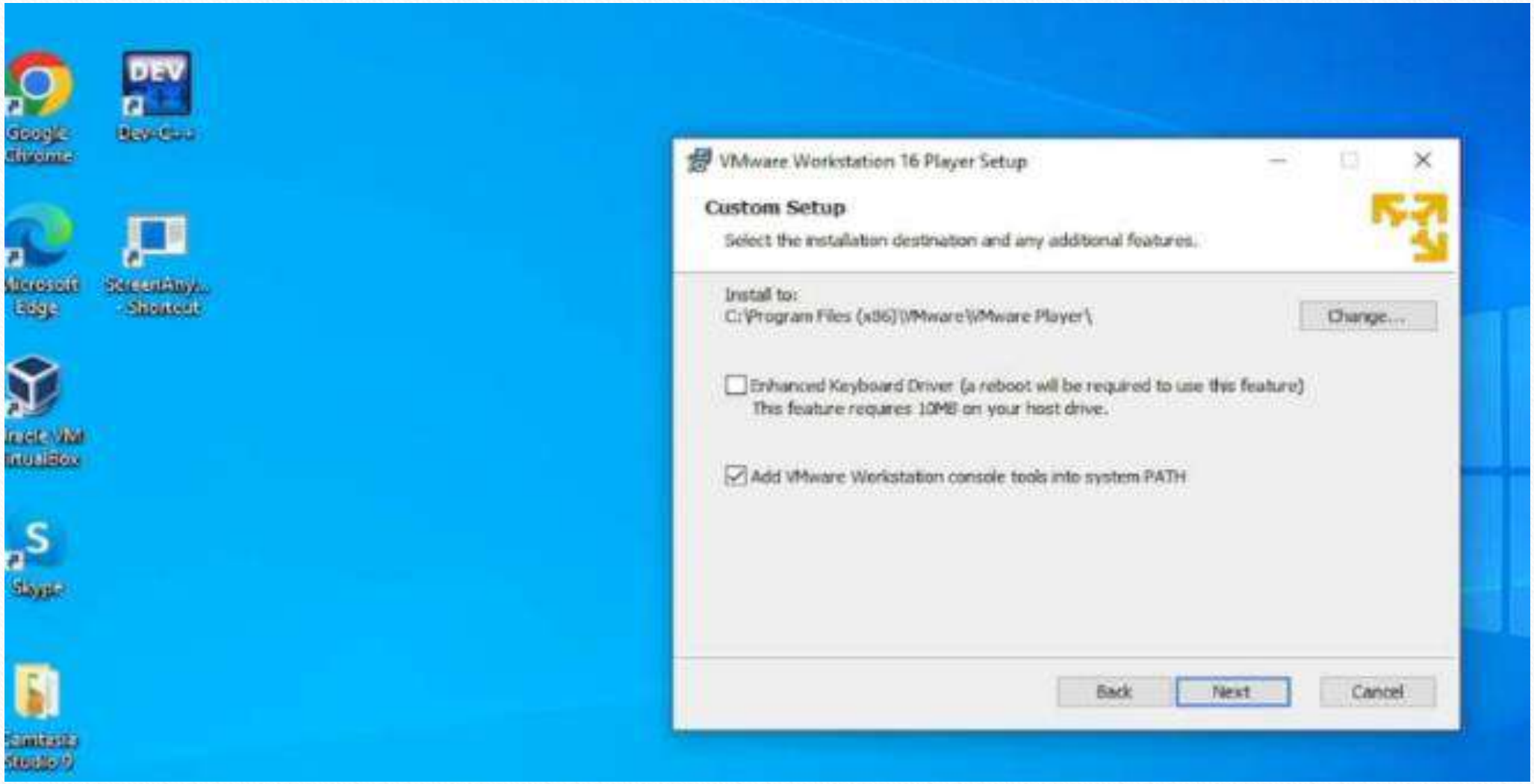
- Downloading and Installing VMware Workstation Player

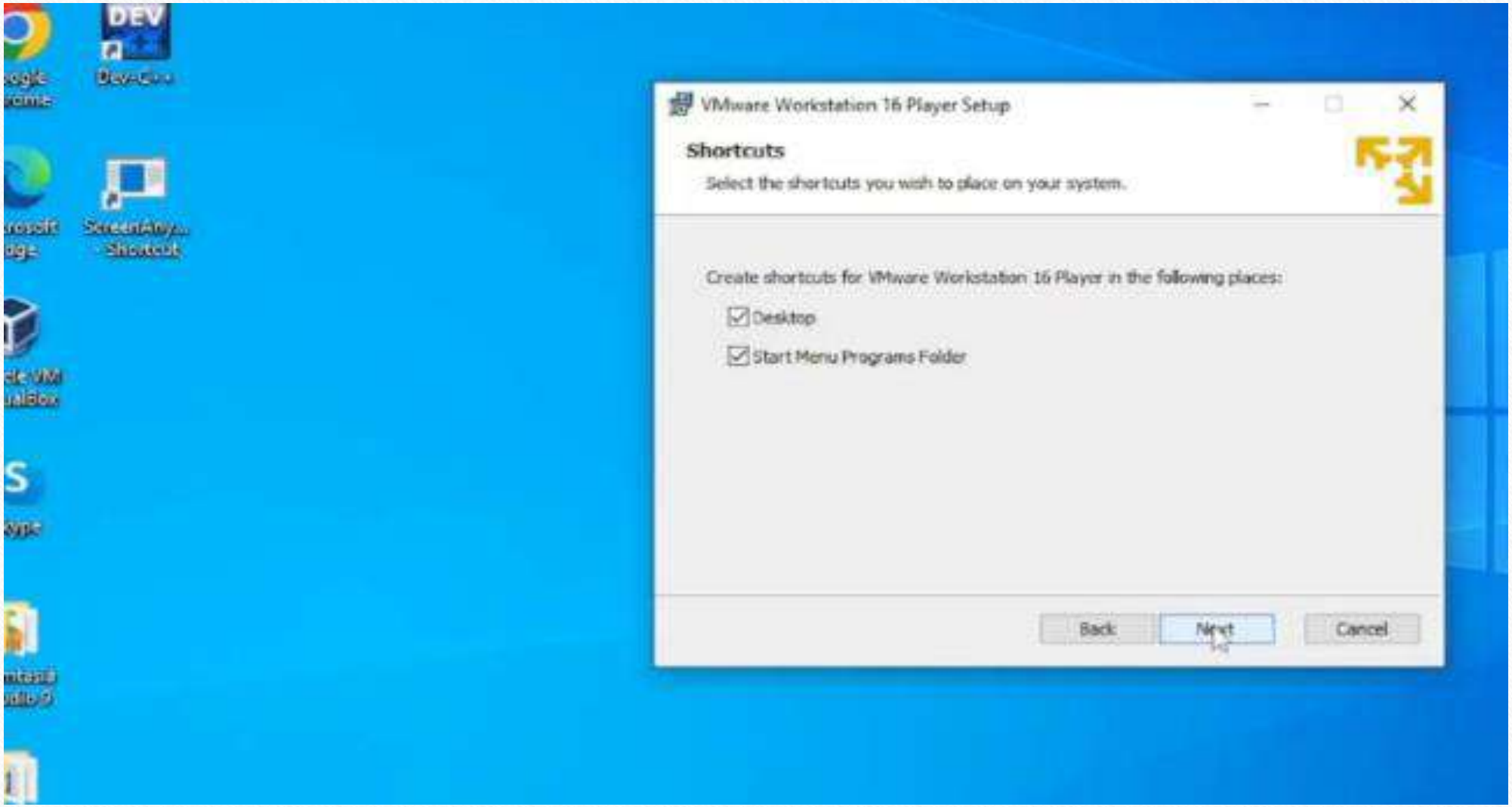

VMWare Workstation player - G
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vmware.com/products/workstation-player.html
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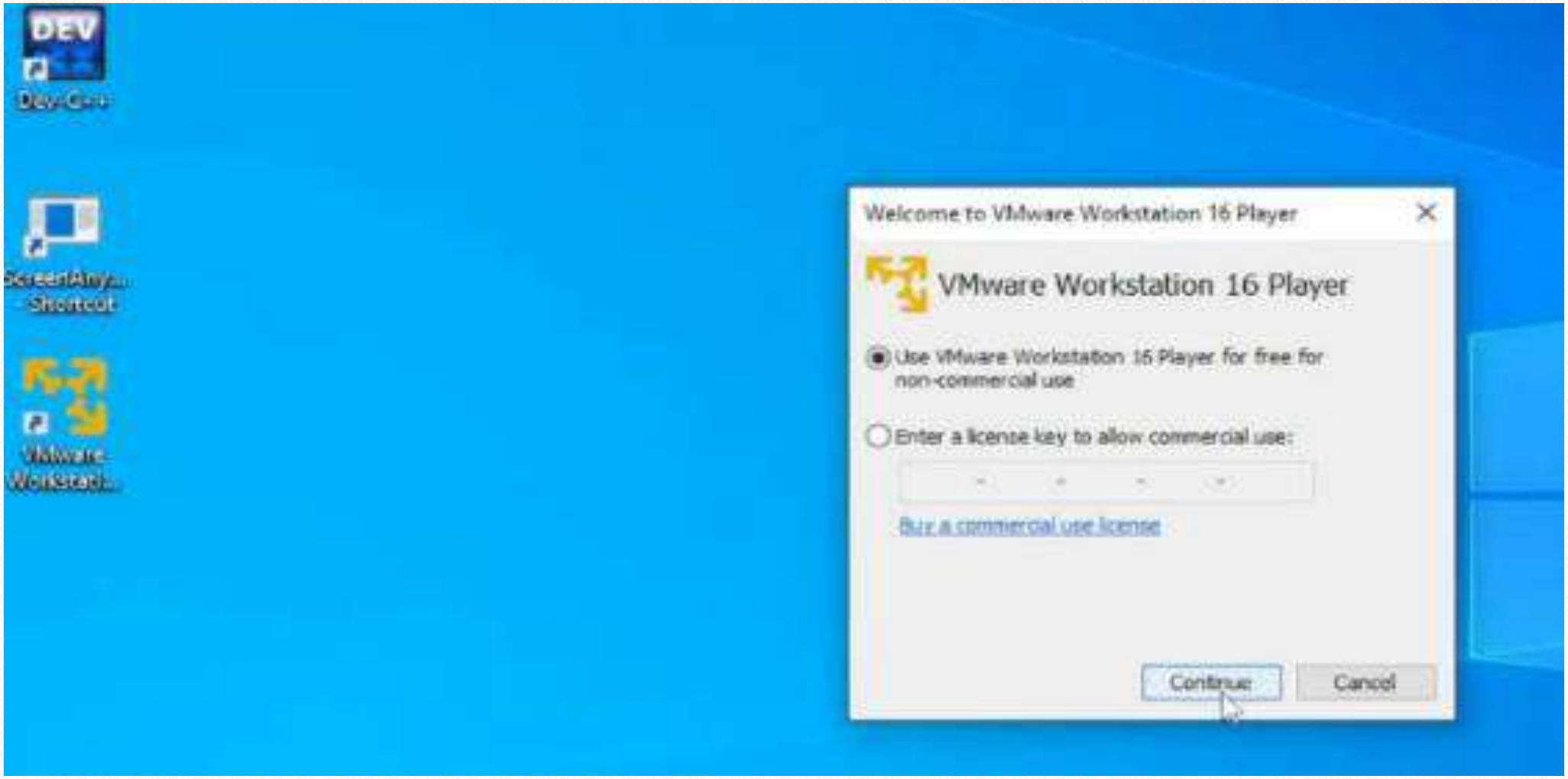

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VMware Workstation 16.2.5 Player for Windows 64-bit Operating Systems
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Information about MD5 checksums, and SHA1 checksums and SHA256 checksums

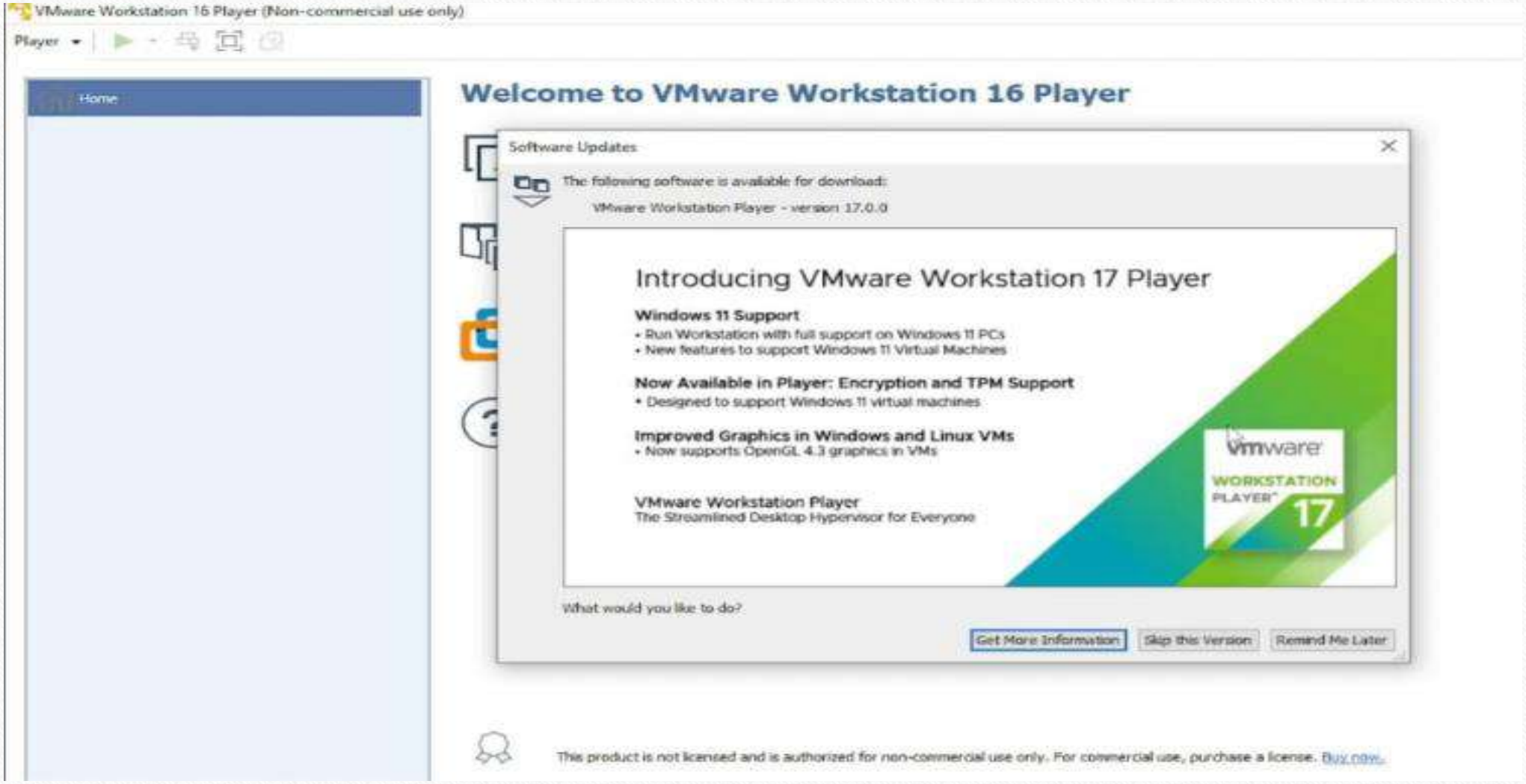

Recycle Bin
Design Doc
Google Chrome
Dev-C++
Microsoft Edge
ScreenAny... - Shortcut
Oracle VM VirtualBox
Skype
VMware Workstation 16 Player Setup
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Last updated: 03 May 2021
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VMware Workstation 16 Player Setup
Custom Setup
Select the installation destination and any additional features.
Install to:
C:\Program Files (x86)\VMware\VMware Player\
Change...
Enhanced Keyboard Driver (a reboot will be required to use this feature)
This feature requires 10MB on your host drive.
Add VMware Workstation console tools into system PATH
Back
Next
Cancel


VMware Workstation 16 Player Setup
Shortcuts
Select the shortcuts you wish to place on your system.
Create shortcuts for VMware Workstation 16 Player in the following places:
Desktop
Start Menu Programs Folder
Back
Next
Cancel


Welcome to VMware Workstation 16 Player
VMware Workstation 16 Player
Use VMware Workstation 16 Player for free for non-commercial use
Enter a license key to allow commercial use:
Buy a commercial use license
Continue
Cancel


VMware Workstation 16 Player (Non-commercial use only)
Player
Home
Welcome to VMware Workstation 16 Player
Create a New Virtual Machine
Create a new virtual machine, which will then be added to the top of your library.
Open a Virtual Machine
Open an existing virtual machine, which will then be added to the top of your library.
Upgrade to VMware Workstation Pro
Get advanced features such as snapshots, virtual network management, and more.
Help
View online help.


VMware Workstation 16 Player (Non-commercial use only)
Player
Home
Welcome to VMware Workstation 16 Player
Software Updates
The following software is available for download:
VMware Workstation Player - version 17.0.0
Introducing VMware Workstation 17 Player
Windows 11 Support
• Run Workstation with full support on Windows 11 PCs
• New features to support Windows 11 Virtual Machines
Now Available in Player: Encryption and TPM Support
• Designed to support Windows 11 virtual machines
Improved Graphics in Windows and Linux VMs
• Now supports OpenGL 4.3 graphics in VMs
VMware Workstation Player
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17
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# week12



# Virtual Networking

# What is Virtual Networking




- **Topics to be Covered**:
  - Creation of virtual versions of networking components
    - Like virtual switches, virtual routers, and virtual firewalls
  - Enables the use of software-based solutions for networking tasks
  - Simplifies network management and improve network agility

# What is Virtual Networking...




- Allows data centers to stretch across different physical locations
- Enables Network Administrators to easily modify the network as required
- No need to switch out or buy more hardware
- Greater flexibility in provisioning the network to specific needs

# Tools of Virtual Network



# Tools of Virtual Network:




- **Virtual switches:** Emulate physical switches and allow virtual machines to communicate with each other and with the physical network.
- **Virtual routers:** Provide routing capabilities between virtual subnets and can also be used to implement network security policies.
- **Virtual firewalls:** Offer firewall protection for virtual networks and can be used to enforce security policies.

# Tools of Virtual Network:

- **Virtual load balancers:** Distribute network traffic across multiple virtual machines to improve performance and availability.
- **Virtual VPNs:** Provide secure remote access to virtual networks from remote locations.
- **Virtual network adapters:** Connect virtual machines to virtual networks, just like physical network adapters connect physical machines to physical networks.

# Tools of Virtual Network:




- **Virtual network security groups:** Provide security for virtual networks by defining access controls for incoming and outgoing network traffic.

- **Virtual network monitoring tools:** Monitor virtual network performance, identify network issues and provide network visibility.

# Virtual Switch & its Types



# Virtual Switch




- Same as physical switches connect physical machines
- Software-based switch that provides network connectivity to VMs
- Allows virtual machines to communicate with each other
- Can be created and managed within a virtualization platform

# Types of vSwitch




- There are there types vSwitches provided by  Hyper-V

- External vSwitch
- Internal vSwitch
- Private vSwitch

# Internal vSwitch




- Provides communication between virtual machines and the Hyper-V host
- Used to isolate virtual machines from each other
- Allow communication within the same host

# Private vSwitch




- Provides network connectivity only between virtual machines running on the same hypervisor host.
- Does not provide access to the physical network
- Used to isolate virtual machines from each other for security or testing purposes

# External vSwitch




- Provides network connectivity between virtual machines and the physical network.
- Used for virtual machines that need access to the physical network

# Standard Switch Architecture



# Standard Switch Architecture...

- Refers to the traditional design and structure of a network switch. The architecture typically consists of the following components:

- Input/Output (I/O) interfaces
- Switch Fabric
- Processor
- Memory
- Power supply
- Operating system
- Management interface

# Standard Switch Architecture...




- **Input/Output (I/O) interfaces**: The ports on a switch where data enters and exits the switch.
- **Switch Fabric**: The internal component of a switch responsible for forwarding and switching packets between the I/O interfaces.
- **Processor**: The component of a switch responsible for managing the switch and its configuration, as well as performing packet forwarding decisions.

# Standard Switch Architecture...

- **Memory**: Used to store the switch's configuration and forwarding information.
- **Power supply**: Provides power to the switch components.
- **Operating system**: The software that runs on the switch

# Standard Switch Architecture...




- **Management interface:** The interface used to configure and manage the switch, such as a command-line interface (CLI) or web-based graphical user interface (GUI).

# Central Aspects of Network Security and Compliance



# Central Aspects of Network Security and Compliance




1. Network Policies
2. Security
3. Traffic shaping
4. NIC teaming
5. Failover

# Central Aspects of Network Security and Compliance

- **Network policies:** A set of rules and procedures that govern the use of the network and help ensure that data and devices are protected.
- **Security:** Measures such as firewalls, intrusion detection and prevention systems, encryption, and access controls to prevent unauthorized access and protect against security threats.
- **Traffic shaping:** Techniques used to manage and control the flow of network traffic to ensure that critical applications receive sufficient bandwidth and to prevent network congestion.

# Central Aspects of Network Security and Compliance




- **NIC Teaming:** The process of combining multiple network interfaces into a single logical interface to provide increased bandwidth, failover protection, and load balancing.

- **Failover:** A backup solution that provides automatic failover to an alternate system or device in the event of a failure or outage.

# Week 13






# Overview of Virtual Storage

# Overview of Virtual Storage




- Refers to the creation of a virtual  representation of storage devices
- a software layer acts as an intermediary between the operating system and the physical storage
- enables the creation of multiple virtual storage devices from a single physical storage resource
- Provides improved resource utilization, enhanced data protection, and simplified data management.

# Overview of Virtual Storage




- **Improved resource utilization:** Virtual storage enables the efficient use of storage resources by pooling physical storage and allocating it to virtual storage devices as needed.

- **Enhanced data protection:** Virtual storage provides features such as data snapshots, replication, and backups that help protect data and ensure its availability.

# Overview of Virtual Storage




- Simplified data management: Virtual storage abstracts the physical storage, making it easier to manage, provision, and allocate storage resources. This reduces the time and effort required for tasks such as adding or removing storage, or resizing storage devices.

- Increased scalability: Virtual storage makes it easier to scale storage resources as needed, allowing for the seamless addition of storage capacity as required.

# Overview of Virtual Storage




- **Flexibility:** Virtual storage enables the creation of multiple virtual storage devices, each with its own characteristics, such as size, performance, and availability, from a single physical storage resource.

- **Improved performance:** Virtual storage can improve the performance of storage-intensive applications by providing features such as caching, data compression, and de-duplication.

# Overview of Virtual Storage

- Improved disaster recovery: Virtual storage makes it easier to create and manage disaster recovery solutions, as virtual storage devices can be quickly moved to alternate locations or stored in the cloud for easy access in the event of a disaster.
- **Simplified data management:** Virtual storage abstracts the physical storage, making it easier to manage, provision, and allocate storage resources. This reduces the time and effort required for tasks such as adding or removing storage, or resizing storage devices.




# Overview of Virtual Storage

# Overview of Virtual Storage




- **Relational databases:** These store data in tables with rows and columns and use structured query language (SQL) to access and manipulate data. Examples include MySQL, Oracle, and Microsoft SQL Server.

- **NoSQL databases:** These databases store data in a non-relational format, such as key-value pairs, document-based, or graph-based. NoSQL databases are often used for handling large amounts of unstructured or semi-structured data. Examples include MongoDB, Cassandra, and CouchDB.

# Overview of Virtual Storage

- **Object-oriented databases:** These store data as objects, which are instances of classes, rather than as tables with rows and columns. Examples include MongoDB and CouchDB.

- **Time-series databases:** These databases are optimized for storing and querying time-stamped data. Examples include InfluxDB and OpenTSDB.

# Overview of Virtual Storage




- **Columnar databases**: These databases store data in columns rather than rows, allowing for faster data retrieval for certain types of queries. Examples include Apache Cassandra and Hbase.

- **Key-value stores**: These databases store data as a collection of key-value pairs, where each key is unique and is associated with a value. Examples include Redis and Riak.

# Overview of Virtual Storage




- **File-based databases:** These databases store data as files on a file system. Examples include JSON or XML files.

# Week 14

Virtual Systems and Services

# Full Virtualization



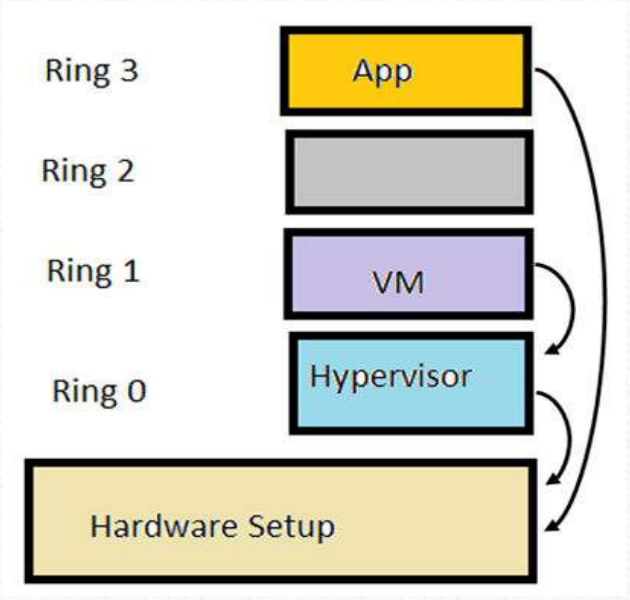
# Full Virtualization




- Full Virtualization is a virtualization technique in which VMs or Virtual Servers are created and run on a single physical server.
- Each VM has its own OS and runs independently
- The software layer providing the virtualization is called as Hypervisor or Virtual Machine Monitor (VMM).
- The hypervisor is inserted b/w the hardware and VMs

# Full Virtualization

- The guest OS is completely decoupled from the underlying hardware
- Each guest OS is unmodified and unaware that it is being virtualized
- The application/user level instructions are directly executed by underlying hardware
- The OS level requests are translated by Hypervisor at run time



# Full Virtualization




- Examples of full virtualization are VMware $ESX_i$, Virtual Box and Microsoft Virtual Server.

# Disadvantages




- The performance of full virtualization may not be ideal as the Binary translation of OS requests takes place at run time
- Any physical server fault or failure can effect all the VMs

# Para Virtualization




Virtaul System and Services

# Para Virtualization




- Para Virtualization is a virtualization technique, used to address the issues found in full virtualization
- Like Full virtualization, this technique allows us to create and run multiple VMs on a single physical server.
- In this technique, the Guest OS in not completely isolated from the under lying hardware and Virtualization layer

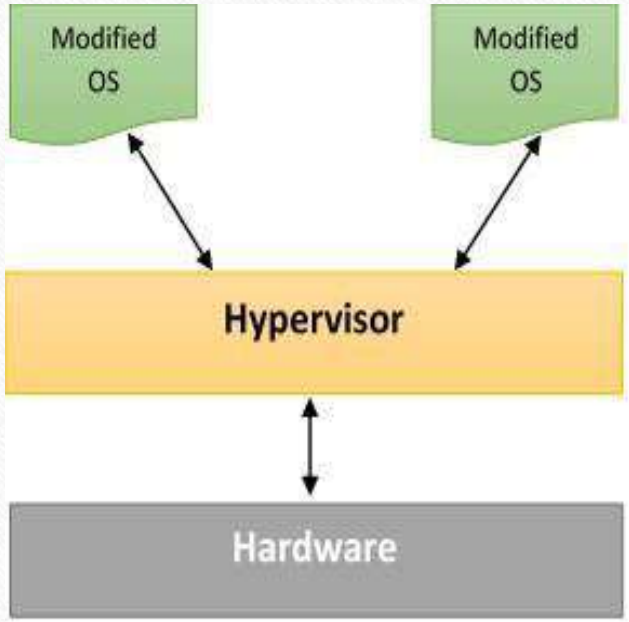
# Para Virtualization




- The Guest OS is modified to work with a VM. A special software called drivers are installed with each guest OS.
- The guest OS is aware that it is running in virtual environment on top of hypervisor
- The hardware resources are shared among the Guest OSs instead of emulating the whole hardware

# Para Virtualization

- Instead of issuing calls to native hardware, the Guest OS issues hyper calls to hypervisor
- The Hypervisor handles hyper calls at compile time and then perform operation on hardware.
- The result is communicated back to VM through hypervisor



# Para Virtualization




- **Advantages**:
  - Virtualization overhead reduced
  - Performance improved
- **Disadvantages**:
  - Modified Guest OS is required to work with hypervisor
  - Limiting support to open source OSs like Linux
  - Less portable and compatible

# Hints




- Para Virtualization cannot work with unmodified OS.
- Xen an open source hypervisor
- **Virtualization overhead reduced**: hypervisor has less responsibility; instead of emulating the whole hardware it receives hyper call & perform operation on hardware

# Full Virtualization vs Para Virtualization



# Full Virtualization vs Para Virtualization

## Full Virtualization

- In Full Virtualization, the Guest OS in a VM is completely decoupled from the underlying hardware and Virtualization layer
- In this technique, the unmodified Guest OS is installed.

## Para Virtualization

- In Para Virtualization, the Guest OS in a VM is partially isolated from the underlying hardware and Virtualization layer
- In this technique, the modified Guest OS is installed.

# Full Virtualization vs Para Virtualization

## Full Virtualization

- It allows the Guest OS to execute independently.
- The Guest OS is unaware of the virtualization
- In this technique, Guest OS issues hardware calls to access hardware

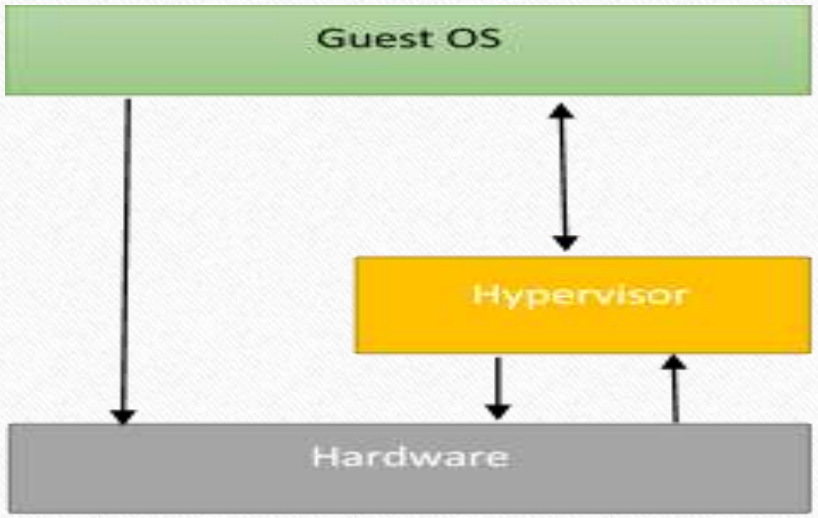
## Para Virtualization

- It allows the guest OS to interact with the hypervisor
- The Guest OS is aware that it is being virtualized
- In this technique, Guest OS issues hyper calls to hypervisor to access hardware

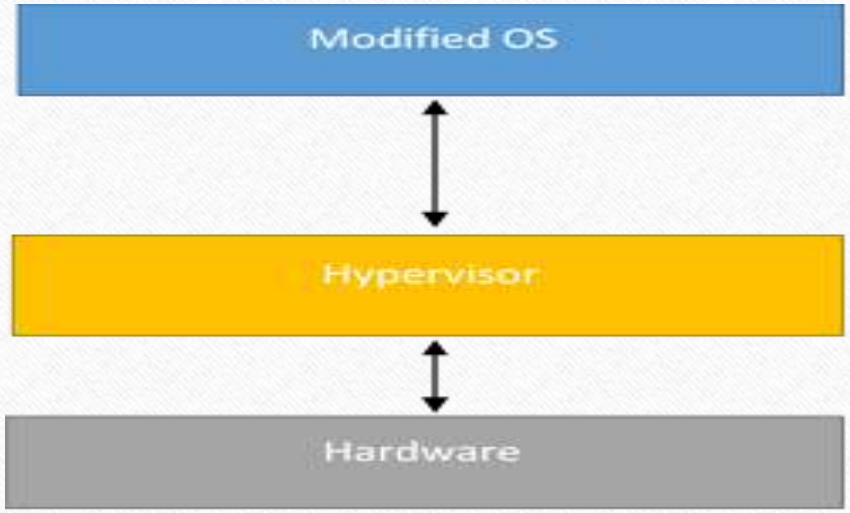
# Full Virtualization vs Para Virtualization

Virtual University

## Full Virtualization




## Para Virtualization



# Full Virtualization vs Para Virtualization




## Full Virtualization

- In Full Virtualization, performance is low.
- Full virtualization is less secure
- It is more portable and compatible

## Para Virtualization

- In Para Virtualization, performance is comparatively high.
- Para virtualization is more secure
- It is less portable and compatible

# Week 15



# Containers




Virtual Systems and Services

# Container



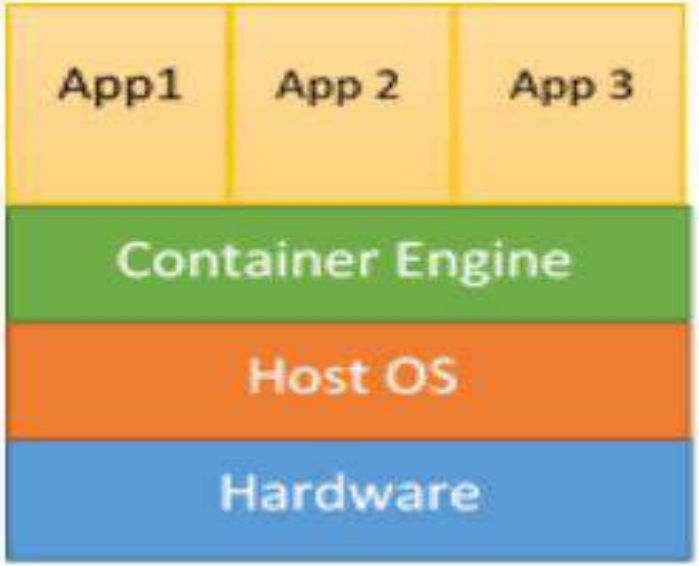

- A Container is a software unit in which Application code, its libraries and dependencies are packaged together
- Unlike Virtual machine, it doesn't include Operating System
- It is a small, fast and light weight unit
- It sits on top of physical server and virtualizes the OS only.

# Container




- It accesses the operating system kernel via container engine
- It uses the physical resources of hosting machine through container engine
- It is portable unit because each container image includes all the dependencies to execute
- Multiple independent containers can run on a single OS
- Cloud applications are built on containers

# Container



# Container Software




- Docker
- Microsoft Azure
- Amazon ECS
- Google Cloud Platform

# Merits and Demerits of Container



# Merits of Container




- **Containers are most light weight**
  - Their image include only the application related data (in MB instead of GB)
- **Less overhead**
  - Requires less system resources to deploy, run and manage
- **Boot Fast:** takes less time in boot up due to less overhead
- **Operational Efficiency**
  - Operationally efficient than VM

# Merits of Container




- **Portability**
  - Container encapsulates all dependencies, own runtime environment; runs on different platforms. It is write once and runs anywhere
- **Isolation**
  - Containers run in isolation
- **Security**
  - Security improve due to isolation, each container runs in self-contained environment

# Demerits of Container




- **Need Dependencies**
  - Packaging workload with supporting dependencies is one of the limitations.
- **Issue with Stateful Applications**
  - It is difficult to containerize stateful applications. They need parallel system to run like database access.
- **Security**
  - Since application layer and host OS is shared across the containers, it opens the door to security breaches across containers

# Virtualization vs Containerization



# Virtualization vs Containerization




- Virtualization makes use of Virtual Machines to convert a traditional hardware into a virtual environment
- Virtualization allows us to run multiple virtual servers on top of a single physical machine

- Containerization makes use of containers to convert a traditional hardware into a virtual environment
- Containerization allows us to run multiple applications on top of a single physical machine

# Virtualization vs Containerization




- Virtualization enables us to efficiently utilize the computer's hardware resources
- Virtualization uses Virtualized software to create an abstract layer over hardware

- Containerization enables us to efficiently utilize the computer's host operating system
- Containerization uses Container Engine to create an abstract layer over hardware

# Virtualization vs Containerization




- Virtualization allows developers to create and deploy VMs faster & more securely
- In Virtualization, an OS is a part of VM
- Use of Virtual Servers or VMs reduces the hardware cost

- Containerization allows developers to create and deploy applications faster & more securely
- In containerization, an OS is not a part of container
- Use of Containers reduces the software cost

# Virtualization vs Containerization




- VMs take more time to boot up
- Containers take less time to boot up